



JOSÉ RIZAL UNIVERSITY
College Division

**Academic Resource and Knowledge (ARK): An AI-Driven
Platform for Enhanced Learning and
Knowledge Management**

**A Project Study
Presented to the Faculty of the
College of Computer Studies and Engineering**

**In Partial Fulfillment of the Requirements
for the Degree of Bachelor of Science in Information Technology**

by

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CERTIFICATE OF ORIGINALITY

We hereby declare that the thesis entitled, **ACADEMIC RESOURCE AND KNOWLEDGE (ARK): AN AI-DRIVEN PLATFORM FOR ENHANCED LEARNING AND KNOWLEDGE MANAGEMENT** is our own original work carried out as Bachelor's student at Jose Rizal University except to the extent that assistance from others in the thesis' design and conception or in style, presentation and linguistic expression are duly acknowledged.

All sources used for the thesis have been fully and properly cited. It contains no material which to a substantial extent has been accepted for the award of any other degree at JRU or any other educational institution, except where due acknowledgement is made in the thesis.

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ABSTRACT

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The Academic Resource and Knowledge (ARK) system is a digital library platform powered by AI, aimed at facilitating more effective information access, resource discovery, and user engagement through the use of a variety of advanced technologies. The platform incorporates an intelligent smart search using NLP, machine learning, and semantic reasoning to understand the queries and fetch the relevant documents along with a hybrid recommendation engine that offers personalized suggestions based on semantic similarity, category, trend, popularity, and authorship. The event management module allows proposal



submission, activity tracking, and automated notifications, and administrators receive simplified tools for review and publication. The system quality was measured against the ISO/IEC 25010:2023 software quality model criteria, where forty respondents rated functional suitability, reliability, usability, security, and performance efficiency at a high level, with an overall mean of 4.51 (Strongly Agree). The results show that ARK is a reliable, secure, and friendly-user platform that supports the digitization of library services and academic engagement.

Keywords: Academic Resource Management, AI-Driven Library System; Smart Search, Machine Learning, Natural Language Processing, Hybrid Recommendation System, Event Management, Library Analytics, ISO/IEC 25010, Digital Library Platform



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Chapter 1

INTRODUCTION

This chapter discusses important details about the study, including the inspiration and background of the study, the objectives of conducting the research, scope and system limitations, an enhanced conceptual framework, the significance of the study, and the definition of specific terms.

Background of the Study

Technology's rapid advancement, particularly in AI, cloud computing, and digital platforms, had transformed education and information management. Libraries, which were once reliant on manual cataloging, had begun embracing digital repositories, AI-powered indexing, and e-resources to enhance accessibility [1]. With the widespread use of smartphones and high-speed internet, students and researchers had increasingly demanded faster and more structured access to information, making it essential to modernize academic resource systems [2]. However, traditional OPACs often fell short of expectations, as several studies noted that although users were aware of these systems, their actual usage remained low due to limited training and unfamiliarity with advanced features [3], [4].

To address these challenges, the Academic Resource and Knowledge (ARK) system was proposed as an AI-driven platform focused on intelligent search, natural language processing (NLP), and personalized recommendations for the Kaban ng Hiyas Congressional Library. In many areas across the Philippines, libraries still offered offline-only access, which significantly restricted



research productivity [5]. By integrating AI techniques such as smart search assistance, automated tagging, and personalized recommendations, ARK sought to guide users more efficiently through academic resources [8].

Furthermore, ARK supported mobile learning through cross-device accessibility and cloud-based storage [9], while also addressing scalability to meet evolving research demands [10]. It also aimed to build a vibrant academic community by introducing feedback and resource rating systems that encouraged user participation and collaboration [11]. Through the combination of AI technologies and community-driven contributions, ARK ultimately sought to modernize academic resource management and enhance the overall research and learning experiences of students, researchers, and educators.

Statement of the Objectives

This study aims to create a web-based system for the Kaban ng Hiyas Congressional Library to improve access to resources through smart search and personalized recommendations.

General Objective

The goal of this study is to design an AI-driven library system for the Kaban ng Hiyas Congressional Library that improves access to and organization of library materials through smart search and personalized recommendations. The system



will also enable users to view upcoming library events and submit event proposals while providing administrators with insights to identify areas for improvement.

Specific Objectives

1. To develop a system with a smart search capability using NLP, AI, and Machine learning that interprets user queries, improving search experience and making it easier to find relevant resources.
2. To develop a data-driven recommendation system using machine learning method that suggests popular books based on usage trends and personalizes recommendations based on recent user interactions. Include an analytics dashboard to help administrators optimize frequently used categories.
3. To integrate an event calendar to display upcoming library events and activities. Allow users to submit event proposals or requests for library-related activities. Support library outreach and program promotion by increasing user engagement with events.
4. To evaluate the developed system using ISO 25010:2023 in terms of:
 1. Functional Suitability
 2. Performance Efficiency
 3. Compatibility
 4. Interaction Capability
 5. Reliability
 6. Security



7. Maintainability
8. Flexibility
9. Safety

Conceptual Framework

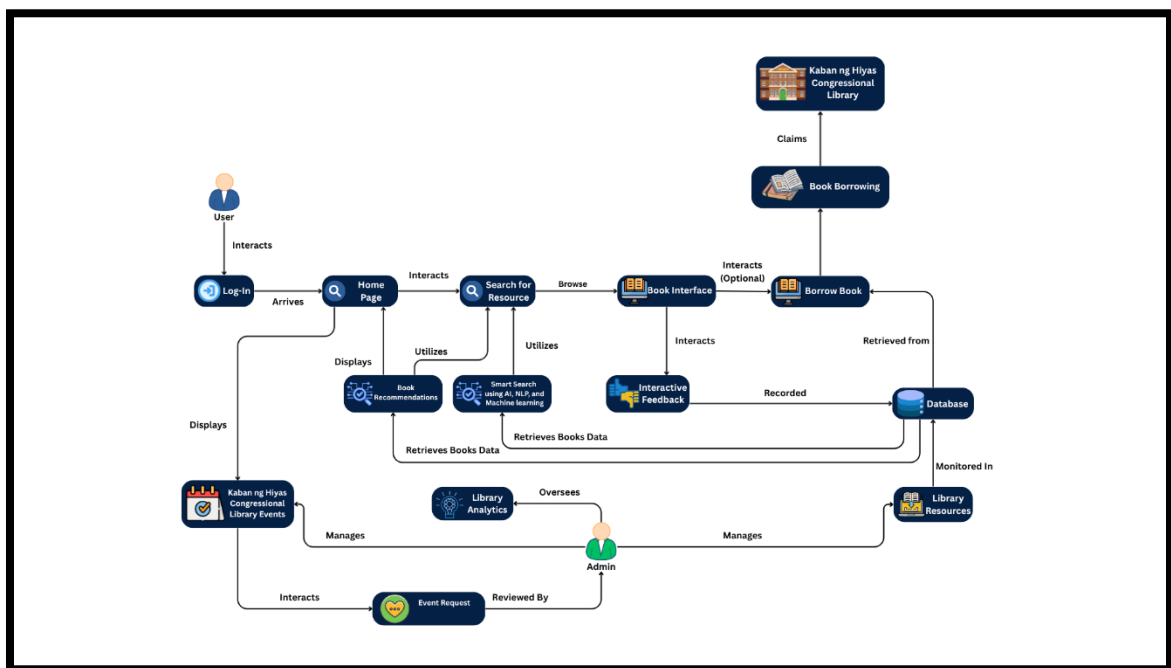


Figure 1.1. Conceptual Framework

The conceptual diagram (Figure 1.1) illustrates how the ARK Library System bridges the gap between users and administrators in the overall management of library resources. In order to take advantage of all services, users must first log in and be established as a user. When users log in, the first thing they see is a dashboard that contains books and other features of the library. Users can browse for books, rate them, leave comments, and save their favorites. If the



user is not logged in, they can browse, but cannot perform any interactive action such as commenting or rating.

There are two main search options for the system that enhance how users find information. The first is a normal search function that a user can look for books by entering certain keywords such as book names, names of authors, and subjects. The second is an advanced search option that is powered by Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML). This advanced intelligent search option allows users to ask a question in a more intuitive or natural way, such as “Who is Jose Rizal?”, “How to predict weather?”, or “What is Python programming?”. Then the system will read the question and provide a direct, response or recommendation for reading material that relates to the question being asked. After the user performs a search, users can borrow a book, use it from the library, and engage with the book by rating and leaving comments. Additionally, the book recommendations use a user's last view activity and the books that the user has marked as liking.

In addition to features oriented toward library materials, the system also provides for event interaction in the library. Users can see the library's upcoming events, make requests for events, and participate in events approved by the admin or person in charge. An administrator is in charge of managing all aspects of administration, including adding, modifying or deleting any books stored in the database, reviewing event requests, and monitoring library analytics. These features together enable the ARK Library System to promote and engage users to



actively use library programs and to promote intelligent and efficient book searching and recommendation methods using AI technologies.

Significance of the Study

This study aims to create a web-based library system with smart recommendation and easy navigation. The following could potentially benefit from the research:

The Municipality of Mandaluyong City may benefit from updating its public library services, which would help people learn how to use technology better and give them better access to information. The system will support the city's efforts in fostering technological advancement and enhancing public services.

Kaban ng Hiyas Congressional Library may experience more efficient library management and user engagement. The smart recommendations, automated book search, and availability check will make the user flow run more smoothly, reducing manual work and raising the quality of service.

Library administrators may utilize the system to streamline library processes. The system features may reduce redundant tasks, and administrators may concentrate on developing the library's resources and user experience.

Mandaluyong City students may utilize easier access to library resources that enable them to search for books in a timely manner, view available books, and receive recommendations. This system may support their academic and personal learning needs by offering a convenient and user-friendly library experience.



Researchers may gain easier access to literature and references, which may enhance their research efficiency. The smart recommendation system will help them discover related works, while remote access will allow them to retrieve resources anytime, anywhere.

Future researchers may use this study as a reference for similar or related projects. The findings and contents of this study could serve as a guide in conducting future studies on web-based library systems or related technological advancements.

Scope and Limitations of the Study

This study focuses on the development of a web-based AI-driven library system for the Kaban ng Hiyas Congressional Library to enhance access to academic resources and improve library management processes. The system, titled Academic Resource and Knowledge (ARK), integrates smart search capabilities that interpret user queries to deliver accurate and relevant results. It also features a personalized recommendation module that suggests popular books based on usage trends and individual user activity, aiming to increase user engagement with library materials.

In addition to resource management, the system incorporates an event management component. This module displays upcoming library events and activities and allows users to submit event proposals, supporting community engagement and library outreach efforts. An administrative analytics dashboard is



included to provide insights into user interactions and resource utilization, helping library staff optimize services and identify areas for improvement.

The system is designed for evaluation based on the ISO/IEC 25010:2023 quality standards, focusing on functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety.

Limitations

The ARK system, while offering advanced digital features, had certain functional limitations during its development. It did not include advanced capabilities such as a scan-to-search feature for identifying books through barcodes or covers, a speech-to-search option for voice-based queries, or a user content-sharing module that would allow students and faculty to upload and share academic materials. Additionally, the system lacked a mobile application version and an AI-powered chat assistant for real-time support and user guidance.

Definition of Terms

This study provides operational definitions for the following terms:

ARK (Academic Resource and Knowledge). It is an AI-driven web-based system designed to improve academic research by offering intelligent search, personalized recommendations, and user-shared insights.

Artificial Intelligence (AI). In this research, it refers to the implementation of machine-learning algorithms and data-driven methods within the ARK platform to



boost system functionality. AI is utilized to drive functionalities such as smart recommendations and search assistance, making it possible for the platform to provide more precise, personalized, and effective access to academic resources.

Features. This term refers to the functional components of the ARK platform, including AI-powered search assistance, smart recommendations, and discussion forums.

Knowledge Sharing. Knowledge sharing is when users contribute their insights, comments, or feedback on a specific book or academic resource to help others.

Natural Language Processing (NLP). Natural Language Processing (NLP) is a field of AI that enables computers to understand and process human language.

Physical Library Integration. This term refers to the seamless connection between the library's physical collection and the ARK platform, allowing users to access information about on-site books, materials, and reservation options online.

Resources. In this study, it refers to the academic resources provided by the library and external sources, such as textbooks, research papers, journals, and other educational content available through the ARK platform.

Semantic Search. Semantic search is a method that focuses on understanding the meaning and intent behind a user's query to deliver more relevant results.

Smart Recommendation System. In this study, it refers to the feature that suggests relevant academic resources, such as textbooks, research papers, and journals, based on the user's interests, search history, and study patterns.

Traditional OPAC. A basic online catalog system for searching library resources by title, author, or subject.



User Experience (UX). This term refers to the overall interaction and satisfaction of the users with the ARK platform, including ease of use, efficiency, and accessibility.

User Interface (UI). In this study, it refers to the visual and interactive design of the ARK platform, including the layout, navigation, and accessibility features that enhance usability.

Web-Based System. This term refers to an online platform that users can access through web browsers on different devices using the internet, enabling them to search, interact, and navigate academic content.



Chapter 2

REVIEW OF RELATED LITERATURE

Utilization, Awareness, and Impact of Digital Library Systems

This theme explores the utilization and digital transformation of library services in academic settings, focusing on how systems like Online Public Access Catalogs (OPACs), Learning Management Systems (LMS), and integrated digital platforms are accessed, perceived, and used by students and faculty. Studies highlight user awareness, satisfaction, frequency of use, and the academic benefits of these tools, while also examining innovations such as AI-driven searches and enhanced user interfaces. Research, particularly from the Philippines, reveals challenges like limited digital literacy, infrastructure gaps, and the need for better user training and support services to ensure equitable and effective access to digital library resources.

The transition of libraries to digital platforms was explored through the study titled “The Prospects of Libraries in the Digital Era”, finding that digital libraries enhance accessibility and engagement through e-books, online journals, and cloud-based catalogs. AI-driven search functions and interactive portals improve information retrieval, though challenges such as cybersecurity risks, digital literacy gaps, and funding constraints remain. Data vulnerabilities pose security concerns, while limited digital literacy hinders effective resource use. The study suggests enhancing cybersecurity, implementing literacy programs, and forming technology



partnerships to sustain digital library services. Its relevance highlights both the benefits and challenges of digital library transformation, supporting our aim to develop a secure, user-friendly academic resource platform with improved accessibility and engagement. [13]

A study about empowering literacy among students emphasizes that digital libraries significantly enhance reading literacy by providing easier access to diverse materials, addressing the declining interest in conventional libraries. Utilizing the waterfall method, the project included phases such as problem definition, analysis, design, development, testing, implementation, and maintenance. Key findings reveal that features like online reading capabilities, personalized book recommendations, and a user-friendly interface boost student engagement, resulting in a 100% success rate in functional testing and a 95% usability score. The study also points to the role of digital libraries in promoting independent learning and critical thinking, and mentions challenges such as the requirement for continuous updating and encouraging digital literacy. The study proposes future improvements could widen the reach and influence of the library on student literacy, underlining the potential of digital libraries to transform educational resources and enable the creation of an accessible academic resource platform. [14]

The following study titled "From Card Catalogs to Clicks: The Modern Library Website's Influence on Information Access." points out the pivotal contribution of library websites to enabling efficient and user-friendly access to a



collection of digital resources, including books, scholarly articles, and multimedia. Findings reveal that contemporary library websites have deeply enhanced user experiences through many different features by increased search capabilities make users able to find useful information effectively, and accessible design guidelines ensure interfaces remain intuitive and feasible. Interactive functionalities, such as incorporating chat assistance and user comments, build enhanced interaction and community creation among users. In addition, the study values accessibility to all user groups with the observation that library websites have leveled the field in terms of access to information by disregarding geographical boundaries. Nevertheless, there are also limitations such as calls for constant digital literacy programs, privacy issues where user information could be at risk, and security needs to be strong to preserve sensitive data. The authors concluded that library websites are crucial knowledge gateways whose importance lies in their role of keeping pace with the changing requirements of users within a more electronic society. [15]

Ayo et al. conducted a study to tackle the issues of libraries in adapting to the digital age. The study highlights the importance of an integrated library management system to improve library services and user experiences. Major findings confirm that the system was designed through a thorough needs assessment of 98 participants, which highlighted some of the painful issues such as challenges in using virtual libraries, accessing resources, and with the need for better user instruction on technology use. The study established that 56% of the



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students would be willing to use new technologies if they were well instructed, highlighting the need for educating users on technology adoption. Further, the study found inadequate modern technologies like 3D printing facilities as barriers to efficient library services. The iLMS developed has features such as an online public access catalog, feedback systems by users, and automated reminders for overdue materials, which as a whole increase user interaction and consolidate library operations. The study concludes that the iLMS not only solves existing problems but also positions the library to meet the future changing needs of their constituents with more emphasis on ongoing improvement and adjustment in library services. [16]

The response of university libraries during the COVID-19 pandemic was examined in the study entitled "Transition of University Libraries to Adapt to the New Normal: A mid- and post-COVID-19 Response." The study emphasizes how libraries evolved their services by shifting from physical to online materials and adopting new technologies to keep serving users even in physical closures. Major findings show that libraries effectively involved users through innovative practices such as the creation of virtual reference services that enabled users to request services remotely, enabling ongoing support of research and information requirements. Online training sessions were also carried out by libraries to enlighten users about how to explore digital resources as well as properly use new technology, which played a significant role in improving students' and teachers' digital literacy. Furthermore, other libraries expanded their digital collections,



giving people access to e-books, online journals, and databases that were very useful for schoolwork during the pandemic. Libraries also used interactive platforms to encourage people to talk to each other and allow people to access materials, have discussions, and work on projects from home. In spite of these successes, the study also documented limitations, including the digital divide in reaching some users, especially those who do not have consistent internet or technology, and the problem of upholding service quality in a dynamic environment. The study recommends establishing robust digital infrastructures, enhancing user education on technology, and ensuring equal access to resources as potential solutions to these issues. The study found that libraries can better meet the changing needs of their communities with the new concepts created during the pandemic. This shows how important it is for library services to be flexible. [17]

Riady conducted a study with a purpose to model the factors of the usage, user satisfaction, net benefits, and user loyalty towards digital library services. Using Delone and McLean's model of information system success, the researchers established a framework and tested a survey instrument that collected data from 2,105 Indonesian Open University (IOU) students. The results verified all hypotheses, with service quality being the most powerful predictor of user satisfaction, which was the strongest predictor of actual usage. Moreover, the study established that net benefits were mainly driven by usage, with user loyalty largely depending on net benefits. The study aids useful information about quality



criteria that stakeholders can employ to build better digital library services, ultimately enhancing user satisfaction, usage, net benefits, and loyalty. The study highlights the need for ongoing monitoring and enhancement of digital library services to address changing user needs in a more digital environment. [18]

Lagas III and Isip conducted a study that describes the maturity of digital projects in Philippine academic libraries based on the sustainability of digitization activities. The study used a descriptive-comparative approach to gather information on digitization efforts from research and higher education institutions. It found that the majority of libraries encountered significant challenges, including insufficient personnel, resources, and enough space or equipment. The study found that most digitization initiatives were reactive, primarily triggered by external events like the COVID-19 outbreak, rather than being part of a planned framework. Recommendations stated by the study involve planning consultancy programs, enhancing digitization training in library schools, and developing communities of practice for resource sharing. The findings reinforce the need for Philippine academic libraries to revisit their short- and long-term objectives toward digitization, making sure that digital services are well-integrated to address changing user needs. The topic remains relevant after the pandemic, as people still need accessible digital information, which affects library services and what users expect in an academic setting. Locally, the study pinpoints the need for libraries in the Philippines to evolve with technological developments and user



interests in order to remain relevant and useful in delivering educational materials and assistance to students and academics in an ever more digital world. [19]

The study titled "Data Security Mechanism in Digital Library: Issues and Challenges" addressed the security mechanisms necessary for safeguarding digital libraries. The study identifies essential elements of information security, which involve the availability, integrity, and confidentiality of information. The study emphasizes the need to have strong identification and authentication measures to avoid unauthorized access and to ensure that only authorized personnel have access to critical information. The paper explains a range of security practices, including encryption, safe communication protocols, and routine data backups, to protect digital assets from threats such as data breaches and cyberattacks. It also emphasizes the importance of adequate standards and policies in controlling access and management of digital data and ethical implications in designing and applying security measures. Beneficial and ethical security measures are essential for protecting the privacy and integrity of information in digital libraries [20].

A following study also analyzed the knowledge and challenges that librarians experience in conducting HyFlex library operations. The findings showed that most of the respondents were female, 31–40 years old, with 11–15 years of tenure, and demonstrated high levels of knowledge and preparedness for HyFlex operations. Although they were proved prepared, librarians faced serious obstacles, such as insufficient funding, slow internet speed, and limited user



interaction with library resources. The discussion emphasized that although librarians have the required competencies for HyFlex operations, outside factors hinder effective implementation. In response to these issues, the study suggested an action plan designed to improve librarians' competency through ongoing professional development, with the creation of detailed plans for accessible print collections, and ensuring sufficient budget and support from school administration. This action plan is applicable in that it brings empirical evidence which can guide library and school administrators regarding the distinct needs of Higher Education Institutions (HEIs) in the Calabarzon area. Locally, the results are especially pertinent given the current effects of the COVID-19 pandemic, as they highlight the need to evolve library services to cater to changing demands of students and instructors in a blended learning context. Through a local focus on challenges and opportunities, the study contributes to the general library innovation and adaptation conversation in reaction to shifting educational contexts. [21]

Dharmaraj authored a literature that focuses on the development of library services with the use of the KOHA Integrated Library System (ILS) by the Pangasinan Provincial Library. This open-source system, made available for free by the National Library of the Philippines, allows users to conduct online searches, browse e-resources, and discover digital collections remotely, expanding the library's coverage beyond its walls. KOHA was described as a "game-changer," as it facilitates dynamic services that enable users to retrieve information at any time,



which is vital to keeping libraries relevant in the digital era. The article highlighted that 60% of those who visit the library use the Internet for research, whereas others interact with hard copies or social media, showing multifaceted user interactions. The study also emphasized the need for modern librarians to work with digital and physical resources and highlighted the necessity of cybersecurity skills for safeguarding digital files. Ongoing professional development and challenges youth to study Bachelor of Library and Information Science (BLIS) are advocated to cater to the increased need for professionally qualified librarians who can make a connection between traditional services and digital innovation. The Philippines' dedication to inclusiveness in libraries was also discussed, which advocates the conversion of libraries to adapt to changes in technology and emphasizes their status as vital learning centers and hubs for community participation. [22]

Student preferences during educational technology integration at San Sebastian College - Recoletos de Cavite were examined, focusing on the use of digital and physical books in academic settings. The study, from a survey of senior high school and college students, indicated a major preference for digital resources with 60% frequently using digital materials and 40% for physical resources. Statistical analyses, including ANOVA, showed discipline-specific variations, particularly that Bachelor of Science in Criminology students accessed physical resources more often. A strong positive relationship was found with a value of $r = 0.707$ between digital resource availability and academic performance, which means access to digital material improves learning achievement. Limitations



involved the singular nature of the cultural setting within the institution and using self-reported measures, which compromise generalizability and accuracy. The study highlights the importance of aligning library services with student preference, suggesting customized interventions to enhance resource delivery and ensure equitable access. The study also correlates with the Philippines' commitment to Sustainable Development Goals (SDGs) and provides useful insights for future policy and program development in library services. [23]

The study titled “Knowledge, Attitudes and Practices Towards Open Educational Resources in Higher Education Institutions in the Philippines”, examined academic librarians' awareness and usage of Open Educational Resources (OER). Emphasis was placed on the librarians' awareness, attitudes, and practical application based on the 4Rs model: reuse, revise, remix, and redistribute. While librarians had a positive attitude towards OER, the results showed that their practical application of OER was moderate. Most librarians reported awareness of OER; however, they hardly applied it in school libraries. Identified challenges were lack of enough exposure, limited access to digital tools, and poor institutional support. Moreover, the study showed that demographic variables like gender, level of education, and years of professional experience had no significant influence on librarians' use of OER. On the other hand, the availability of both print and non-print resources in libraries was found to have a positive effect. The authors recommended more training programs, more funding, and better institutional support to facilitate librarians acquiring skills related to the



use of digital resources. This study supports our study by showing that the successful technology utilization of libraries requires the coexistence of expert professionals and supportive institutions. [24]

In their 2024 study, Estrada et al. investigated the impact of library resource usage on academic performance at a public university in the Philippines. The research, titled "Enhancing Academic Success through Utilization of Library Resources in a Public University in the Philippines," aimed to explore how the frequency of library use influenced undergraduate students' academic outcomes. Analyzing data from 800 students between the academic years 2015–2016 and 2018–2019, the study employed a correlational descriptive research design. The results revealed a strong positive correlation between library usage and students' General Weighted Average (GWA) (0.769), a very high correlation with achieving dean's list status (0.819), and a moderate correlation with passing board exams (0.592) [25]. The study emphasized that students primarily used library resources for reading, research, and exam preparation, which played a significant role in their academic success. The authors concluded that enhancing library resources and services is crucial for improving student performance and recommended further investment in these areas to support academic achievement.

Roa et al. explored the effectiveness of Learning Management Systems (LMS) in the University of Science and Technology of Southern Philippines [26], focusing on the impact of e-learning on student motivation and learning outcomes.



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The study involved 157 university students and utilized a questionnaire based on the Technology Acceptance Model (TAM) to assess perceived usefulness, ease of use, course content, student motivation, and faculty capacity. Results indicated that user-friendly LMS interfaces and adequate technical support significantly enhance the learning experience. However, challenges such as digital infrastructure and the need for faculty training were identified as barriers to effective LMS implementation. The authors emphasized the importance of continuous improvement in LMS design and faculty support to foster a more engaging and effective online learning environment—elements that are essential to better outcomes in higher education.

The study conducted by Doliente et al., "Exploring Students' Utilization of Online Public Access Catalog in the Learning Commons," focused on the usage of the OPAC system at a higher educational institution in Pagadian City. A total of 240 students participated in the survey, which was analyzed using a qualitative case study design [6]. Findings showed that while some students expressed a consistent need for library resources, most rarely utilized OPAC services due to lack of awareness. Students also reported dissatisfaction stemming from poor internet connectivity, limited computer access, and outdated references. Interestingly, even those who were aware of OPAC functions often underutilized the system, reflecting the broader issue of digital literacy and access. The authors stressed the importance of better promotion, user training, and technical support to raise OPAC engagement and satisfaction levels.



[27] Nahotko's study, "Knowledge Organization Affordances in a Faceted Online Public Access Catalog (OPAC)," which appeared in *Cataloging & Classification Quarterly*, examines the function of faceted navigation in enhancing the organization and retrieval of information in OPACs. The research indicates that faceted catalogs enable users to narrow down search results according to certain attributes, like author, subject, and format, making the search more effective. It also emphasizes how the system ensures improved knowledge organization and user satisfaction. Additionally, it suggests that further investment is made in user-centered design for improved functionality and usability of library catalog systems.

In "The Level of User Engagement of the Online Public Access Catalog (OPAC) According to the Role of Cognitive Absorption and Computer Self-Efficacy Variables" [28], the researchers explored how students engage with online library systems and what factors affect their usage. The research focused on users of the Ferdowsi University Central Library and found that engagement with the OPAC was generally low. Two main reasons were identified: students' interest and focus while using the system, and their confidence in using computers. The study showed that students who found the OPAC feasible and helpful were more likely to use it regularly. On the other hand, those with limited digital skills or who felt the system was too plain or hard to use were less interested. The study suggests that improving user experience and building digital confidence can help increase the use of digital library systems. This supports our project's aim of creating ARK, a platform that is easier to use and more engaging for students and librarians alike.



According to Atanda and Ekeh, many students and staff from selected tertiary institutions in Lagos State knew about OPAC but used it only when needed or rarely. Their study titled “Utilization of Online Public Access Catalogue (OPAC) Among Library Users in Some Selected Tertiary Institutions in Lagos State” [29] revealed that users usually experienced difficulties in constructing search queries, absence of support, system errors, and not very user-friendly OPAC interfaces. The majority of the users were not aware of all the capabilities of OPAC. The study highlighted the need for enhancing OPAC design, training sessions, and staff support to facilitate frequent utilization. This study is applicable to our study since it requires user-centric digital systems, advocating for the ARK platform's mission to enhance accessibility, usability, and user experience in academic libraries.

The study “Awareness and Perception of Library Users Towards the Use of Online Public Access Catalogue (OPAC) in Bayero University, Kano, Library” [30] looked into how students perceived and used OPAC. The study revealed that a majority of the students were familiar with OPAC, but a small number were aware that it was available at their university library. Fewer still utilized it. The majority of users applied OPAC intermittently or monthly, and access was limited to in-library terminals. Students were hampered by low awareness, limited access, low ICT ability, and unreliable power supply. Because of these issues, a majority of students did not know how to gauge the value or ease of use of OPAC. The authors suggested more vigorous promotion of OPAC through orientation and library



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notices, and more staff training and technical equipment. This study supports the necessity for awareness and access improvement of digital library resources, which is ARK's objective to make scholarly materials more apparent, accessible, and efficient to all.

Adjei and co-researchers investigated the use of OPAC at Catholic University College, Ghana and discovered that despite good awareness, many users still relied on manual searching. Their study titled "Usage of Online Public Access to Catalogue (OPAC) by Library Users in Catholic University College, Ghana" [3] concluded that while most students were aware of OPAC, most of them still used manual searching due to limited IT knowledge, lack of access to computers, and usability issues with the system. The result also showed that students found it challenging to use the interface of OPAC and tended to compare OPAC with modern search engines like Google, leading to frustration. The researchers recommended increasing the number of devices, improving training in system use, and addressing usability issues to promote increased use. Its relevance to our study is in showing the importance of an easy-to-use and accessible digital library system, which is in line with our goal of developing ARK as a wiser, AI-driven system that improves both user interaction and information access.

Users at the Olusegun Oke Library in Nigeria were studied to understand their interaction with OPAC. The study "Online Public Access Catalogue (OPAC)



Among Library Users: A Case Study of Olusegun Oke Library, LAUTECH, Ogbomoso, Oyo State, Nigeria” [4], pointed that OPAC was used mostly to search for journals, check book availability, and view new arrivals. Most users accessed OPAC daily or every other day and reported being satisfied with the service. However, many faced issues such as system errors, slow searching, and a lack of guidance from library staff. The researchers suggested improving OPAC features, offering user training, and developing simpler search functions. This study supports our project by showing that even when digital tools are available, proper instruction, user awareness, and system efficiency are necessary for their full impact.

In a study about digital access in modern library settings, Nwobu and George discussed how OPAC helps users retrieve information efficiently. The study titled “Access to Information Resources Using the Online Public Access Catalogue (OPAC) in the 21st Century” [31], described how OPAC empowers users to search across many points including author, title, subject, and keywords. It is easier and quicker to use than the earlier card catalogues. The research also, nonetheless, discovered that most students fail to utilize OPAC due to poor digital capabilities, inadequate training, poor internet connectivity, and old computers. It emphasized user training and carrying out frequent information literacy sessions so that users will utilize OPAC appropriately. It was the advice of the researchers that library administration should invest in improved infrastructure and ensure user satisfaction by fulfilling information needs. It is relevant for the development of the



ARK platform because it emphasizes the value of user-friendly systems and the right digital support to use systems effectively.

The research conducted by Puse Jr., Maceda, and Balada focused on how students and staff used electronic library resources at Camarines Norte State College. Their study, “Utilization and Perception of Library Users in the Use of Electronic Resources: An Input Towards Digitalization of COTT Library” [32], the study revealed that 82% of the respondents knew of the accessible e-resources, and the faculty members were most satisfied. E-books were the most used resource, accessed heavily for assignments and research. Despite this, issues such as slow internet speed, narrow availability of e-resources, and insufficient digital skills were revealed. The majority of users accessed e-resources in the campus library, and the majority knew of them through orientations and the library's Facebook page. The study concluded that upgrading internet facilities, increasing e-resource subscriptions, and keeping information campaigns can improve digital library usage. This is consistent with the mission of the ARK platform in placing emphasis on accessibility, training of users, and system upgrade to improve satisfaction and use of digital academic tools.

In the study titled “Status of Blended Library Services at the University of Baguio” [33], Papay and Colas examined how students, faculty, and staff made use of both traditional and digital library services. The survey indicated that a majority of the users, mainly faculty and staff, appreciated highly the services



provided by CANVAS LMS, online document delivery, and reference by email. On the contrary, some of the features such as the OPAC and the online want list form were afflicted with underuse, mainly due to the insufficiency of awareness or digital literacy. Improvement in user training, refinement in promotional activities, and addition of more resources on learning platforms such as CANVAS were recommended. This study coincides with our ARK platform's objectives in proving the necessity for an improved, user-oriented digital library system that effectively services the varied demands of different users.

Dahuya et al. conducted a study entitled "Client's Assessment on the Accessibility, User-Friendliness, and Relevance of E-Resources of a Public Library in the Philippines" [5] was undertaken with the clear purpose of assessing the user perception of digital library services through a local public library. The study utilized surveys to gather feedback from 100 clients with specific attention to three key dimensions: accessibility, user-friendliness, and relevance. Results showed that clients have high ratings to the e-resources along all parameters studied, with the highest positive scores going to relevance (4.34) and user-friendliness (4.33), and accessibility a slightly lower but impressive score of 4.24. In spite of largely positive scores, the researchers identified areas for improvement, i.e., offline accessibility and e-print services. Recommendations were given to enhance these areas and to make sure the platform is continuously updated to ensure it remains relevant and attains user satisfaction. This study is pertinent to our study as it highlights the growing need for digital libraries not only to be comprehensive and relevant but



also user-friendly and easily accessible. It highlights the need for developing an intuitive, AI-based academic platform like ARK to provide more interaction and access to digital sources.

[34] Pacatan-Gomonit carried out a study entitled "Students' Awareness and Usage of Koha – Online Public Access Catalog (OPAC)" that investigates the interaction of students from La Salle Academy, Iligan City, with the Koha-OPAC system during the school year 2019–2020. The research reveals that there was a moderate level of awareness among Junior and Senior High School students but usage was low with the majority of students hardly interacting with the system. The study identifies some of the barriers to use as low user capacity, lack of guidance, and inadequate availability of Koha-OPAC terminals among library units. The study also finds technical issues such as system failure in displaying book availability as well as low levels of connectivity. The study recommends training workshops, enhanced accessibility, as well as system upgrading to support more active and efficient use of OPAC tools by school libraries. In addition, it emphasizes acquiring digital literacy so that they will be able to access and utilize automated catalog systems independently.

Development, Innovation, and Digital Transformation of Library Systems

This theme encompasses research that centers on the design, development, and digital transformation of library systems. It includes studies—both local and international—that aim to modernize library operations through the



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creation of web-based platforms, integration of artificial intelligence, and implementation of user-centered technologies. These works address institutional adaptation, system usability, and the broader role of technology in evolving library services, especially in response to challenges like the COVID-19 pandemic and digital transition in education.

The creation of a Learning Management System (LMS) designed to enhance the learning experience for senior high school students in the Philippines was explored by Delos Santos et al. [35]. This study focused primarily on Grade 11 and employed a descriptive-survey method to collect input from 3,072 stakeholders, including principals, teachers, and students. Significant challenges were revealed, such as overcrowded classrooms, underprepared teachers, and a lack of sufficient learning materials for Grades 11 and 12. To address these issues, the researchers adopted an agile development approach for the LMS, aiming to improve access to learning resources, track student progress, and facilitate various forms of assessment. The findings underscored the importance of an LMS in improving educational outcomes and recommended further evaluation of the system's acceptability, functionality, and scalability to ensure its success in the educational environment.

In a different context, the study titled "Library Systems Development: Utilization, Usability and Impact for Private Schools in Negros Occidental, Philippines" was conducted by Lampaza et al. [36]. This research focused on the development and evaluation of a Library Management System (LMS) at Colegio



de Sta. Ana de Victorias (CSAV). Aimed at addressing the inefficiencies of traditional manual library systems, the study implemented a web-based LMS designed to automate library tasks, optimize information retrieval, and enhance overall operations. Findings revealed significant improvements in library functions, reduced processing times, and increased user satisfaction among librarians, students, and faculty. However, challenges such as resistance to change, technical issues, and inadequate training were identified during implementation. The study concluded that the LMS has the potential to significantly enhance library services in private schools and recommended further expansion of system features, comprehensive training for staff, and collaboration among schools to share best practices.

Malabanan et al. [37] conducted research titled "Challenges and Strategies of Academic Libraries During COVID-19 Crisis Among NOCEI Member Schools in the Philippines." This study aimed to identify the challenges and strategies faced by academic libraries during the COVID-19 pandemic. Utilizing a descriptive method, the researchers conducted an online survey completed by 31 head librarians from member schools of the Network of CALABARZON Educational Institutions (NOCEI). The findings revealed that the primary challenges included barriers in transforming physical collections to digital formats, budget cuts, and a decrease in the purchasing and usage of print materials. In response, libraries implemented various strategies, such as enhancing online library services and providing real-time assistance through chat services. The study concluded that it



is essential for higher education institutions to ensure the sustainability of library services during and beyond the pandemic, highlighting the need for adaptive measures to meet the evolving landscape of education and knowledge sharing.

The integration of large language model (LLM) technology, specifically ChatGPT, into book recommendation systems is explored in the study titled "BookGPT: A General Framework for Book Recommendation Based on a Large Language Model" by Zhiyuli et al. [38]. This research presents the BookGPT framework, which aims to enhance book understanding and personalized recommendations through three key tasks: book rating recommendation, user rating recommendation, and book summary recommendation. Experimental results demonstrate that BookGPT achieves competitive performance, particularly in scenarios with limited information, such as zero-shot and one-shot learning tasks, often surpassing traditional recommendation algorithms. Additionally, the framework excels in generating personalized and interpretable content recommendations based on user attributes. The study identifies challenges such as the need for improved prompt engineering and user feedback integration, recommending future research directions, including fine-tuning LLMs for specific domains and enhancing user interaction through multi round conversations.

In a different context, the personalized book recommendation system presented by Yu et al. [39] focuses specifically on students, taking into account their grade levels and subject preferences. The research emphasizes the importance of tailored recommendations to enhance students' learning



experiences and foster reading habits. By integrating user feedback mechanisms, the system allows students to express their preferences through "Like" and "Dislike" options, which helps refine the recommendations over time. The development process involved multiple iterations, including user-centered design, low-fidelity prototype testing, and high-fidelity prototype evaluation, ensuring that the system effectively meets the unique needs of students. Findings indicate that this personalized approach significantly improves user satisfaction and addresses common challenges faced by traditional book recommendation systems, such as irrelevant suggestions and complex navigation. The study concludes that implementing a personalized book recommendation system can greatly enhance educational outcomes by providing students with relevant and accessible study materials.

Ying Lin [40] constructs a digital library information integration system in the paper titled "Computer Information Technology Applied to the Design of Digital Library Information Integration Service System." This system is divided into an information retrieval module and a personalized service module according to user demand, utilizing WebServices to realize data integration and library information retrieval. Providing 24-hour service, the average response time of each operation is within 0.2 seconds. Performance testing verifies the effectiveness of the system, showing that the application of computer information technology in digital libraries can provide users with personalized services and enhance their interests. The analysis includes user requirements, system architecture, and a Chinese word



classification module that is closely related to the information query and indexing modules. The conclusion emphasizes that integrating computer information technology improves the indexing efficiency of digital libraries and enhances the overall user experience.

The necessity of developing a shared digital library model for Vietnamese universities is analyzed in the study by Nhung Thi Nguyen [41], titled "Development of Shared Digital Library Model for Vietnamese Universities During the Digital Transformation Process." This research highlights the current operational status of shared digital library models in Vietnamese universities and identifies key factors affecting their effectiveness, such as policy and management mechanisms, infrastructure, and data standardization. The importance of collaboration among universities is emphasized to create a robust national digital information resource that can enhance educational quality. The proposed shared digital library model aims to unify digital resources and improve access for students and faculty. Findings suggest that a well-structured shared digital library can significantly contribute to the educational landscape by facilitating resource sharing and enhancing the overall learning experience in Vietnamese higher education institutions.

The study titled "Innovative Transformation of Libraries in the Digital Environment" edited by Dr. Upasana Yadav and Dr. G. Stephen [42] examines the profound changes libraries are experiencing due to digital advancements. It



highlights the integration of modern technologies, particularly artificial intelligence (AI), which is revolutionizing library services by automating cataloging, enhancing user experiences with personalized recommendations, and streamlining routine tasks. The volume also discusses best practices for managing open research data, ensuring equitable access to electronic resources, and incorporating e-learning solutions to support education. Key themes include the development of adaptable content management systems, the importance of digital literacy initiatives, and the role of social media in engaging patrons. Ultimately, this study serves as a valuable resource for library professionals and policymakers, offering insights and strategies to navigate the challenges and opportunities of the digital age.

The study titled "User Experience and Information Architecture of the National Digital Library and Archives of Iran: A Usability Investigation and Card Sorting" by Azam Najafgholinejad [43] aims to evaluate the usability and information architecture of the digital library website of the National Library and Archives of Iran. The research employs an exploratory mixed-methods approach, including interviews, observations, usability tests, and card sorting to gather data from both initiator users (public users) and expert users (librarians). Findings reveal that a significant portion of users faced challenges in completing tasks, with only 30% of initiator users and 10% of expert users expressing satisfaction with the website. The study identifies 12 categories of usability problems, with vague perceptions of concepts and labels being the most frequent issue. The card sorting results suggest changes in the organization of information items, leading to



recommendations for redesigning the website to better meet user needs and improve overall user experience.

In their 2021 study, Rivera investigated the deployment of a Mobile Library Resources Application (MoLiRA) in an academic institution, titled "Case Study of the Deployment of Mobile Library Resources Application (MoLiRA) in Academic Institution." The research aimed to explore the effectiveness of MoLiRA in enhancing library resource utilization and academic performance among students [2]. Analyzing data from 371 respondents, including students and IT professionals, the study employed a descriptive research design and utilized the e-SERVQUAL questionnaire to assess the service quality of MoLiRA. The results revealed a high overall service quality rating of MoLiRA ($M = 4.24$, $Sd = 0.63$), indicating that the application is highly efficient, reliable, responsive, designed, and secure. The study found that MoLiRA improved library information finding, search engine capability, and user comfort in sharing personal information. The authors concluded that MoLiRA is a valuable tool for enhancing library resource utilization and academic performance, and recommended further development and implementation of similar applications in academic institutions.

[44] The study, "Online Public Access Catalogues and Library Discovery Systems," appearing in Knowledge Organization examines the history of the ancient library catalogs to more modern, interactive, and user-friendly discovery systems. It examines how Online Public Access Catalogues (OPACs) have



developed to provide keyword searching, filtering, and ranked relevancy results in an effort to stay abreast of the needs of today's users. The research indicates that discovery systems provide more enhanced information seeking and user satisfaction by combining several sources into a single interface. It also indicates the requirement of continuous development to fulfill the continually increasing users' expectations of the present day.

AI-Powered Library

This theme addresses research in the area of digital transformation of library services, which investigates innovations in library services such as integrated library systems, artificial intelligence-based searches, and deployment of these platforms and user experience analysis. The literature examined related user participation and access, satisfaction, and overall success to library digital services in academic settings. Several studies, notably coming from the Philippines, show that there are challenges such as narrow digital literacy, infrastructure issues, and issues regarding the maintenance and education of users to achieve efficient and equitable access to digital library content.

The study by Ming and Niazi, "Digital Innovations and Future Trends in Libraries," addresses the future of libraries in the context of digital technologies [1]. The study recognizes that cloud computing, artificial intelligence, mobile technologies, and virtual reality increase access and improve user experience. All of these technologies also facilitate future trends such as greater technology



vendor coordination, user-controlled services, information-based decision-making, and environmental stewardship. Cost constraints and digital divides still exist, however. The research implies that libraries must evolve with these advances to continue in alignment with changing community needs and preserve opportunity against the challenges of the digital revolution. The result emphasizes adopting the proactive and innovative attitude inside the library organizations. Acceptance of these electronic tendencies can possibly help make the library relevant and proactive in the information economy.

[10] This study by Khan and Ahmad, "A Study on the Impact of Artificial Intelligence (AI) on Library Services," outlines how technology on artificial intelligence dictates the manner and process for adapting the library service. According to the study, AI raises the working efficiency to a greater extent, enhances the quality of the user experience through personalized services, and enhances information retrieval. These advantages notwithstanding, challenges in the form of staff training requirements, AI algorithm bias, and data privacy remain. The report recommends that libraries invest in the training of AI for employees and develop policies on how to approach ethical challenges. It focuses both on the benefits and obstacles to be overcome if libraries are to fully utilize AI's capabilities. Addressing these challenges would require a strong policy framework and proactive involvement of administrators and IT professionals. The study promotes an even-handed strategy that prioritizes innovation alongside safeguarding ethical concerns.



[7] Ali and Rahman's research, "Integrating AI-Powered Library Systems to Enhance Research and Learning in English Language Departments: A Case Study of Faculty and Student Perceptions," supports the fact that AI-powered library systems are key stakeholders in learning and research through the delivery of customized content and enhanced information access. Both scholars and students positively perceive AI as able to contribute to scholarly activity. However, the study lays a focus on proper training and teaching to maximize the use of such technologies. In its view, the addition of AI in libraries can facilitate more interactive and improved learning environments. It also highlights the need for academic institutions to reassess their library strategies to incorporate intelligent tools more effectively. Providing structured onboarding for new users could increase the benefits derived from AI platforms.

[8] The research conducted by Bassey and Daniel, "AI Application in Library Management: The Process and Prospects," discusses various uses of AI that automate library processes, improve user services, and optimize resource utilization. The research identifies the innovation potential of AI in library management but claims that challenges such as infrastructure limitations and workforce training requirements must be addressed. The writers emphatically cite that with well-planned strategy, AI implementation will lead to extremely high efficiency of operations and enhanced customer satisfaction for libraries. The authors recommend investing in the development of human capacity as the first



step in achieving sustainable integration of AI. They also recommend strategic planning workshops to identify AI applications specific to libraries.

[45] Montoya and Sison's work, "Navigating the Future of Technologies: Lived Experiences of TUP Librarians on the Use of Artificial Intelligence Tools," finds librarians to be beneficial in services provision and management of resources using AI tools but encounter difficulties such as embracing new technologies and how to handle complaints regarding AI usage. The authors recommend training sessions for specialists to allow librarians to acquire skills required to use AI effectively. They emphasize ongoing support to allow librarians to stay updated on technological advancements. There is also a need for peer-sharing platforms to allow librarians to exchange insights and strategies. Embracing collaborative learning can enhance AI implementation across institutions.

[46] The study by Armilla, "The Challenges and Strategies in Digitizing Library Collections," discusses challenges and strategies for digitization of library collections. The study lists typical challenges like insufficient resources, technical capacity, and resistance from staff. The author advises a strategic approach to digitization, such as technology alliances and employee training investment. The study advocates for breaking down barriers to digitization to ensure greater access to library collections and enhance the user experience. Besides, cloud-based repositories could assist in overcoming storage and accessibility challenges.



Institutional commitment and policy alignment are essential to successful digitization programs.

[47] The study by Abrigo and Labangon, "The Library as a Digital Scholarship Hub: Opportunities for Leveraging Learning Support" in *Qualitative and Quantitative Methods in Libraries*, determines that the library can facilitate digital scholarship through access to tools, resources, and staff. It also recommends collaborating with the academic departments in order to enhance learning and develop research culture. The research emphasizes the role of libraries to remain current in the changing climate of scholarship and be proactive to participate with the academic community. Libraries also need to set up digital literacy labs tailored for the needs of students and faculties. These facilities can be used in order to implement a better learning environment and greater innovation in studies.

[48] The de Leon, Flores, and Alomo article, "Artificial Intelligence and Filipino Academic Librarians: Perceptions, Challenges and Opportunities" in the *Journal of the Australian Library and Information Association*, examines perceptions and interactions among librarians in the Philippines and AI in academic library practice. The authors say that despite librarians understanding the potential contribution of AI towards enhancing library services, they face challenges like limited resources and needs for training. The document underlines the importance of overcoming such obstacles in order to make the most out of AI



in academic libraries, which eventually would lead to improved services delivery and user satisfaction. The study suggests institutional investments in AI training workshops and cross-institutional collaborations. Encouraging a mindset of adaptability and learning among librarians is key to navigating digital transformation effectively.

Book Recommendation System

This is a topic that deals with the development and efficiency of recommendation systems within e-library systems. These systems aim to enhance user experience through recommending content personalized based on individual interests, preferences, and history of interactions. By leveraging algorithms for studying user behavior, such as browsing records and rating patterns, these systems provide personalized book recommendations that ease the process of resource identification and render it more effective. The inclusion of personalized elements, such as adjustable bookshelves, also assists users in organizing and maintaining their reading materials, making the platform more interactive and engaging. The theme also delves into the importance of ongoing enhancements to these systems through the inclusion of user feedback to ensure that recommendations are still relevant and effective as user preferences change.

[49] Patel and Smith's "An E-Library System Integrated with Bookshelf and Recommendation Components" discusses the influence of integrating



recommendation functionality and personalized bookshelves in e-library systems on user activity. The system suggests to users according to their interest, and the bookshelf allows them to organize their study material in an effective way. The research indicates that integration has the effect of not only increasing user satisfaction but also that of imparting a boost in use frequency of libraries. Also, it focuses on the requirement of regular system updating and integrating the users' feedback within the integration in order to ascertain that recommendation mechanisms improve with the evolving user preference. Such an ongoing development, apart from adding to the users' satisfaction, also makes for long-term commitment, making the system more helpful as it discovers individual needs over time. By promoting a more personalized and dynamic experience for users, the system becomes an instrumental tool in inspiring users to read new materials, further enriching their academic and intellectual endeavors. Moreover, such a customized response generates a perception of ownership and pertinence among users that results in improved retention rates as well as in general platform allegiance. As the system adapts to user behaviors, it regularly generates an expanded sense of contact between the user and the resources of the library so that the platform becomes a necessary tool in their learning as well as in their research.



Knowledge and Experiences of Librarians Towards Technology Utilization

Libraries are changing because of new technology today, and the librarians play an important role in the change. Research has shown that even though most librarians understand and accept new technologies like AI and cloud systems, they still lack the skills to utilize them to their full capacity. This theme discusses how librarians' experience and understanding affect the way they use technology, which is pertinent to our goal of designing a system like ARK that is useful for both the users and the librarians.

The study “Exploring Librarians’ Knowledge, Technology Acceptance, and Utilization of Emerging Technologies” examined the experience and knowledge of academic librarians. With a focus on the new technologies such as AI, big data, cloud computing, and IoT and how the librarians' knowledge and familiarity with the technologies were perceived in Caraga and Northern Mindanao. The study deduced that librarians were largely knowledgeable and accepting of the technologies, especially AI and cloud computing, which made a positive effect on their application. Training deficiencies and uneven familiarity among the technologies were present, though, as cloud computing was applied most, while AI least. The study noted that librarians are inclined to adopt the technologies that they find useful and easy to utilize, and presented the cases of training, assistance, and simplicity of use. Its application in our study is the determination of how librarians' knowledge and familiarity with the technology can accelerate or hinder



the adoption of the smart systems, in accordance with our goal of designing ARK to effectively meet both user and librarian needs. [9]

Tano conducted research that studied the preparedness, experiences, and challenges of academic librarians, with emphasis on how libraries transform in adopting current technologies. The study indicated that academic libraries have been doing well in the area of digital transformation, with the right integration of digital systems and services. However, librarians have challenges in adjusting to rapidly changing technologies, managing information overload, satisfying users' expectations, and effective communication in technologically driven environments. The study highlighted the need for professional development, training, and collaboration to enable librarians to adjust to changing roles. The relevance of this study to our research is that it indicates that technology integration is not merely an issue of systems used but also of people behind them, which aligns with our objective of developing a platform that caters to both technical innovation and librarians' evolving needs. [11]

A literature review from 2025 examines how academic libraries adjusted to the difficulties created by the COVID-19 pandemic using Information and Communication Technology (ICT). The study points out that libraries that were already using digital solutions were in a better position to switch to full online operations, with others having to quickly change to continue offering their services. The main results show that digital repositories, virtual reference services, and



learning management systems played a big role in keeping academic activities going. The study also highlights the changing role of librarians, who had to learn new digital skills to perform ICT-based services efficiently. In spite of the progress, the digital divide, cybersecurity risk, and improper training for the staff remained a major challenge. To build future academic sustainability, it is crucial to establish measures to boost digital library services with a view to promoting continuous investment in digital infrastructure as well as user-focused services in a post-pandemic era. The study continues to be relevant after the events of the COVID-19 pandemic, as the development of digital resources and remote access continues to define the academic library environment as more responsive to future needs and challenges. [50]

The impact of library use frequency on student satisfaction was evaluated at Baguio Central University, Philippines, with the objective of assessing student usage and levels of satisfaction regarding library resources, services, and facilities. Based on a sample of 300 randomly chosen students, the study used quantitative data collection, such as frequency counts, mean scores, and ranking, by a questionnaire with a structured format. Results indicated that 42.66% of the students visit the library regularly, and expressed high satisfaction with library resources, services, and facilities, recording an average weighted mean rating of 3.94 using a 5-point Likert scale. The study identifies the importance of libraries to enhance students' learning processes and ensure academic success, calling for sustained investment in library services and materials, regular users' feedback,



and active promotion of libraries. The limitations of the study include the exclusive Baguio Central University environment that might constrict the generalizability of the results and the use of self-reported measures, which can influence accuracy. The study offers practical recommendations to schools looking to upgrade library services and highlights the value of academic libraries in ensuring student success. [51]

Table 2.1 *Comparisons/Differences of Previous and Present Study*

Title	Author	Date	Key Findings	Key Gaps Present	Present Study
The Prospects of Libraries in the Digital Era	Paul, P.	2024	Digital libraries enhance accessibility and engagement but face cybersecurity risks.	Need for enhanced cybersecurity and literacy programs to address digital literacy gaps.	Enhances digital literacy and AI-assisted access to resources.
Empowering Student Readers:	Dellia, P. et al.	2024	Digital libraries improve	Need for continuous updating and	Improves engagement through interactive features.



Digital Library Development and Literacy Enhancement			reading literacy and student engagement through diverse materials.	encouragement of digital literacy to maintain user engagement.	
From Card Catalogs to Clicks: The Modern Library Website's Influence	Rathee, S. & Sharma, P.	2024	Library websites enhance user experiences but have accessibility and privacy concerns.	Need for constant digital literacy programs and strong security measures to protect user data.	Upgrades user accessibility and privacy features.
Development of an Integrated Library Management System (iLMS)	Ayo, E. B. et al.	2023	iLMS improves library services and user experiences,	Inadequate modern technologies hinder efficient library services.	Implements smart tagging and content organization.



			with a willingness to adopt new technologies .		
Transition of University Libraries to Adapt to the New Normal	Jeyaraj, W. J.	2021	Libraries shifted to online materials during COVID-19, improving digital literacy among users.	Digital divide affecting access for some users, especially those with inconsistent internet.	Ensures access across devices and environments.
How can we assess the success of information technologies	Riady, Y. et al.	2023	Service quality predicts user satisfaction and actual usage of	Need for continuous evaluation and adaptation of digital library services to	Utilize user feedback to improve content recommendations.



in digital libraries?			digital library services.	meet evolving user expectations.	
Challenges to Digital Services in Philippine Academic Libraries	Lagas, S. & Isip, J.	2023	Libraries face significant challenges, including insufficient personnel and resources.	Need for proactive planning and resource allocation for digitization efforts in libraries.	Scalable digital infrastructure for long-term use
Data Security Mechanism in Digital Library: Issues and Challenges	Kumar, A.	2022	Strong identification measures and ethical security practices are essential for digital libraries.	Lack of comprehensive standards and policies for data management and security.	Utilize role-based access for security.



Knowledge, Preparedness and Challenges in HyFlex Library Operations	Navarro, M. R. V. et al.	2023	Librarians are knowledgeable about HyFlex operations but face funding and internet speed issues.	Need for better funding and support from administration to implement effective HyFlex operations.	Sustainable system rollout and continuity.
Tradition and Innovation: Digital Librarianship in the Philippines	Dharmaraj, S.	2024	The KOHA Integrated Library System expands library coverage through online searches	Need for ongoing professional development to equip librarians with necessary skills for digital innovation.	Trains librarians for digital system management.



			and e-resources.		
Knowledge, Attitudes and Practices Towards Open Educational Resources in Higher Education Institutions in the Philippines	Mohillo, L. G. et al.	2024	Academic librarians have positive attitudes towards OER but moderate practical application.	Need for more training programs and institutional support to enhance OER utilization.	Implementation of an intuitive system for beginners.
Enhancing Academic Success through Utilization of Library Resources	Estrada, C. Y. E. et al.	2023	Strong correlation between library usage and academic performance, emphasizing	Need for further investment in library resources to support academic achievement.	Invests in AI-powered tools to strengthen the link between library use and academic performance



			investment in resources.		
Effectiveness of Learning Management System in University of Science and Technology of Southern Philippines	Roa, M. F. V. et al.	2023	User-friendly LMS interfaces enhance learning experiences, but challenges include digital infrastructure .	Need for continuous improvement in LMS design and faculty support to foster effective online learning.	Improves LMS design to enhance online learning and resource management.
Exploring Students' Utilization of Online Public Access Catalog in the Learning Commons	Doliente, C. J. O. et al.	2023	Many students need library resources, but OPAC services are underutilized due to lack	Need for improved user training and technical support to increase OPAC engagement.	Enhances OPAC engagement by integrating user-friendly features and providing continuous support.



			of awareness.		
Knowledge Organization Affordances in a Faceted Online Public Access Catalog (OPAC)	Nahotko, M.	2022	Faceted navigation in OPACs enhances information retrieval and user satisfaction.	Need for further investment in user-centered design for library catalog systems.	Develops ARK based on user-centric principles to enhance information retrieval.
The Level of User Engagement of the Online Public Access Catalog (OPAC) According to the Role of Cognitive Absorption and Computer	Khosravi, R. et al.	2024	Engagement with OPAC is low, influenced by students' interest and confidence in using computers.	Need for strategies to enhance user experience and build digital confidence among students.	Provide personalized recommendations that enhance user engagement on the ARK platform.



Self-Efficacy Variables					
Utilization of Online Public Access Catalogue (OPAC) Among Library Users in Some Selected Tertiary Institutions in Lagos State	Atanda, A. & Ekeh, E.	2021	Users are aware of OPAC but use it infrequently due to difficulties in search queries.	Need for improved OPAC design and user training to facilitate frequent utilization.	Provide AI-powered search assistance and efficient system layout on the ARK platform.
Awareness and Perception of Library Users Towards the Use of Online Public Access Catalogue (OPAC) in	Muhammad, I.	2023	Majority of students are familiar with OPAC but do not utilize it due to low awareness and limited access.	Need for increased awareness and access to digital library resources.	Provide a web-based system that will enhance academic resource accessibility.



Bayero University, Kano, Library					
Usage of Online Public Access to Catalogue (OPAC) by Library Users in Catholic University College, Ghana	Adjei, S. et al.	2024	Users are aware of OPAC but rely on manual searching due to limited IT knowledge.	Need for enhanced training and usability improvements to promote increased OPAC use.	Easy-to-use system that can increase user engagement on the ARK platform.
Online Public Access Catalogue (OPAC) Among Library Users: A Case Study of Olusegun Oke Library,	Adegun, I. A. et al.	2021	OPAC is primarily used for checking journal availability, but users face issues	Need for better user guidance and system efficiency to maximize the impact of digital tools.	Efficient system to enhance user experience and smoother digital research.



LAUTECH, Ogbomoso, Oyo State, Nigeria			like system errors.		
A Comparative Analysis of the Student Preferences for Digital and Physical Books in Library Usage	Caperida, C.	2024	60% of students prefer digital resources, with a strong correlation between digital resource availability and academic performance .	Limitations include cultural specificity and reliance on self-reported measures affecting generalizability .	The present study can address diverse user preferences in ARK's design and offerings.
Access to Information Resources Using the	Nwobu, B. K. & George, E. S.	2024	OPAC helps users efficiently retrieve	Need for improved user training and infrastructure	Optimized platform usability for increased user engagement.



Online Public Access Catalogue (OPAC) in the 21st Century			information, but many fail to utilize it due to poor digital skills.	investment to enhance OPAC usage.	
Utilization and Perception of Library Users in the Use of Electronic Resources: An Input Towards Digitalization of COTT Library	Puse Jr., P. J. et al.	2023	82% of respondents are aware of e-resources, but issues like slow internet speed hinder usage.	Need for better internet infrastructure and more e-resource subscriptions to enhance user satisfaction.	The ARK system provides platform portability.
Status of Blended Library Services at the University of Baguio	Papay, L. C. & Colas, J. B.	2023	Users appreciate services like CANVAS LMS, but OPAC features are	Need for enhanced promotional activities and user training to increase awareness	Implementation of user-contributed content and smart search assistance.



			underused due to low awareness.	and usage of library services.	
Client's Assessment on the Accessibility, User- Friendliness, and Relevance of E-Resources of a Public Library in the Philippines	Dahuya, R. et al.	2023	Clients rated e-resources highly but identified areas for improvement in offline accessibility.	Need for improvements in offline accessibility and e-print services to enhance user satisfaction.	Implement online reservation for physical library resources.
Students' Awareness and Usage of Koha – Online Public Access Catalog (OPAC)	Pacatan - Gomonit, M. J.	2025	Moderate awareness among students but low usage of Koha-OPAC due to	Need for enhanced training and guidance to increase student engagement	Training is provided to ensure effective use of ARK.



			inadequate guidance.	with the OPAC system.	
The development of a proposed learning management system for senior high schools in the Philippines	Delos Santos, M. S. M. D. et al.	2023	Significant challenges exist in education, prompting the proposal of an agile LMS to improve access.	Need for further evaluation of the LMS's acceptability, functionality, and scalability.	Offers an agile platform to improve access, with a focus on evaluating its functionality and scalability
Library Systems Development: Utilization, Usability and Impact for Private Schools in Negros	Lampaza, S. L. et al.	2025	Implementati on of a web-based LMS improved library functions and user satisfaction.	Need for comprehensive training for staff and collaboration among schools to share best practices.	Improves library functions and user satisfaction through a web-based platform.



Occidental, Philippines					
Challenges and strategies of academic libraries during COVID-19 crisis among NOCEI member schools in the Philippines	Malabanan, E. D. et al.	2021	Libraries enhanced online services during COVID-19 but faced challenges in digitizing collections.	Need for sustainable library services and adaptive measures for future challenges.	Provides sustainable, adaptive library services to address challenges in digitizing collections
BookGPT: A general framework for book recommendation empowered by large language model	Zhiyu Li, Z. et al.	2023	The BookGPT framework enhances personalized recommendations but faces challenges in	Need for fine-tuning LLMs for specific domains and improving user interaction.	Fine-tunes recommendations and improves user interaction to address feedback challenges



			user feedback.		
Book recommendati on system	Yu, Z. et al.	2025	Personalized book recommenda tions improve user satisfaction through tailored suggestions.	Need for continuous refinement of recommenda tions to address common challenges in traditional systems.	Continuously refines recommendations to overcome challenges in traditional systems and improve user satisfaction.
Computer information technology applied to the design of digital library	Ying Lin	2024	The digital library system enhances user experience with efficient indexing and retrieval.	Need for ongoing updates and improvements to maintain system effectiveness and user satisfaction.	Ensures continuous updates to maintain system effectiveness and enhance user satisfaction.
Development of shared	Nhung Thi Nguyen	2024	Collaboratio n among	Need for effective policy	Fosters collaboration with external libraries to



digital library model for Vietnamese universities			universities can improve access to digital resources and educational quality.	and management mechanisms to support the shared model's implementation.	improve access to digital resources.
Innovative transformation of libraries in the digital environment	Stephen, G. & Yadav, U.	2024	Integrating AI in library services improves cataloging and user experiences.	Need for strategies to navigate the challenges of digital transformation in libraries.	Integrates AI to enhance cataloging and user experiences, addressing digital transformation challenges
User experience and information architecture of the national digital library	Najafgholinejad, A.	2023	Significant usability issues were found in the digital library website, leading to	Need for a comprehensive redesign to improve usability and meet user	Addresses usability issues with a comprehensive redesign to improve user satisfaction.



and archives of Iran			low user satisfaction.	needs effectively.	
Case study of the deployment of mobile library resources application (MoLiRA) in academic institution	Rivera, J. N. D.	2021	The MoLiRA application improved library resource utilization and academic performance among students.	Need for ongoing development and implementatio n of mobile applications in academic settings.	ARK develops an AI- driven platform to ensure ongoing access and engagement across devices
Online public access catalogues and library discovery systems	Wells, D.	2022	OPACs must continuously evolve to meet modern user needs and enhance satisfaction.	Need for ongoing innovation in discovery systems to keep pace with user expectations.	Includes innovative features that enhance user interaction on ARK.



Digital Innovations and Future Trends in Libraries	Ming, M. Y. & Niazi, M.	2023	Emerging technologies like AI and cloud computing are essential for libraries to remain relevant.	Need for proactive strategies to overcome cost constraints and digital divides.	Ensures accessibility development strategies and makes ARK usable for every user.
A Study on the Impact of Artificial Intelligence (AI) on Library Services	Khan, S. & Ahmad, N.	2022	AI improves library service efficiency and user experience but presents challenges like data privacy.	Need for a strong policy framework and training programs to address ethical challenges.	Develops policies for proper use of AI and offers training to ARK.
Integrating AI-Powered Library	Ali, M. & Rahman, S.	2022	AI-powered systems enhance	Need for structured onboarding	Implement formal training sessions for



Systems to Enhance Research and Learning in English Language Departments			research and learning but require proper training for effective use.	and training programs to effectively integrate AI tools.	successful AI integration in ARK.
AI Application in Library Management: The Process and Prospects	Bassey, M. R. & Daniel, H. K.	2024	AI can automate processes and improve user services, but infrastructure limitations pose challenges.	Need for strategic planning and investment in human capacity development for AI integration.	AI integration planning into the ARK system.
Navigating the Future of Technologies: Lived Experiences of	Montoya, M. B. & Sison, L. R. C.	2025	Librarians find AI tools beneficial but face challenges in and	Need for continuous professional development	Offers continuous training and opportunities for librarians using ARK.



TUP Librarians on the Use of Artificial Intelligence Tools			adapting to new technologies .	collaborative learning opportunities for librarians.	
The Challenges and Strategies in Digitizing Library Collections	Armillá, R.	2025	Digitization efforts face challenges like insufficient resources and staff resistance.	Need for institutional commitment and policy alignment to support successful digitization initiatives.	Advocates for institutional support to maximize ARK's digitization.
The Library as a Digital Scholarship Hub: Opportunities for Leveraging	Abrigo, C. & Labangon, D. L.	2021	Libraries can enhance digital scholarship through collaboration	Need for tailored digital literacy labs to meet the specific needs of students and faculty.	Integrates digital literacy programs to effectively support ARK users.



Learning Support			and access to resources.		
Artificial Intelligence and Filipino Academic Librarians: Perceptions, Challenges and Opportunities	de Leon, L. C. R. et al.	2024	Librarians see AI's potential but face challenges like limited resources and training needs.	Need for cross-institutional collaborations and a culture of adaptability among librarians.	Collaboration among librarians to enhance ARK's performance.
An E-Library System Integrated with Bookshelf and Recommendation Components	Patel, R. A. & Smith, L. H.	2021	Integrating personalized recommendations in e-library systems enhances user satisfaction.	Need for ongoing development to adapt to evolving user preferences and ensure long-term engagement.	Updates ARK continuously in accordance with user feedback and preferences.



Exploring Librarians' Knowledge, Technology Acceptance, and Utilization of Emerging Technologies	Maraon, M. I. & Agcito, M. R.	2024	Librarians are knowledgeable about AI and cloud computing but face training deficiencies, especially with AI.	Need for targeted training programs to enhance familiarity with AI and other emerging technologies.	Implementation of a beginner-friendly system.
Revolutionizing Librarianship: Navigating the Digital Transformation in Academic Libraries	Tano, I. & Cajala, V.	2024	Academic libraries are advancing in digital transformation but struggle with rapidly changing technologies and user	Need for ongoing professional development to support librarians in adapting to new technologies.	Implementation of a system designed to be clear and easy to understand for all users.



			expectations .		
ICT in Libraries: Ensuring Continuity of Academic Services During the Pandemic	Suman, S. K.	2025	Libraries that used digital solutions adapted better during the COVID- 19 pandemic, but face challenges like the digital divide.	Need for continuous investment in digital infrastructure and user- focused services post- pandemic.	Emphasizes on user engagement and user experience.
The impact of library use frequency on student	Fagyan, C. K. et al.	2023	Many students visit the library regularly and	Limitations include the study's focus on a single	User feedback to improve library resources and services



satisfaction: An evaluation of resources, services, and facilities			are satisfied with its resources and services.	institution, affecting generalizability and reliance on self- reported measures.	
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Chapter 3

METHODOLOGY

This chapter delves into crucial aspects of the study, encompassing the research design, software development methodology, sources of data, research locale, instrumentation and data collection, and the statistical tools that will be used for data analysis.

Research Design

The proponents of the study adopted a developmental research design, which involves the systematic creation and evaluation of systems or strategies aimed at addressing specific problems and improving current practices. This research design is particularly relevant in the fields of education and information technology, where the demand for innovative and user-centered solutions continues to grow. Developmental research supports iterative development and testing, ensuring that the resulting system is aligned with user needs and technological advancements [52].

In this study, the developmental approach is suitable for the design and implementation of the Academic Resource and Knowledge (ARK) system, an AI-powered academic platform. This approach provides a structured process for the continuous refinement of system features such as intelligent search, personalized recommendations, and user-generated content management. The developmental research framework ensures that ARK effectively addresses the limitations of

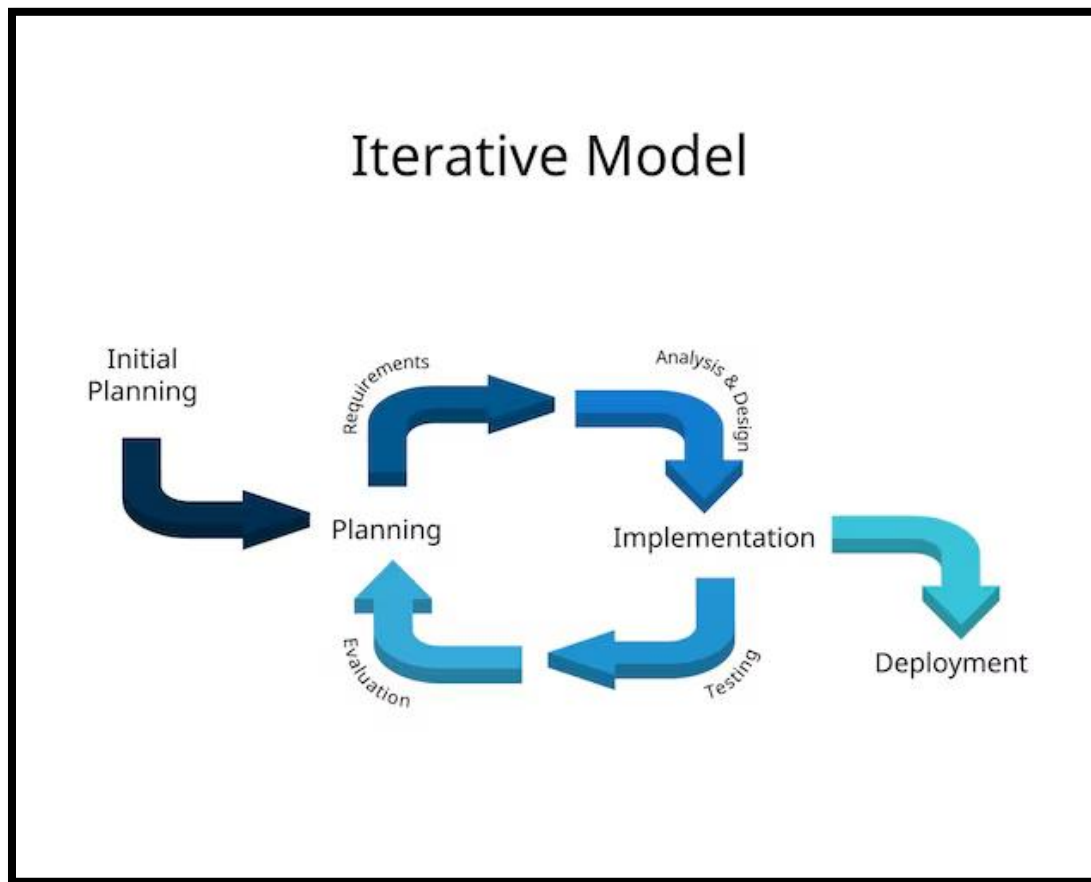


traditional OPAC systems and evolves as a scalable and user-friendly academic resource platform.

Software Development Methodology

The Iterative Model (Figure 3.1) is the software development methodology that will be applied. Through repeated cycles of planning, analysis, design, implementation, testing, and evaluation, the Iterative Model aims to build a system incrementally [53]. Development moves forward in smaller steps rather than producing a finished product all at once, which enables early problem detection and ongoing enhancements. This approach encourages gradual improvement based on feedback received following each iteration and facilitates adaptability to changing requirements [54].

The Iterative Model will be used to direct the development of the Academic Resource and Knowledge (ARK) system. This strategy will make it possible to continuously improve sophisticated features like citation management systems, academic recommendation engines, and search engines driven by AI. The ARK system will gradually adapt to the changing needs of academic users by incorporating user feedback and technical evaluation into each cycle, guaranteeing that the finished product is dependable, effective, and user-centered.



https://www.freepik.com/premium-vector/software-development-framework-iterative-model_399770912.htm

Figure 3.1. Iterative Model chart

Stages of Development

Initial Planning

The development process began with an initial planning stage, which focused on defining the objectives, scope, and key features of the ARK system. This phase served as the foundation for determining what



technologies and AI-driven components would be integrated to meet the needs of academic users.

During this stage, the researchers identified essential features such as AI-powered search, recommendation systems, and citation management tools. Consultations with library staff were conducted to align system goals with the requirements of the Kaban ng Hiyas Congressional Library, ensuring that the planned design met both user and institutional needs.

Iterative Development Cycles

Each iteration of development will involve the following phases:

Requirements Gathering:

This phase involved collecting detailed information to define the system's requirements. It ensured that both functional and non-functional aspects were aligned with real-world academic use, providing a clear understanding of how the system should operate.

The researchers conducted consultations with students, librarians, and faculty members to gather feedback on current library practices and challenges. These insights were used to refine requirements for intelligent search, tagging, and recommendation features, making the system more responsive to user needs.



Analysis and Design:

In this phase, the refined requirements were translated into a structured system design. The focus was on defining the overall architecture, database schema, and interface layout to support the AI components of the system.

The researchers created the database structure, designed workflows, and developed user interface prototypes for easier navigation. Special attention was given to how user interactions such as searches and likes would be stored and processed to support personalized recommendations.

Implementation:

This phase focused on constructing and integrating the system modules according to the finalized design. The main objective was to ensure that all components worked together seamlessly to achieve the intended functionality.

The researchers developed the ARK system using PHP and Python, incorporating features such as AI-enhanced search, recommendation tools, and event management. They configured the system to collect and organize user interaction data, laying the groundwork for adaptive improvements through AI.



Testing:

Testing ensured that all modules functioned correctly and met user expectations. This phase verified the accuracy, usability, and stability of the developed features through multiple testing methods.

The researchers conducted unit testing, integration testing, and user acceptance testing to evaluate system performance. A system demonstration was presented to librarians to gather initial reactions and usability feedback, while a Google Forms survey was distributed to collect structured responses from participants. The insights gathered from these evaluations were used to refine the interface, correct errors, and improve search functionality and overall user experience.

Evaluation:

This phase involved assessing system performance, functionality, and user experience after each testing cycle. The goal was to identify weaknesses and guide improvements before deployment.

The researchers gathered feedback from users and technical evaluators, analyzed their responses, and refined the system's features accordingly. This process enhanced the accuracy of recommendations, improved search results, and optimized the system for academic use.



Deployment

Deployment was the phase where the system became fully operational and accessible to users. It involved preparing the hosting environment, configuring the system, and ensuring that it was ready for real-world academic use.

The researchers deployed the ARK system using Hostinger as the web hosting platform, ensuring stable accessibility and performance. They conducted training sessions for librarians and selected users to familiarize them with the system's functions. All modules—such as intelligent search, recommendations, and event management—were verified to be functioning correctly and ready for academic implementation.

Maintenance

Maintenance focused on keeping the system up-to-date, secure, and responsive after deployment. This phase ensured long-term stability, continuous improvement, and adaptability to new technological developments.

The researchers monitored system performance, fixed bugs, and implemented minor updates based on user feedback. They optimized the system's search and recommendation functions to enhance speed, usability, and reliability, ensuring that the ARK system remained efficient and relevant to academic needs.



Population and Sample

For data collection, a total of 40 people participated in the survey which included 36 library users and 4 library employees who were also users of the developed system. The researchers were deliberate in selecting these groups because the system prototype was designed specifically for active library employees and users, who directly interacted with the library's services. This improved the description of user experience and usability from an actual end-users perspective.

	Total Respondents	Total Target Respondents
Library User	200	36
Library Employee	15	4
Total	40	40

Table 3.1. Target Respondents

Simple random sampling will be utilized in the selection process so that all library users have equal chances of being part of the sample. This approach reduces the possibility of selection bias and makes the study results as representative of the population as possible. It also enhances the validity and generalizability of the results to the whole user population of the library. Moreover, through the use of random sampling, the research ensures that findings obtained from the sample will better guide the development, implementation, and testing of the suggested AI-based, web-based academic resource and knowledge



management system to respond to the actual needs, habits, and likes of the users of the library.

Sources of Data

In order to ensure that the system was based on the real needs of its users and compatible with the workflows of the Kaban ng Hiyas Congressional Library, the developers employed both primary and secondary sources of data in the planning, design, and evaluation phases of the study. The sources gave necessary information on present library conditions and user expectations that were used in developing the web-based platform.

Population Insights

Primary data will be collected using a mix of interviews, survey questionnaires and stakeholder consultations with the librarians, library staff, and student users of the Kaban ng Hiyas Congressional Library.

Interviews, questionnaire surveys, and consultations with stakeholders will be held to determine operational challenges, evaluate satisfaction with existing library services, and obtain expectations of the digital platform. Utilizing a Likert scale, respondents will evaluate system aspects such as service effectiveness and book availability. During system development, validation of findings and ensuring the features match the vision of the library, operational capability, and user



requirements through consultations with the library administrator and selected personnel will be conducted.

Book Records

Primary data also consists of existing and manually coded book records. Various records have been digitized previously, and will be reviewed and reformatted to ensure these align with the standardized structure to be used for the system.

The book records, provided by the Kaban ng Hiyas Congressional Library, contains, but not limited to, the following attributes: Title, Author, Accession No., and Call Number. The researchers aim to format the records and add new attributes such as Abstract, where it will contain the abstract and specific keywords of each book record, and Recommendation Score, which will be utilized to help the recommendation feature of the system.

These records were utilized to support the key features of the system such as AI-driven search, content management, and smart recommendations, to assist users locating and engaging with relevant academic resources in the digital platform.



Research Locale



Figure 3.2. Kaban ng Hiyas Building [69]

The research was carried out in Kaban ng Hiyas Congressional Library (Figure 3.2), within Kaban ng Hiyas building with reference to Maysilo Circle, Mandaluyong City, Metro Manila, Philippines. The setting plays an important role in supporting community access and allows for academic resources for students, researchers and general readers.

Since the library's existing operations are carried out primarily using manual processes, and the library being the hub of information with a dense user population, it is an ideal setting to nurture and further develop the web-based platform. By carrying out the study within the library, the researchers have a direct interface with the end users and library staff, gaining useful insights into the operational challenges as well as expectations. The environment also provided a real-world situation to implement the platform's features such as AI search, content management, and user interactions with the actual needs and work process of the library.



Instrumentation and Data Collection

Survey

A useful method for collecting extensive information from a diverse group of respondents is through surveys. They are effective tools for reaching a large number of participants and gathering a wide range of perspectives and preferences. Although challenges such as managing self-reporting biases and ensuring the accuracy of responses may arise, surveys remain one of the most efficient ways to collect data from a broad audience in a relatively short time [56].

In this study, surveys were administered through Google Forms to students, regular library users, and researchers. The form was distributed after participants had experienced the on-site system demonstration at the Mandaluyong City Library, allowing them to provide informed feedback based on actual system interaction. The survey focused on key quality attributes of the proposed ARK system, such as security, reliability, maintainability, and usability.

Using Google Forms facilitated the online distribution of the survey, enabling automatic collection, organization, and analysis of responses. This method ensured efficiency and accessibility for both participants and researchers. The gathered results provided valuable insights into users' perceptions of the ARK system's performance, their overall satisfaction, and areas that could be further enhanced for improved user experience.



Developmental Tools

Web Hosting Services

The process involved a web hosting provider storing and maintaining website files and applications on a server to make the customers' websites accessible on the internet. Web hosting required the use of one or more physical or virtual servers to serve as the web host, as well as a domain name system (DNS) for name configurations and pointing to the hosting service [58]. The developers used web hosting services—specifically Hostinger VPS Hosting—for the deployment of the system and to make the product accessible as a website.

Front End Development Software

The process involved designing and building the graphical user interface (GUI) of the website. Often called “client-side development,” front-end development referred to the creation of the parts of the website that users interacted with, such as buttons and text boxes [59]. The developers utilized front-end development technologies such as HTML5, Tailwind CSS v4, JavaScript, SVG, and CSS3 to develop a fully functional user interface and user experience.



Back End Development Software

Back-end development focused on creating the server-side components that handled operations not visible to users. Emphasizing databases, back-end logic, APIs, architecture, and servers, developers ensured that the website functioned smoothly and efficiently [60]. The team utilized technologies such as PHP, Python, Flask, the Sentence_Transformers MiniLM (DistilBERT variant), and OpenAI GPT as the framework for intelligent processing and system functionality.

Database Management System

A software system for creating and managing databases. A DBMS enables end users to create, protect, read, update and delete data in a database. It also manages security, data integrity and concurrency for databases [61]. The developers used DBMS such as MySQL and phpMyAdmin as the method of data storage for the proposed system.

Entity Relationship Diagram (ERD)

A type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research [62]. The developers used ERD to illustrate the structure of the system’s database.



Data Flow Diagram (DFD)

A diagram that maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled [63]. The developer used the information collected from the Kaban ng Hiyas Library to create a comprehensive Data Flow Diagram.

Entity Relationship Diagram (ERD) (Figure 3.3)

The users table includes registered accounts and personal information for each user to access the system. Users can leave many comments on books in the system, track their Advanced Search behavior per day, submit multiple book reservations, and set up multiple favorites. The reservations table tracks a borrow request from a user, indicating the reserved book being borrowed in the books table that holds all of the library materials available. The favorites table holds bookmarks or favorite books from users, allowing multiple favorites per user. The book feedback table tracks likes and dislikes by users for each book. Additionally, users can propose events through the proposed event table to track those user requests. An event report table holds approval or denial of these proposed events, which actions would then be taken by an administrator. An event that has received



approval or was posted will be saved in a post event table, which indicates an actual event at the library that is available to the users.

The Page Visits table chronicles website interactions, including visits made by unregistered users. The Comments table enables users to submit multiple comments for a particular book. The admins table retains administrator accounts authorized to view all Advanced Search Logs and supervise system activity. Any admin permissions or modifications made to sub-admins are captured in the admin_access table. The advanced_search_log table tracks the number of advanced search attempts on a daily basis, and the advanced_search_config table enables an administrator to either enable or disable the function and set a daily limit. The Announcements table contains library announcements and enables an administrator to post, sort, and delete announcement posts. Admin levels are initialized in the Role table. The Education Edit Access table retains requests from users to update education levels, which administrators can then approve or deny as structured oversight of user provided data updates.

The relationships between these entities enable the ARK system to efficiently handle and deliver academic resources, customized content, and user-provided data, all of which are at the core of optimizing learning and research activity. These entities enable the ARK system to optimize its user experience by ensuring easy and straightforward interaction with academically-produced content and generating high degrees of personalization by offering accessible and easy interaction with learning content and a high degree of personalization and interaction.

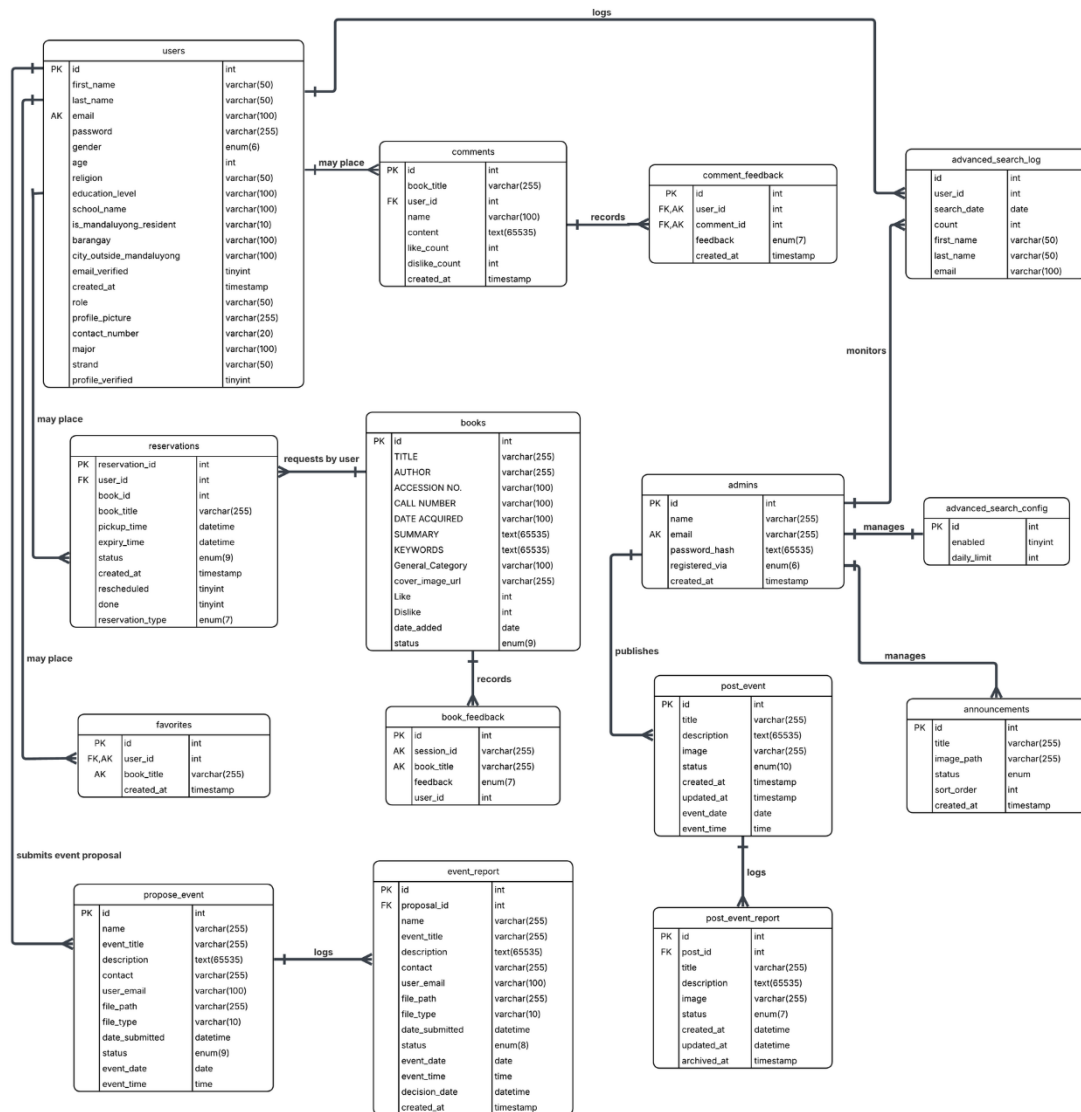


Figure 3.3. Entity-Relationship Diagram

UML Use Case Diagram

Use Case Diagram (Figure 3.4) is a visual description of interactions among actors and a system to achieve specific goals. It uses established symbols such as ovals (use cases), stick figures (actors), and lines (for specifying relationships)



along with specialized connectors such as include and extend. The diagram helps in identifying the functional requirements of the system and the relationships among various use cases [64]. The conceptual framework employing the Kaban ng Hiyas Congressional Library system shall be used by the developers in developing an in-depth UML Use Case Diagram focusing on user actions, administrative operations, and system interactions.

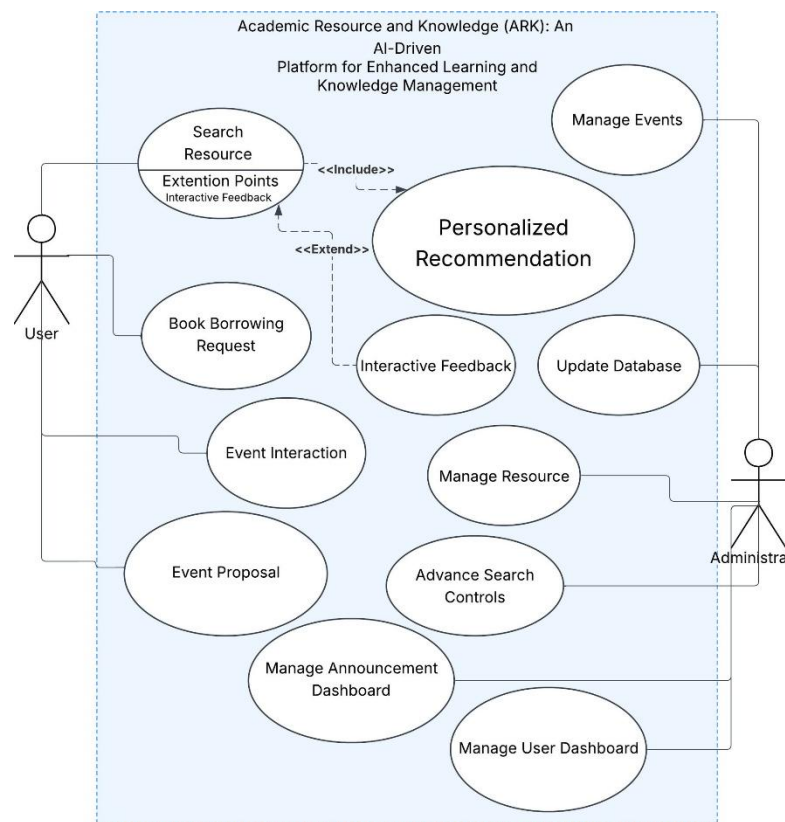


Figure 3.4. Use Case Diagram

The UML Use Case Diagram demonstrates the interaction between the two main actors of the ARK Library System: the Administrator and the User. The User may perform fundamental User functionality such as Sign In, Search for resources,



Reserve books, Request for events to be held in the library, Messaging concerning event content, and provide feedback in an interactive manner. The Search Resource use case is enhanced by Interactive Feedback, meaning it is optional, but associated with preceding the primary functionality of Search Resource (and beneficial where practical) for improving user experience. The system also includes a Personalized Recommendation engine in each of the searching and feedback processes, using an include relationship to highlight its important role in the improve processes. Next, the Administrator has the responsibility to carry out everyday operations of the system such as maintaining the database, managing library resources, and managing event-related actions from the Manage Events use case so that everything functions correctly and accurately. The system also provides the Administrator with Advanced Search Controls to securely find and filter books, users, reservations, or events. Overall, the use case diagram captures the actions of both the User and Administrator in an orderly way that provides quick access to resources and effective management of content. In addition, documenting the use case diagram provides valuable references for planning future changes and new features, and can document design ideas and functionality changes as well, while also assisting in maximizing user satisfaction and improving overall system functionality.

Data Flow Diagram (DFD)

Figure 3.5 is the diagram that maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows,



plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled [63]. The developers will use the information collected from the Kaban ng Hiyas Congressional Library to create a comprehensive Data Flow Diagram.

Context Level Diagram

A high-level data flow diagram, which shows the external entities with which an information system interacts. It shows the boundaries of the system and the major information flows between the system and the external entities [65]. The developers will utilize the Context Level Diagram to display the flow of data between the process and the external entities of the system.

The Context Level Diagram shows how the ARK Library System connects with the two external entities: the User and the Library Administrator. It outlines the information that flows between the external entities and the system. The diagram will be utilized to visualize the flow of data within the system.

External Entities

User. A person who uses the ARK Library System to search, reserve, donate, and review books.



Library Administrator. The library staff who manages the system's book records and handles donations and reservations.

Data Flows Between the User and the ARK Library System

Log-in Credentials (User to System). The user sends their login information to the system to access their account. The user must login using a Google account.

Book Information (User to System). The user looks for books, articles, and other materials by searching in the ARK system.

Book Reservation (User to System). The user reserves a book or other materials online, which they can later use at the physical library.

Book Donation Information (User to System). When a user wants to donate a book, they send donation details to the system. These are then forwarded to the Library Administrator.

Book Review (User to System). Users can rate and write reviews about the books they've read, which can be later on used to enhance the personalized recommendation system.

Book Recommendation (System to User). The system suggests books to the user based on their search history, recently viewed resources, and reviews from all users of the ARK system. This helps users discover popular or relevant books.



Library Event Information (System to User). The system displays the upcoming and current events, as well as past events conducted inside the Kaban ng Hiyas Congressional Library.

Library Event Request (User to System). The user may also request events to be conducted inside the Kaban ng Hiyas Congressional Library, which is further assessed by the Library Administrator.

Data Flows Between the Library Administrator and the ARK Library System

Log-in Credentials (Admin to System). The Library Administrator logs into the system using their credentials to manage records. The Library Administrator must login using a Google account.

Library Reports (From Admin to System). The Administrator can create and look at library reports, including the count of books checked out, user activity, and summaries of donations in order to track and evaluate library operations.

New Book Records (Admin to System). The Library Administrator inputs details of new books that have arrived, making them available in the system.

Book Reservation Data (System to Admin). The system provides the Library Administrator with the list of reserved books made by users so they can prepare them for the users when they will use the books at the physical library.



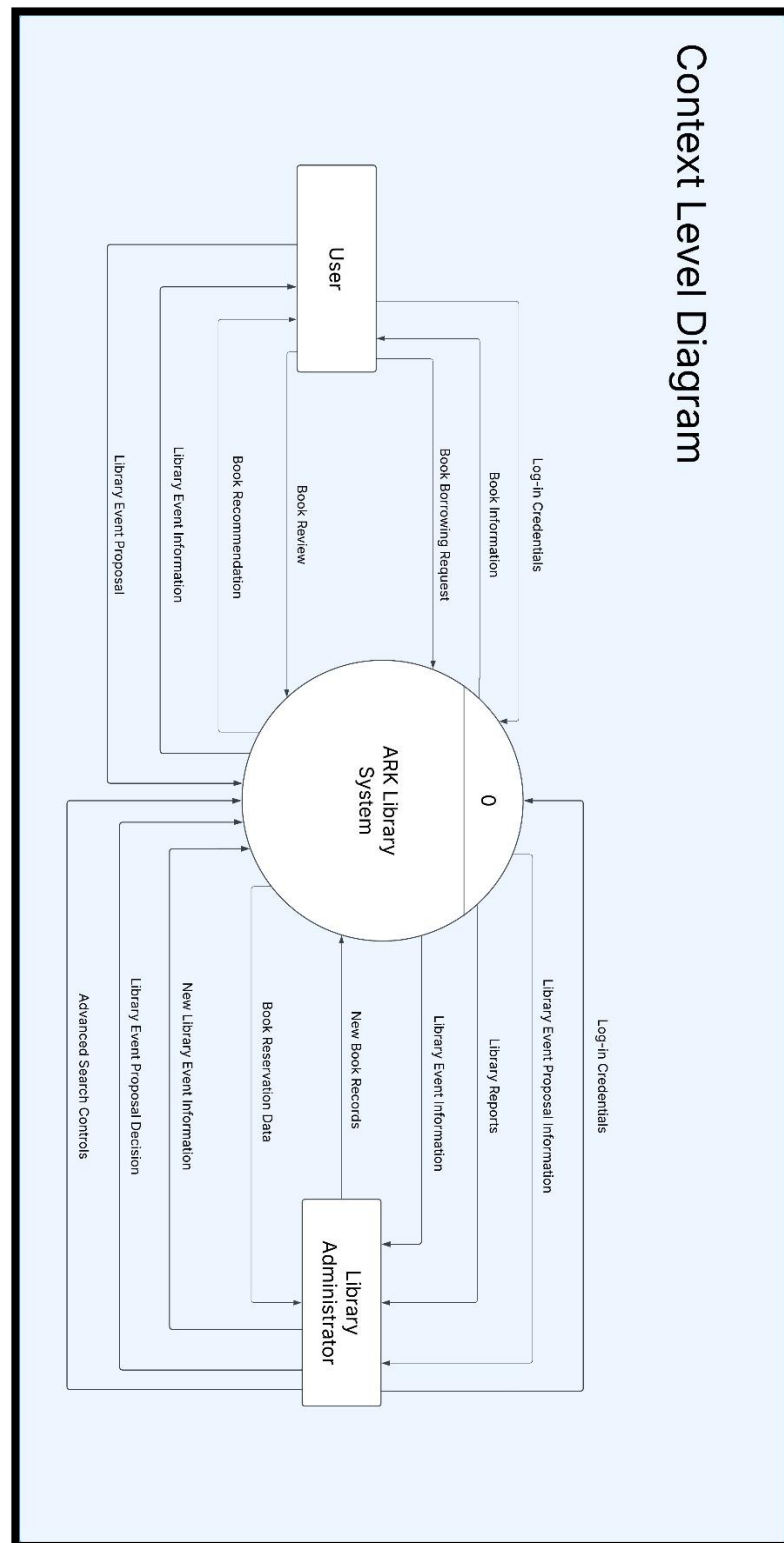
Library Event Information (System to Admin). The system provides the Library Administrator with the list of upcoming, current, and past events.

Library Event Request Information (System to Admin). The system provides the administrator with the existing library event requests, in order to be assessed whether the requested event will be approved or rejected.

New Library Event Information (Admin to System). The Library Administrator may insert new events' information that will be conducted inside the Kaban ng Hiyas Congressional Library.

Library Event Request Decision (Admin to System). The Library Administrator also has the ability to approve or reject the requested library events.

Advanced Search Controls (From System to Admin). The system provides advanced search features to the Administrator in order to filter and find specific books, users, reservations, or events more easily.

**Figure 3.5. Context Level Diagram**



Flowchart

A type of diagram that represents an algorithm, workflow or process. The flowchart shows the steps as boxes of various kinds, and their order by connecting the boxes with arrows. This diagrammatic representation illustrates a solution model to a given problem. Flowcharts are used in analyzing, designing, documenting or managing a process or program in various fields [66]. The developers utilized flowchart to provide an organized visual representation of the functions for the users and administrators during the interaction with the system. The process flow, decision points, and user experience will all be visually represented through flowcharts.

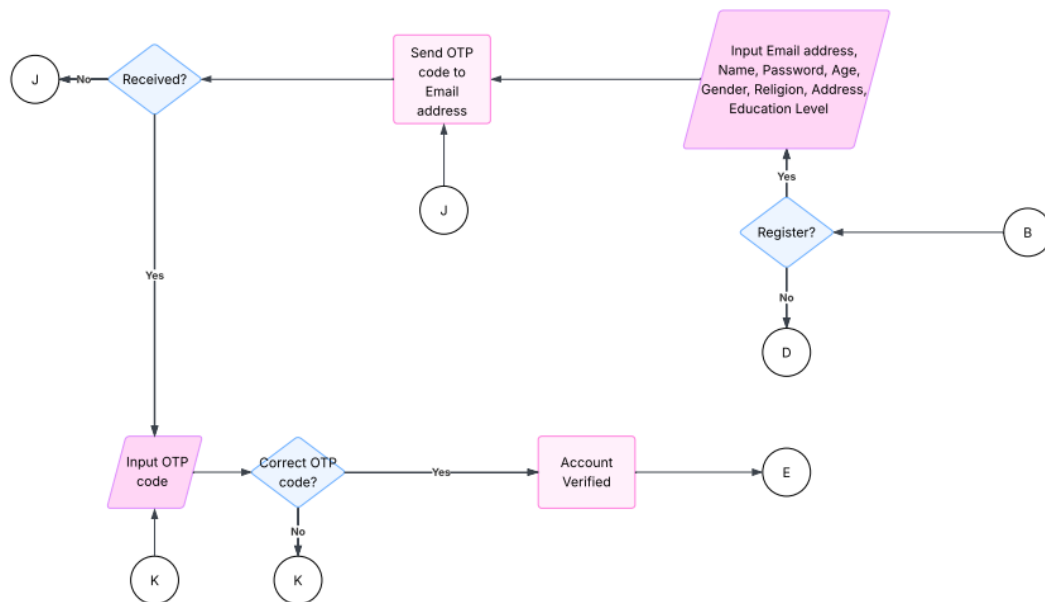


Figure 3.6.1 ARK System Flowchart Part 1

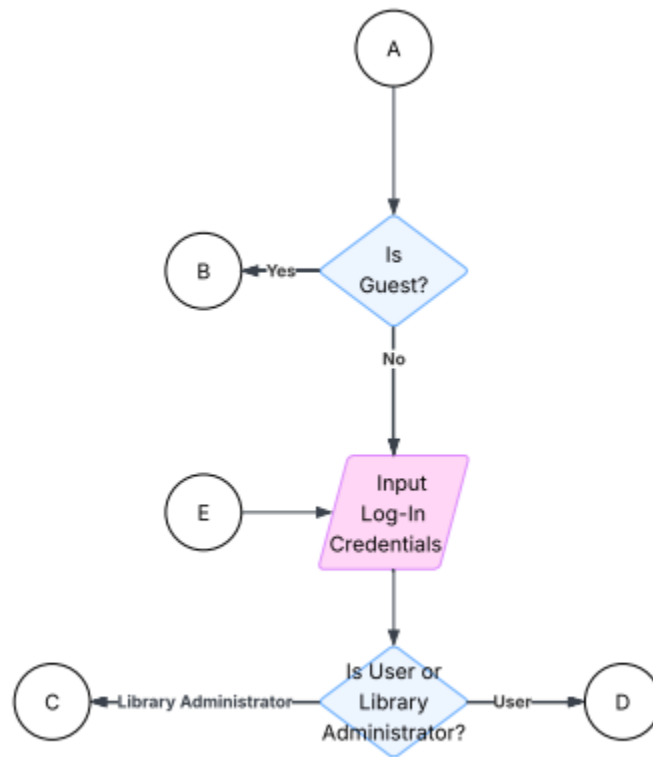


Figure 3.6.2 ARK System Flowchart Part 2

In this flowchart (Figure 3.6.1 to 3.6.5), the start function represents the entry point where the user, guest, and administrator access the system through the user dashboard. If the visitor is a guest, the guest visitor will still be able to use standard search books, view the upcoming library events and announcements, view the about page and contact section of the Kaban Ng Hiyas Congressional Library. The system will offer the guest visitor the option to register. The guest can register using their email address, name, password, age, gender, religion, address, and education level. After registration, the system sends an OTP code to the provided email address for verification. If the guest does not receive the code, they



can request a new one. Once the guest receives the OTP code, they must input it. If the OTP code is correct, the account will be verified, and the guest can log in using the verified account then will be able to access the system as a user. If the OTP code is incorrect, they will return to the OTP input dashboard to retype the correct code. If the guest chooses not to register, they will be directed to the display dashboard, where they can access the system with limitations. However, the system will prompt them to register if they want to use the advanced search, make book borrowing requests, interact with book reviews such as commenting, liking, disliking, or make library event proposals.

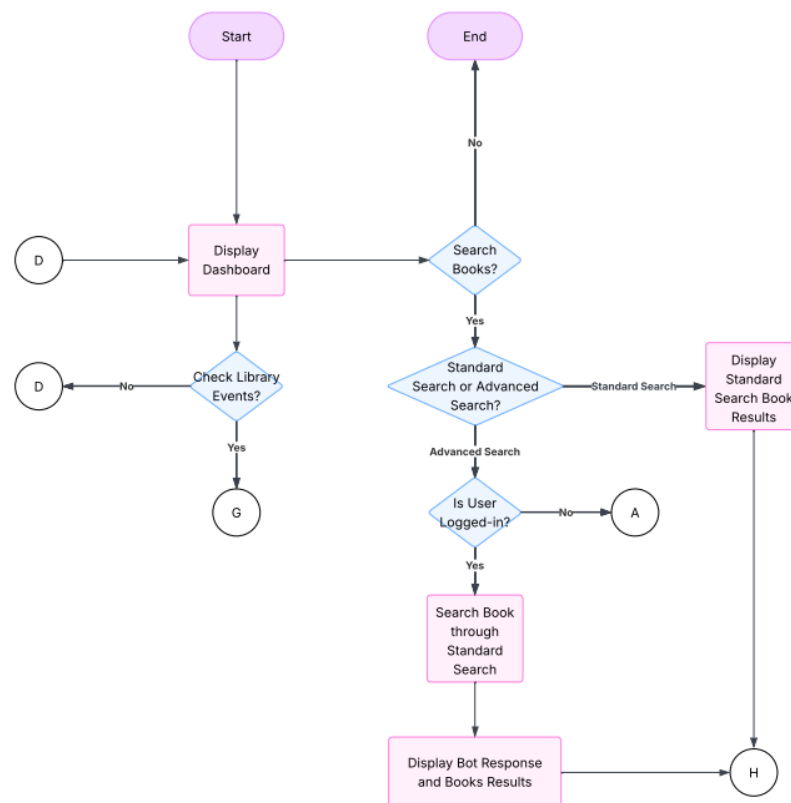


Figure 3.6.3 ARK System Flowchart Part 3



When searching for a book, users can utilize standard book search or advanced search. Based on a standard book search, general book results are displayed. In an advanced search, it will provide an answer even if the user inputs an intuitive search, generally assisted by the model GPT open-a.i on the system.

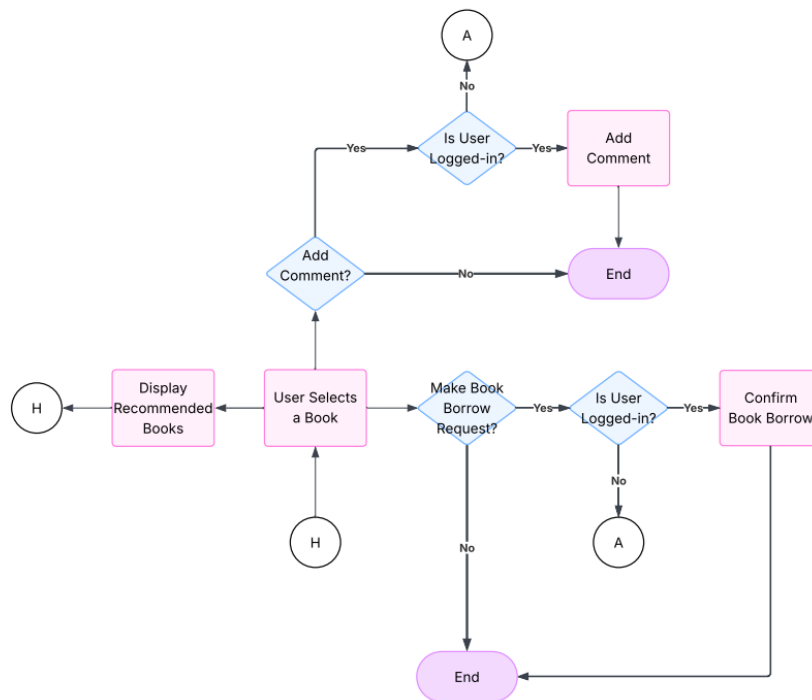


Figure 3.6.4 ARK System Flowchart Part 4

After the search results are displayed, the user can choose a book, request to borrow the book, or leave comments on the book. If the user has requested to borrow a book, an email will then be sent to confirm their book borrow request.

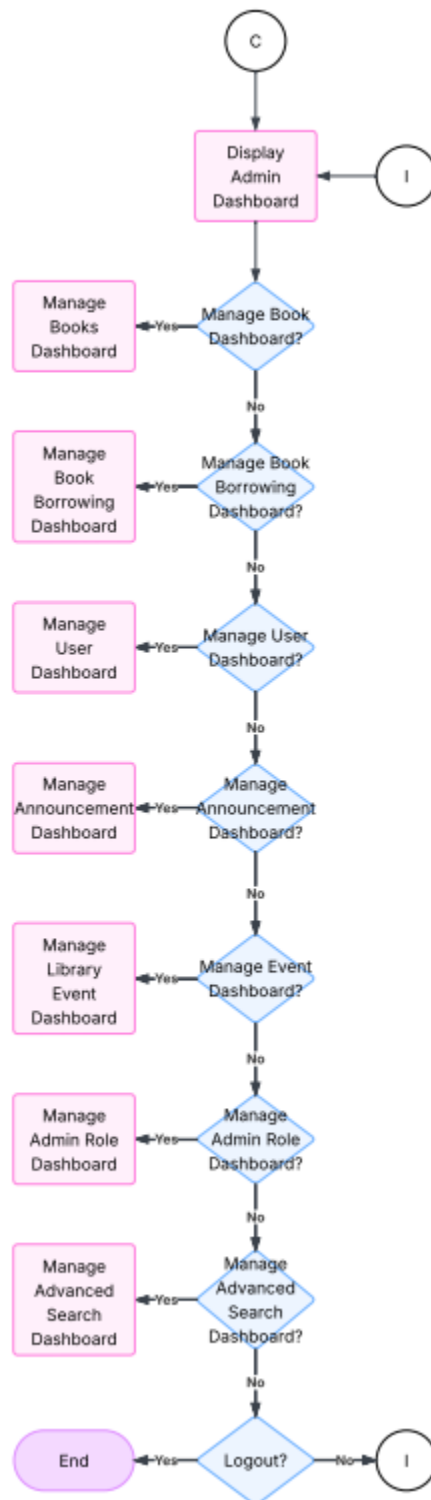


Figure 3.6.5 ARK System Flowchart Part 5



The library administrator can access to manage book records, book borrowing requests, manage user dashboard, library announcements and events, admin roles, and manage advanced search dashboard when logged in and directed to admin dashboard.

Statistical Treatment

The developer utilized statistical techniques to depict and examine the data and processes relevant to the development of the proposed system. The subsequent data analysis tools were utilized in the system's development:

Weighted mean

The weighted mean is a type of mean that is calculated by multiplying the weight (or probability) associated with a particular event or outcome with its associated quantitative outcome and then summing all the products together. It is very useful when calculating a theoretically expected outcome where each outcome has a different probability of occurring, which is the key feature that distinguishes the weighted mean from the arithmetic mean [67]. The weighted mean statistical tool were utilized by the developer to tabulate, analyze, and interpret the data collected from the respondents.

Weighted Mean (figure 3.7): The average obtained by adding together all of the responses to each question and dividing that amount by the total number of respondents is referred to as this.



Formula:

$$\bar{x} = \sum x / n$$

Figure 3.7. Weighted Mean Formula

Where:

$\bar{x}$ = Weighted Mean

$\sum$ = summation of score n

n = number of respondents

Likert Scale

A type of scale used in survey research that measures respondents' attitudes towards a certain subject. Likert scale questions are single-choice, closed-ended questions, and the primary benefit of using a Likert scale is that it provides more granular information on people's attitudes towards a subject than a simple yes/no question type. By using a Likert scale, researchers can assess varying levels of agreement, importance, quality, and other factors [68].

The researchers utilized the Likert scale as a method of data analysis for collecting and analyzing information from the survey. This approach included combining distinct data points to provide a coherent and comprehensible outcome.



Strongly Disagree	Disagree	Moderately Agree	Agree	Strongly Agree
(1)	(2)	(3)	(4)	(5)

Table 3.2. Likert Scale Evaluation

Range	Scale	Interpretation
4.50 - 5.0	5	Strongly Agree
3.50 - 4.49	4	Agree
2.50 - 3.49	3	Moderately Agree
1.50 - 2.49	2	Disagree
1.0 - 1.49	1	Strongly Disagree

Table 3.3. Scale Use for Results

The researchers applied the weighted mean as a statistical tool to determine the acceptability of the system.



Chapter 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the implementation results of the study titled *“Academic Resource and Knowledge (ARK): An AI-Driven Platform for Enhanced Learning and Knowledge Management.”* It discusses how the developed web-based system for the Kaban ng Hiyas Congressional Library addresses the objectives stated in the previous chapter. The presentation includes detailed descriptions and screenshots of the system’s major features such as smart search, personalized recommendations, analytics dashboard, and event management modules.

Objective 1. To develop a system with a smart search capability using NLP, AI, and Machine learning that interprets user queries, improving search experience and making it easier to find relevant resources.

The ARK system’s intelligent search feature integrates Natural Language Processing (NLP), Machine Learning (ML), and Artificial Intelligence (AI) to provide meaningful and contextually relevant responses to user queries. Figure 4.1 illustrates the overall workflow, showing how user queries are processed through these three tightly integrated layers. The workflow begins with a user submitting a query, which is first handled by the NLP and ML components to understand its semantic meaning and identify relevant library resources.

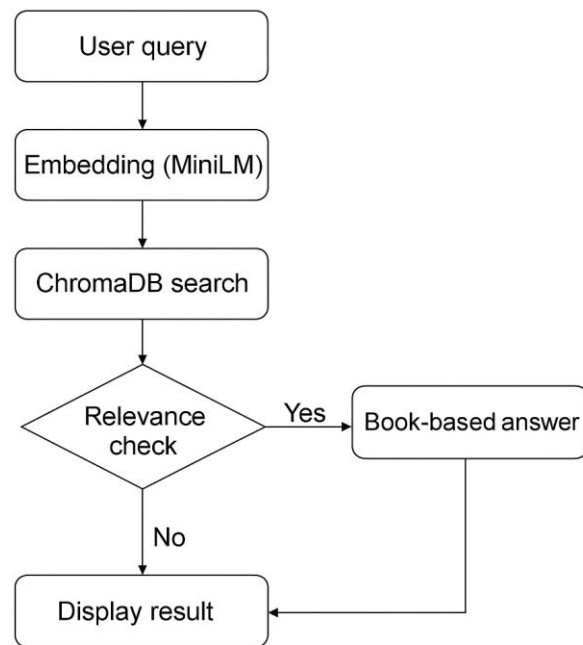


Figure 4. 1. User Input Question

When a user submits a query, ChromaDB automatically converts the text into a high-dimensional vector using the pre-trained MiniLM sentence transformer, capturing the semantic meaning of the query to enable comparison with precomputed embeddings of all library resources. In the implementation, a persistent ChromaDB client is created, and a collection named "library_books" is set up with the MiniLM embedding function. This configuration ensures that whenever a query is executed, ChromaDB internally generates the query embedding, performs similarity comparison against the stored book embeddings, and retrieves the most relevant results. By leveraging this process, the system accurately interprets user intent and identifies relevant books even when exact keywords are not present, as illustrated in Figure 4.2.



```
chroma_client = chromadb.PersistentClient(path=str(DATA_DIR / "chroma_db"))
embed_func = embedding_functions.SentenceTransformerEmbeddingFunction(model_name="all-MiniLM-L6-v2")
collection = chroma_client.get_or_create_collection(
    name="library_books",
    embedding_function=embed_func
)
```

Figure 4.2. Query Embedding Using MiniLM

In the module implementation, this embedding is handled internally within ChromaDB when the query is executed. The single line code both generates the query embedding and retrieves the top-matched books, reflecting the ML component of the system. The returned results include the IDs, similarity scores, and metadata of the most relevant books, enabling the system to determine which resources best match the user's query. The similarity search process is illustrated in Figure 4.3.

```
results = collection.query(query_texts=[question], n_results=5)
```

Figure 4.3. Query Matching and Retrieval Using ChromaDB

Once the most relevant books have been retrieved through ChromaDB, the AI component, powered by GPT, takes over to generate contextually coherent and human-readable responses. GPT first processes the main book content, summarizing it into a concise and clear explanation. It also reviews related works identified in the similarity search, producing brief summaries to provide additional context and recommendations. Using the retrieved metadata, such as the book title, author, and call number, along with similarity scores and the original user query, GPT constructs a final, well-structured answer. This multi-step process ensures that the system delivers responses that are not only accurate but also



easy for the user to read and understand. By integrating semantic understanding from ML with the language generation capabilities of GPT, the ARK system provides a seamless and intelligent search experience, as illustrated in Figure 4.4.

```
# Main answer
system_prompt = f"""You are a helpful, concise library assistant.
Answer in one short paragraph (2-4 sentences) using ONLY the following book summary.
After answering, ALWAYS add:
For a deeper understanding, I recommend reading "{top_title}" by " + top_author if top_author != "Unknown" else "" (Call Number: {top_call_no}). {book_intro}

Summary:
{top_summary.strip()}
"""

gpt_answer = client.chat.completions.create(
    model="gpt-4",
    messages=[
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": f"Question: {question}"}
    ],
    max_tokens=600,
    temperature=0.6,
).choices[0].message.content.strip()
```

Figure 4.4. GPT Integration for Contextual Answer Generation

To illustrate the system's behavior, two test queries were executed. For the query "How to predict weather?", the library contained sufficient information to produce a book-based answer, as shown in Figure 4.5. In contrast, for "What is Python programming language?", the library did not contain enough content, so GPT generated a fallback factual response while still recommending relevant readings, as shown in Figure 4.6.

Bot Response

Predicting the weather involves understanding weather patterns and using meteorological tools, both of which are topics covered in this book. The book explores the science behind weather patterns and introduces the tools used by meteorologists for predictions. For a deeper understanding, I recommend reading "Let's Investigate Science: The Weather" (Call Number: KH500 K468L 1994). "Let's Investigate Science: The Weather" is an educational book that introduces children to the basics of weather and meteorology. It explains weather patterns, meteorological tools, and the impact of weather on daily life, using clear explanations and illustrations to foster a curiosity about the natural world.

Featured Recommendation

Science

Let's Investigate Science: The Weather

Call Number: KH500 K468L 1994

let's investigate science: the weather is an educational book that explores the fundamental concepts of weather and meteorology. It introduces young readers to the science behind weather patterns, the...

[View Details >](#)

Figure 4.5. Answer to "How to predict weather?" – book-based answer

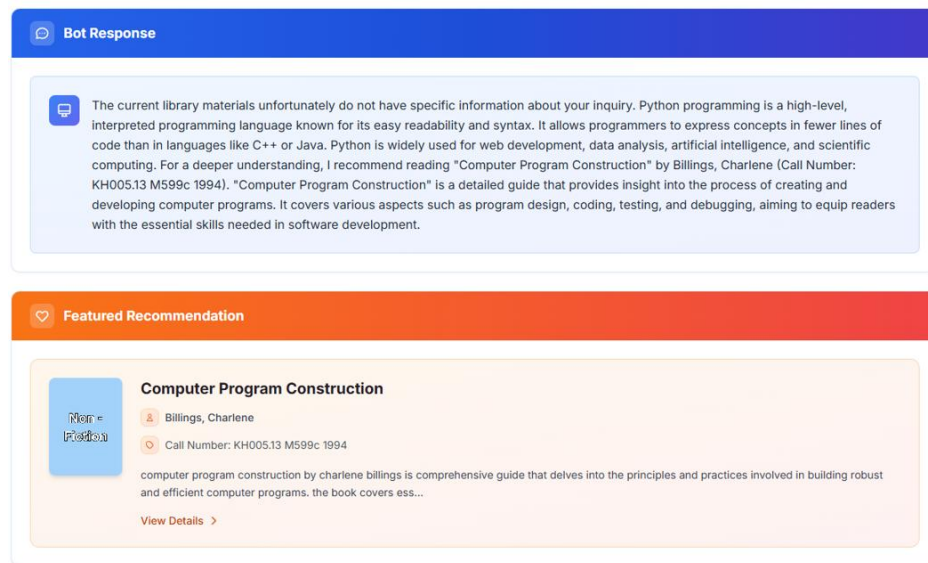


Figure 4.6. Answer to “What is Python programming language?” – fallback answer

In summary, the ARK intelligent search pipeline operates through three integrated layers: NLP (MiniLM) for query comprehension and embedding, ML (ChromaDB) for similarity-based retrieval of relevant books, and AI (GPT) for generating dynamic, human-like responses. Together, these components provide a context-driven search experience that surpasses traditional keyword-based library search by accurately interpreting user intent and recommending relevant readings in a concise, readable format.

2. To develop a data-driven recommendation system using machine learning method that suggests popular books based on usage trends and personalizes recommendations based on recent user interactions. Include an analytics dashboard to help administrators optimize frequently used categories.



The ARK system incorporates a hybrid recommendation engine aimed at providing both personalized and popular book suggestions, thereby enhancing user engagement and enabling more meaningful discovery of library resources. This recommendation engine leverages multiple strategies that analyze book content, user interactions, and category trends to generate contextually relevant recommendations.

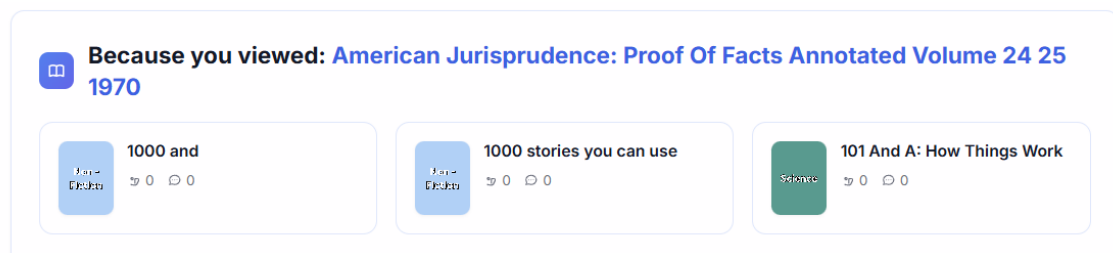


Fig 4.7. Semantic Similarity Recommendation

The first component is shown in figure 4.7, the semantic similarity recommender, focuses on content-based discovery. By transforming each book's summary and keywords into vector embeddings, the system can compute the semantic similarity between a selected seed book and all other books in the collection. Books that are most contextually similar, based on their embedding proximity, are retrieved while excluding low-similarity matches and the seed book itself. This method allows users to explore books that are conceptually related even when titles or keywords differ significantly, expanding the scope of discoverable materials beyond traditional search methods.



The screenshot displays a library interface with a blue header bar. The main content area is divided into three sections:

- Book Details:** Features a purple book cover with the word "Law" in white. To the right, the title "American Jurisprudence: Proof Of Facts Annotated Volume 24 25 1970" is displayed. Below the title, metadata is listed:
 - AUTHOR:** (empty field)
 - CALL NUMBER:** KH347.06 Am35
 - ACCESSION NUMBER:** 4077 à 4078kh
 - PUBLICATION YEAR:** No Records
 - KEYWORDS:** evidence, legal principles, court cases, annotations, procedural tips
- Book Summary:** A light blue box containing a paragraph: "american jurisprudence: proof of facts annotated volume 24 25 is comprehensive legal reference that provides detailed guidance on the evidentiary requirements and legal principles necessary to establish facts in court cases. published in 1970, this work serves as an essential resource for legal practitioners, containing case annotations, procedural tips, and examples to aid in the preparation and presentation of evidence across various legal issues."
- Related Books:** A green-bordered box on the right showing three recommended books:
 - Supreme Court Reports Annotated, Volume 114 by KH348.599 044 P538s v. 114
 - Supreme Court Reports Annotated, Volume 148 by KH348.599 044 P538s v. 148
 - The 1985 Rules On Criminal Procedure Annotated Rule 110-127 by KH345.59905 P19n 1995
- Trending in Law:** An orange-bordered box at the bottom right showing two trending books:
 - Supreme Court Reports Annotated, 1973 Volume 50 by (empty field) with a popularity score of 50
 - Summary Of 1991--- Supreme Court Rulings Ja-Je 1991 by (empty field) with a popularity score of 4.0

Figure 4.8. Category-Based Recommendation

Complementing this is the category-based recommendation (Figure 4.8), which emphasizes thematic relevance. This strategy prioritizes books that belong to the same general category as the selected book. Initially, semantic similarity is attempted to find relevant matches. When the number of similar books is insufficient, the system falls back on filtering by overlapping keywords or library call number prefixes. In all cases, books are ranked by popularity, calculated from user likes minus dislikes. This layered approach ensures that recommendations remain contextually meaningful while accounting for both content similarity and user engagement patterns.

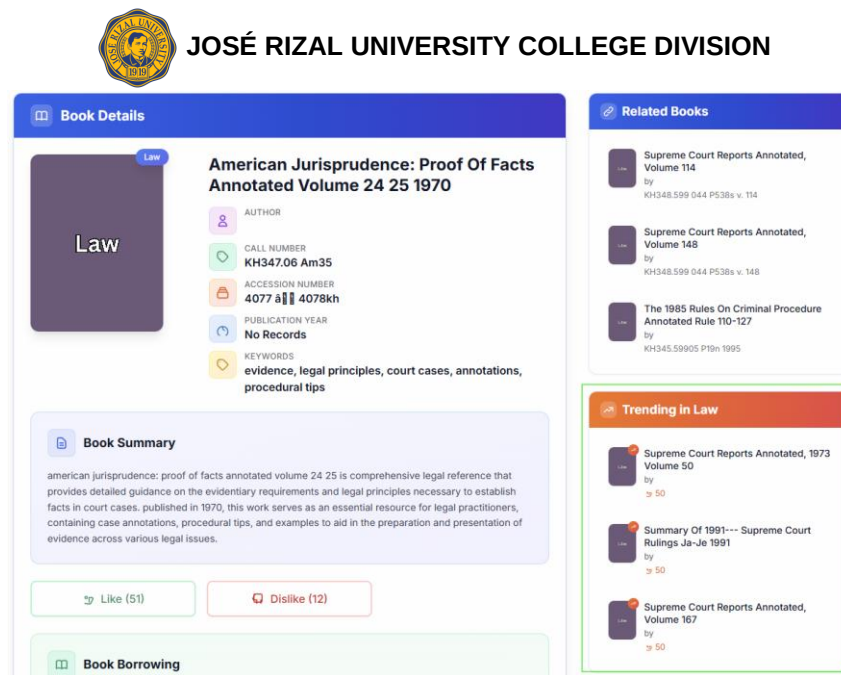


Figure 4.9. Trending Books in Category

The system also incorporates a trending within category recommender (Figure 4.9), which highlights books that are currently popular or highly rated within the same general category as the seed book. By filtering the collection to include only category-matched books and ranking them by a popularity score derived from likes and dislikes, this component directs users toward widely appreciated resources. Books already suggested by semantic or category-based recommendations are excluded to avoid duplication, resulting in a curated set of trending titles that reflects both community interest and user engagement trends.

To further support user exploration, the author-based recommendation (Figure 4.10) identifies other works by the same author as the selected book. After excluding the seed book and any previously recommended items, the system ranks the remaining books by popularity, providing users with access to the full range of an author's works. This approach fosters continuity in reading,



encourages deeper engagement with familiar authors, and helps users systematically explore collections within their areas of interest.

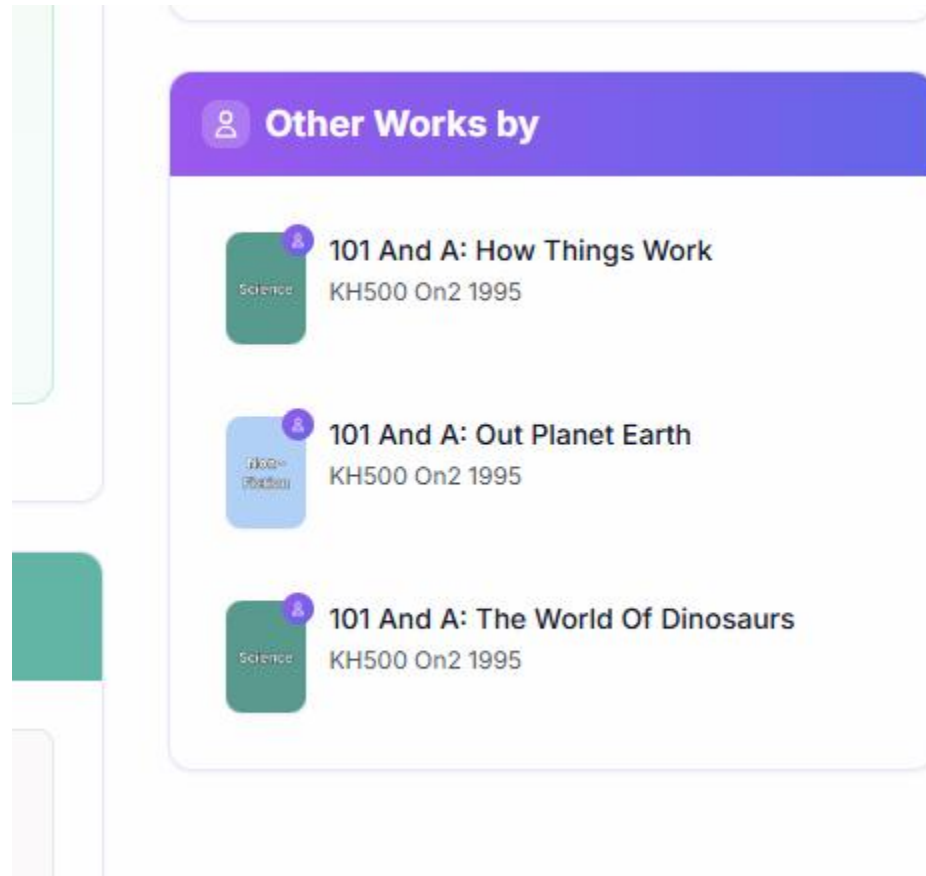


Figure 4.10. Author-Based Recommendation

Finally, the ARK system employs a combined recommendations aggregator, which unifies all individual recommendation strategies into a single, structured output. This aggregator first retrieves books from the category-based semantic method, then adds trending titles within the same category, and finally incorporates author-based suggestions. Any duplicates are removed to ensure a cohesive, comprehensive recommendation set. By integrating semantic similarity, category relevance, trending popularity, and author continuity, this hybrid approach provides users with a balanced, data-driven, and highly personalized set



of suggestions. The combined strategy enhances discoverability, aligns with user preferences, and maximizes engagement with the library's resources, presenting a robust framework for intelligent recommendation in an academic setting.

```
def recommend(self, seed_idx: int, n_similar: int = 5, n_trending: int = 5, n_author: int = 5) -> Dict[str, List[int]]:
    sim_ids = self._category_recs(seed_idx, k=n_similar)
    gen_cat = self.df.at[seed_idx, "General_Category"]
    author = self.df.at[seed_idx, "AUTHOR"]

    trend_ids = self._trending_in_cat(gen_cat, exclude=[seed_idx] + sim_ids, k=n_trending)
    author_ids = self._author_works(author, exclude=[seed_idx] + sim_ids + trend_ids, k=n_author)

    if not trend_ids:
        pop_df = self.df.copy()
        pop_df["pop"] = pop_df["Like"] - pop_df["Dislike"]
        trend_ids = (
            pop_df.drop(index=[seed_idx] + sim_ids, errors="ignore")
            .sort_values(["pop", "Like"], ascending=False)
            .head(n_trending)
            .index.tolist()
        )

    return {"similar": sim_ids, "trending": trend_ids, "author": author_ids}
```

Figure 4.11. Combined Recommendation Aggregator

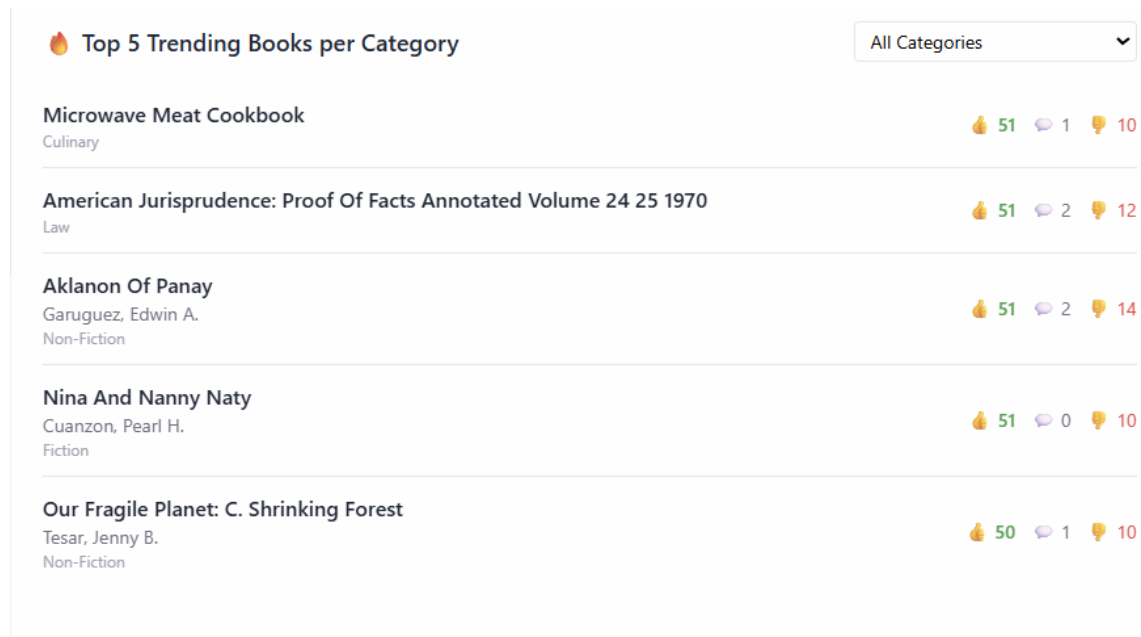


Fig 4.12. Graph for Most Used Categories

Figure 4.12 presents the “Top 5 Trending Books per Category” interface of the ARK system, highlighting how the recommendation workflow captures the most popular books based on recent user interactions. The system calculates a trending



score for each book by weighing user engagement metrics such as likes, dislikes, and comments within a configurable recent activity window (defaulted to the last 30 days). For example, likes contribute positively, dislikes negatively, and comments add a smaller positive weight, producing a final score that dynamically ranks books. The figure illustrates the resulting list, showing each book's title, author, category, and corresponding engagement counts, which allows admins to quickly identify the most relevant and currently popular resources across different categories. Additionally, the interface includes a dropdown menu for filtering by category, enabling focused exploration of trending materials within specific subjects. This visualization emphasizes the practical application of the recommendation logic, demonstrating how ARK integrates user interaction data to enhance content discoverability and guide readers toward books that are both popular and contextually relevant. The display also reinforces the system's real-time adaptability, as the rankings automatically update with new activity, ensuring that users always receive an up-to-date reflection of trending academic and general-interest materials.

3. To integrate an event calendar to display upcoming library events and activities. Allow users to submit event proposals or requests for library related activities. Support library outreach and program promotion by increasing user engagement with events.

The Event Management and Engagement interface of the ARK Library System provides a streamlined and interactive platform where users can view



upcoming library events and submit proposals for new activities. The interface displays event information in a clean, card-based layout that highlights event titles, brief descriptions, and other relevant details, allowing users to easily explore library programs and initiatives. A dedicated submission form enables users to propose library-related events by providing key details such as the event title, description, contact information, and optional attachments (PDF or image). Submitted proposals are stored in the system database and reflected in the admin dashboard for review and approval by library administrators. Designed to be visually consistent with the system's overall layout, this interface enhances accessibility, supports program promotion, and strengthens user engagement through active participation in library events.

SUBMISSION OF EVENT

Your Events

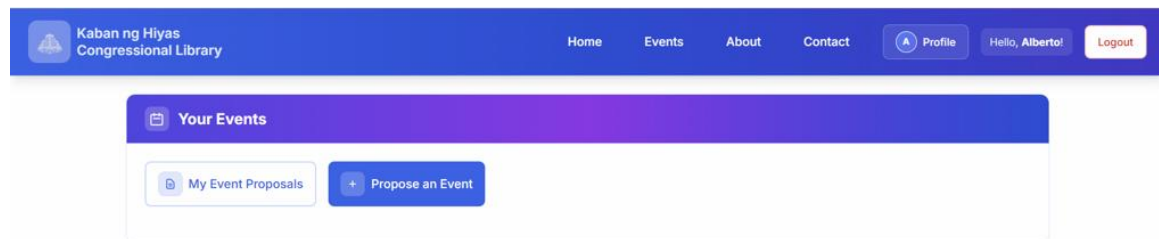


Figure 4.12. Users' Events Dashboard.

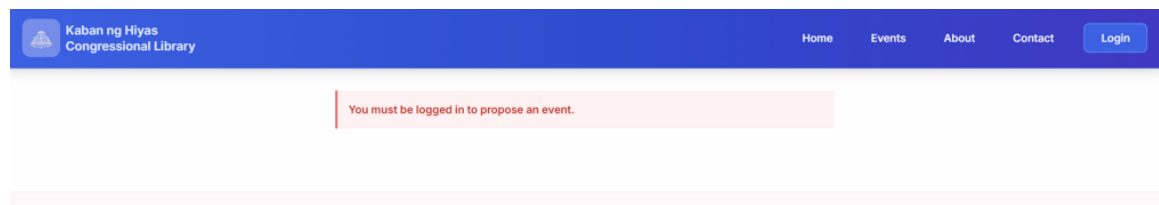


Figure 4.13. Users' Attempt for Proposal without logged in account.



The **Propose Event** section of the ARK Library System provides a dedicated interface where registered users can submit event proposals directly to the library administration (as shown in Figure 12 Users' Events Dashboard). Access to this feature is limited to logged-in users, and any attempt by guests to access the "Propose Event" page without authentication triggers a notification indicating that login is required (as shown in Figure 13), preventing unauthorized submissions.

The form is titled "Personal Information" and "Event Details". It contains several input fields: "Your Full Name *", "Contact Information *" (with a note "We'll use this to contact you about your proposal"), "Event Title * (max 60 characters)", "Short Description * (max 500 characters)" (with a note "Keep it short and descriptive"), "Event Date *" (with a calendar icon and placeholder "dd/mm/2025"), and "Event Time *" (with a dropdown menu showing "9:00 AM" and a range "9:00 AM - 5:00 PM"). There is also a character count "0/500" for the description and a note "Min: 5 days from today (No Sundays)" for the date.

Personal Information

Your Full Name *

Enter your full name

Contact Information *

Email address or phone number

We'll use this to contact you about your proposal

Event Details

Event Title * (max 60 characters)

Enter a concise and clear title

Keep it short and descriptive

Short Description * (max 500 characters)

Provide a brief description of your event

Brief overview of your event 0/500

Event Date *

dd/mm/2025

Event Time *

9:00 AM

9:00 AM - 5:00 PM

Min: 5 days from today (No Sundays)

Figure 4.14. User's Event Proposal Information Form



JOSÉ RIZAL UNIVERSITY COLLEGE DIVISION

Event Documentation

Upload Event Details (optional) (PDF, PNG, or JPG only)

To help us understand and promote your event effectively, please include any relevant documentation, such as:

- Images of posters, banners, or visual materials for the event
- A PDF containing more detailed information like agenda, speakers, or resources
- Any supporting files that will help the library staff prepare for or promote the event

Click to upload
or drag and drop

.PDF, .PNG, or .JPG up to 10MB

Upload Valid ID *

For verification, upload a valid Philippine government-issued ID (Driver's License, Passport, SSS, UMID, Voter's ID, etc.).

Click to upload
or drag and drop

File Size to 5MB

Privacy Notice: Uploaded documents, including valid IDs, will only be used for event verification purposes and will be kept confidential. By submitting, you consent to the use of your information for verification and communication regarding your event proposal.

← Cancel

Figure. 4.15. Event Proposal Upload

The Event Proposal Submission form in the ARK system is divided into three main sections: Personal Information, Event Details, and Event Documentation. As shown in Figure 4.14, users must provide their name, contact details, event title, and a short description (up to 500 characters), with validations ensuring all required fields are properly filled. Date selection allows only dates at least five days ahead and excludes Sundays, while time selection is restricted to the library's hours, 9:00 AM to 5:00 PM. Figure 4.15 illustrates the optional Event Documentation section, where users can upload supporting files in PDF, PNG, or JPG format. This section also includes a field for submitting a valid ID for verification, ensuring the authenticity of the event proposer. The form provides



feedback on file size and type, and the “Submission Guidelines” from Figure 4.16 remind users of proper scheduling and documentation requirements. This structured layout ensures clarity, compliance, and a smooth submission process for event proposals.

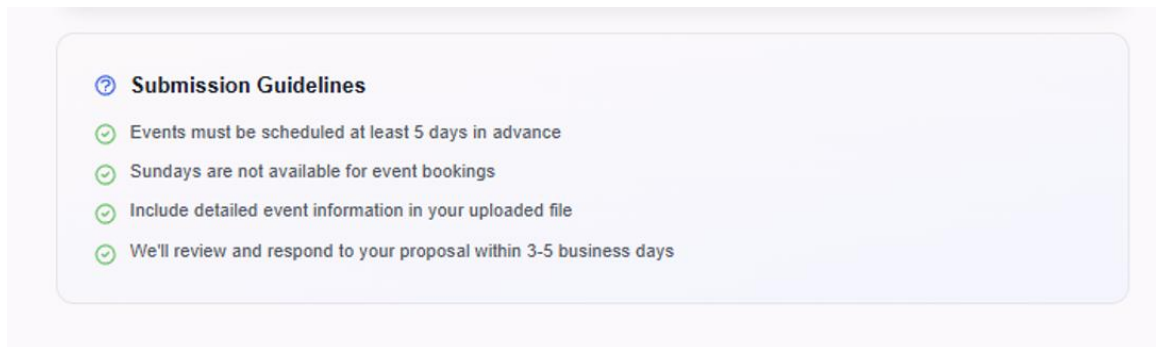


Figure 4.16 User’s Event Proposal Guidelines

After completing and submitting the event proposal form, a **loading popup** (Figure 4.17) appears on the screen to indicate that the system is processing the submission. This brief transition ensures that users are aware their input is being validated and saved into the database. After a few seconds, the loading indicator disappears, and a confirmation modal is displayed, informing the user that the event proposal has been successfully submitted.

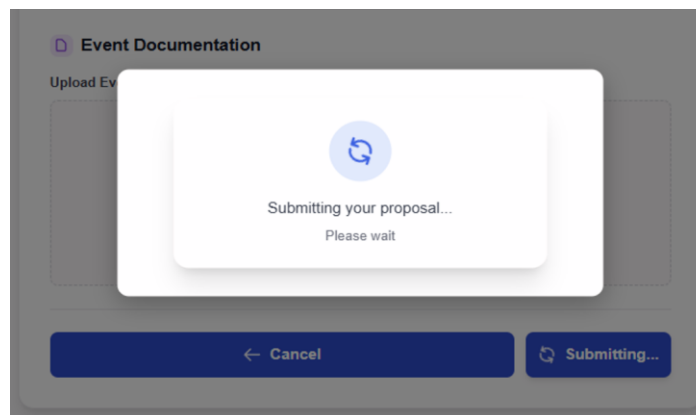


Figure 4.17. User’s Event Proposal Submission



The confirmation window (Figure 4.18) presents a clear summary of the submitted details, such as Name, Title, Short Description, and a button labeled “View Uploaded Valid ID”, which allows users to verify the attached document. A red “Close” button is provided at the bottom of the modal to return to the main event page. This feedback mechanism ensures transparency and user confidence by confirming that the proposal has been received and recorded successfully.

✓ Event Proposal Submitted Successfully!

Name: Uno Yoro

Title: **Bayanihan Through Books: Celebrating Filipino Nationalism**

Short Description: Test Function optional event documentation and required valid ID

Date: 2025-10-31

Time: 9:00 AM

Contact: unoyoro@gmail.com

Email: yoroleo10@gmail.com

Valid ID: [View Uploaded Valid ID](#)

File: None uploaded

[Close](#)

Figure 4.18. User’s Event Proposal Submission Confirmation Window

Kaban ng Hiyas Congressional Library					
Home Events About Contact Profile Hello, Antonio! Logout					
My Event Proposals					
Filter by Status: All					
Bayanihan Through Books: Celebrating Filipino Nationalism Date Submitted: 2025-10-25 15:52:24 Event Date: 2025-10-31 Event Time: 9:00 AM Status: ACCEPTED	Test file and id Date Submitted: 2025-10-25 15:43:09 Event Date: 2025-10-30 Event Time: 9:00 AM Status: PENDING	Engineering Books Turnover Date Submitted: 2025-10-25 15:14:54 Event Date: 2025-10-30 Event Time: 1:30 PM Status: REJECTED	Bayanihan Through Books: Celebrating Filipino Nationalism Date Submitted: 2025-10-25 15:09:10 Event Date: 2025-10-30 Event Time: 10:00 AM Status: ACCEPTED	Pop Culture Book Pantry Date Submitted: 2025-10-23 15:39:43 Event Date: 2025-10-31 Event Time: 10:00 AM Status: PENDING	Book signing event Date Submitted: 2025-10-23 08:32:48 Event Date: 2025-10-30 Event Time: 12:00 PM Status: ACCEPTED

Figure. 4.19. User’s My Event Proposal Dashboard



After a user successfully submits an event proposal, the system automatically records the submission in the My Event Proposals section, where users can conveniently view and track the status of their submitted events. As shown in *Figure 4.19*, each submitted proposal is displayed in a structured card format containing essential details such as Event Title, Date Submitted, Event Date, Event Time, and the Status of the request. The system initially marks newly submitted proposals as Pending, indicating that they are awaiting review or approval from the library administration.

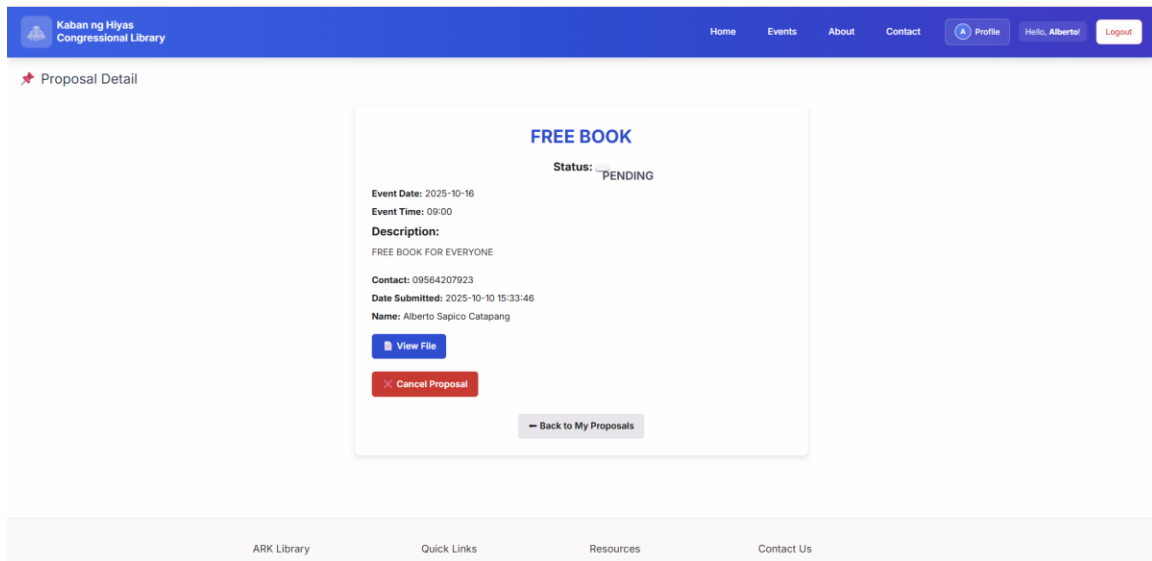


Figure 4.20. Event Proposal Details

When the user selects a specific proposal, the system redirects to a detailed view page (see *Figure 4.20.*), which presents the complete event information, such as Title, Status, Description, Contact Information, Date Submitted, and Proposer Name. This page also provides actionable options, such as a “View File” button to open the attached document and a “Cancel Proposal” button that allows the user to withdraw the request if necessary. Additionally, a “Back to My Proposals” button



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is positioned at the bottom for easy navigation. This interface ensures transparency and control for the user, enabling them to monitor proposal progress and manage submissions efficiently within the system.

Event Confirmation

After a user successfully submits an event proposal, the system automatically sends a confirmation email to the registered email address of the proposer, as shown in *Figure 4.21*. This email serves as formal acknowledgment that the event proposal has been received and recorded in the system database. The confirmation message includes all the details entered during submission, such as Event Title, Description, Event Date, Time, Contact Information, and Submitted By, allowing the user to verify the accuracy of their submitted data.

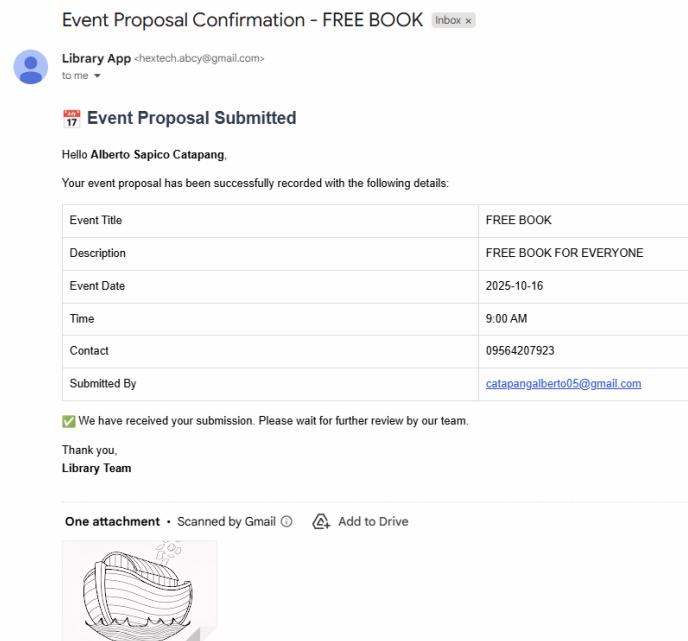


Figure 4.21. Event Proposal Confirmation Email



The email also contains a message indicating that the proposal is now under review and that the user should wait for further evaluation or communication from the library team. This notification provides assurance that the submission process was completed successfully and reinforces transparency between the system and its users. Additionally, any uploaded file or attachment, such as a poster or event plan, is included in the email for reference. This automated response enhances user trust and communication efficiency by confirming that the proposal is being processed by the library administration.

ADMIN SIDE

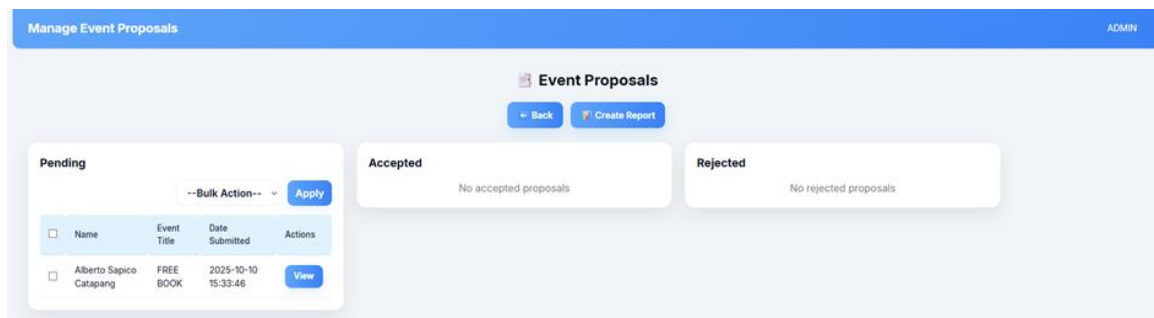


Figure 4.22. Admin Manage Event Proposal Dashboard

In the admin interface, all pending event proposals submitted by users are displayed in the Event Proposals section, as shown in *Figure 4.22*. This interface serves as the main workspace for administrators to review, evaluate, and take action on user-submitted event requests. Each proposal card presents essential information such as the event title, proposer's name, event date, submission date,



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and current status, allowing the admin to quickly identify proposals that are awaiting review. Once an admin selects a specific pending event, the full proposal details appear, including the description, contact information, and access to the uploaded file for verification. This detailed view allows the admin to assess whether the proposed event aligns with the library's objectives and operational policies.

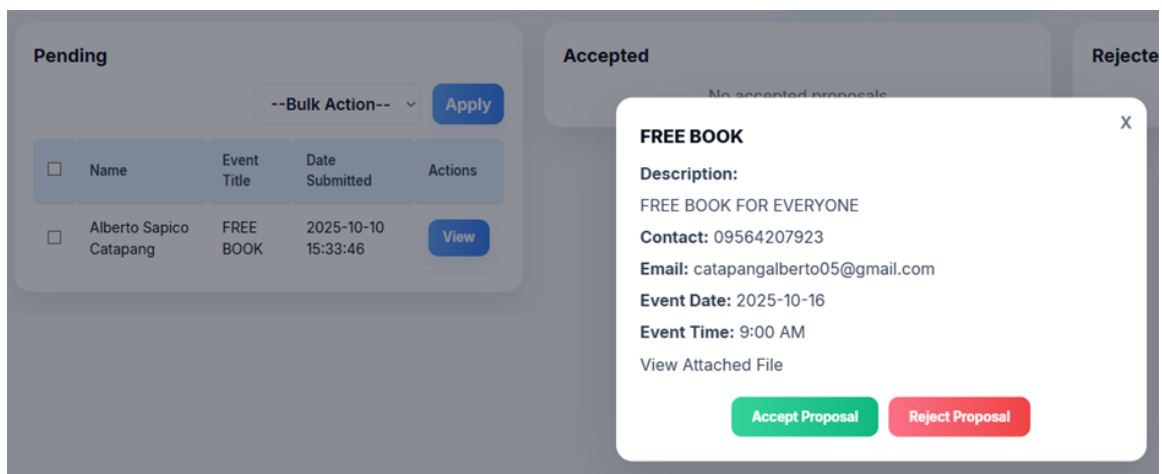


Figure 4.23. Pending Event Proposal View Details Popup

As illustrated in *Figure 4.23.*, when the admin decides to proceed with an action, clicking the “**Accept**” button triggers a confirmation popup saying, “*You are about to accept the proposal.*” This serves as a verification step to prevent unintended approvals and to confirm the admin’s decision, see Figure 24. Once confirmed, the proposal status is updated from **Pending** to **Accepted**, and the event is automatically moved from the pending list to the accepted proposals section, see Figure 25.

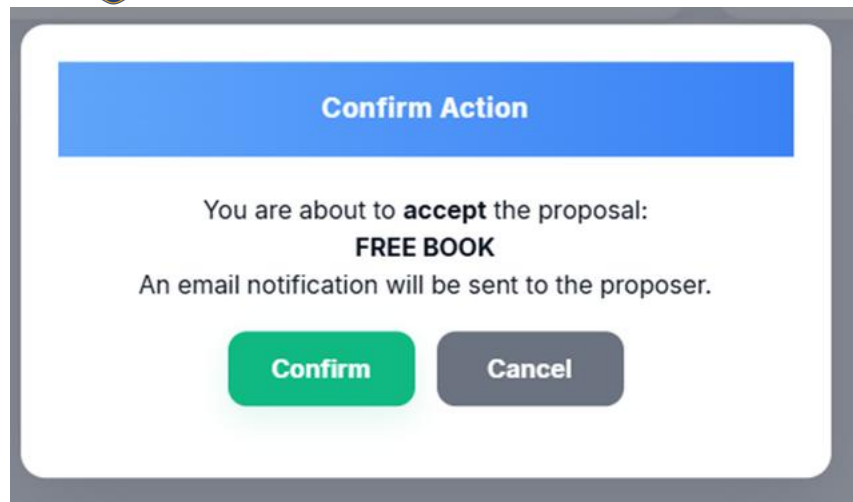


Figure 4.24. Pending Event Acceptance Confirmation Popup

Acceptance at this stage indicates that the proposal has passed the initial review and that the library administration is open to further discussion with the proposer. However, it does **not** mean that the event is officially approved or scheduled. Instead, it signifies a preliminary agreement to meet and coordinate details at the library, such as event logistics, schedule confirmation, and compliance with guidelines. This process ensures that each event proposal is carefully reviewed and that communication between users and library staff remains structured, transparent, and efficient.

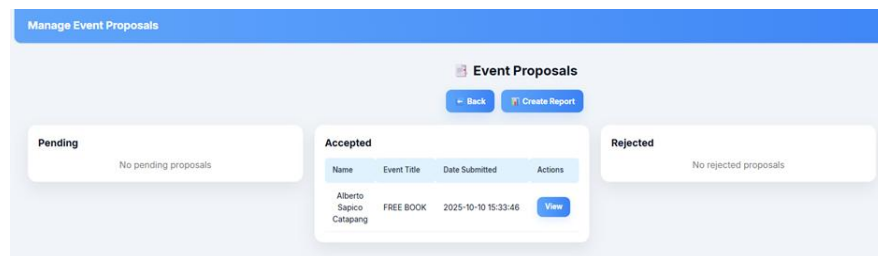


Figure 4.25. Event Proposal Accepted by Admin



EMAIL OF INVITATION

Event Proposal Accepted - FREE BOOK Inbox xLibrary App <hextech.abcy@gmail.com>
to me ▾

✓ Event Proposal Accepted

Hello Alberto Sapico Catapang,

Your event proposal has been accepted:

Event Title	FREE BOOK
Description	FREE BOOK FOR EVERYONE
Event Date	2025-10-16
Event Time	9:00 AM
Status	ACCEPTED

Important: Please go to *Kaban ng Hiyas Congressional Library* as soon as possible to discuss the event with the officer in charge.

Thank you for submitting your event proposal! We look forward to your event.

Library Team

Figure 4.26. Event Proposal Accepted Email

Once the administrator accepts a submitted event proposal, the system automatically generates and sends an acceptance email to the user's registered email address, as illustrated in *Figure 4.26*. This message formally informs the proposer that their event submission has been marked as **Accepted** and includes a detailed summary of the event information such as Event Title, Description, Event Date, Event Time, and Status. The inclusion of these details ensures that both the user and the library administration maintain consistent and transparent records of the accepted proposal.

The email also serves as an invitation notice, instructing the proposer to personally visit the Kaban ng Hiyas Congressional Library to meet with the officer in charge and discuss the proposed event in detail. This discussion stage allows the library to evaluate the event's feasibility, logistical requirements, and alignment



with institutional policies before confirming it as an official library activity. The message concludes with a courteous acknowledgment thanking the proposer for their submission and expressing anticipation for the potential event collaboration. This automated confirmation and invitation workflow reinforces communication efficiency between users and administrators while maintaining the professionalism and structure of the event proposal process.

POSTING NEW EVENT

The screenshot displays the 'Manage Events' interface for an administrator. At the top, there is a blue header bar with the text 'Manage Events' on the left and 'ADMIN' on the right. Below the header, a button labeled '- BACK TO EVENTS' is visible. The main section is titled '+ Add New Event'. It contains several input fields: 'EVENT TITLE' with the placeholder text 'Free Book'; 'DESCRIPTION (MAX 200 WORDS):' with a text area containing a paragraph about a 'Free Books for All' event; 'EVENT DATE:' with a date picker showing '10/18/2025'; 'EVENT START TIME:' with a dropdown menu showing '8:00 AM'; 'EVENT END TIME:' with a dropdown menu showing '1:00 PM'; and 'PUBLICATION STATUS:' with a dropdown menu showing 'POSTED'. At the bottom left of the form is a blue button labeled '+ CREATE EVENT'. On the right side of the form, there is a section titled 'EVENT POSTER:' with a 'Choose file' button and a preview of an image showing a boat on water.

Figure 4.27. Admin Posting Events

Figure 4.27. shows the administrator interface where approved event proposals are finalized and posted to the main library system. After an event proposal has been accepted, the administrator proceeds to this section to input or refine the event details before publication. The interface includes fields such as the event title, description (limited to 200 words), event date, start and end time, and



a dropdown for **publication status**. The admin can also upload an **event poster**, which will visually represent the event on the user dashboard once published.

Existing Events								
TITLE	DESCRIPTION	IMAGE	EVENT DATE	START TIME	END TIME	STATUS	ACTIONS	
Free Book	Join us for an inspiring and heartwarming event, as we celebrate the spirit of community, knowledge...	 View Image	2025-10-16	9:00 AM	1:00 PM	 POSTED	 EDIT	 DELETE
Read and Rise: Community Book Pantry (Afternoon Session)	Celebrate the love of reading at our Book Pantry this November 28–29, 2025! Wind down with fellow...	 View Image	2025-11-28	4:30 PM	6:30 PM	 POSTED	 EDIT	 DELETE
Read and Rise: Community Book Pantry (Morning Session)	Celebrate the love of reading at our Book Pantry this November 28–29, 2025! Start your day surro...	 View Image	2025-11-28	9:30 AM	11:30 AM	 POSTED	 EDIT	 DELETE

Figure 4.28. Library Existing Events on Admin

At this stage, the administrator has full control over the event's visibility (Figure 28). The **publication status** option determines whether the event is made public or kept hidden. When the status is set to **POSTED**, the event becomes visible to all users in the library's Upcoming Events section. Conversely, if the status is changed to **NOT POSTED**, the event remains unpublished and will not appear on the user side. This feature allows administrators to review and manage event visibility before or after publication, ensuring accuracy and coordination with library schedules.

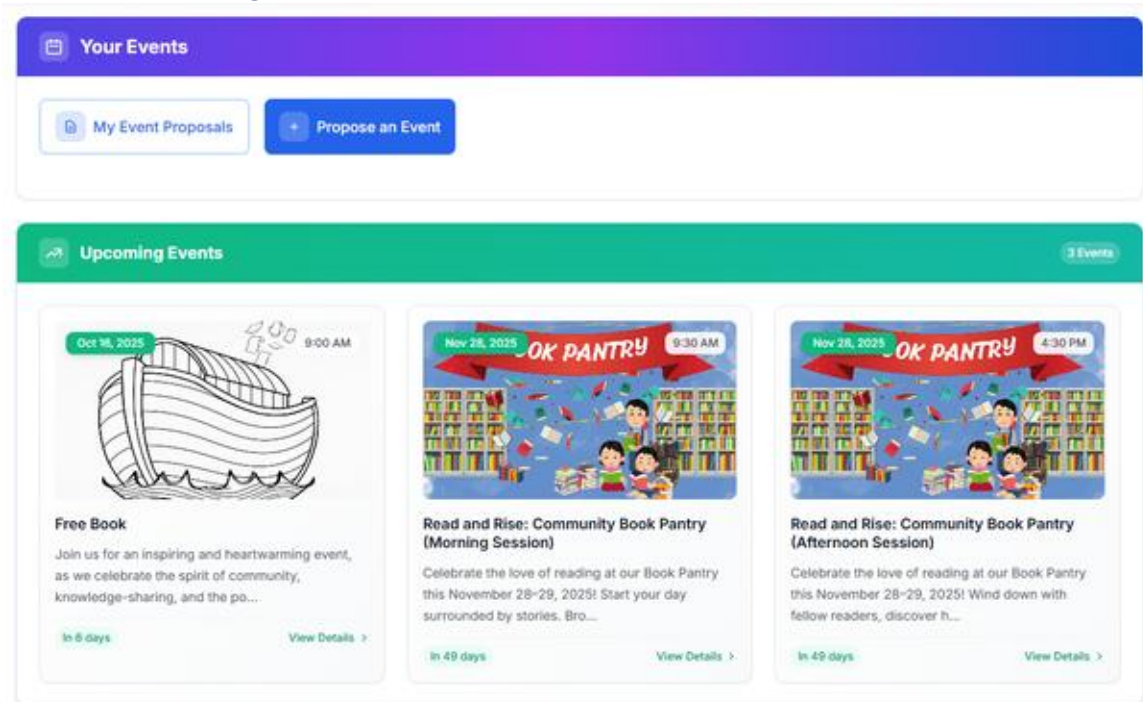


Figure 4.29. User's Preview for Upcoming Events

In the Figure 29, the event titled “*Free Book*” is being reviewed for posting. Once all details are confirmed and the status is set to **POSTED**, the event automatically becomes accessible on the main dashboard where users can view the event information and schedule.

PAST EVENT

The **Upcoming Events** section in the ARK Library System displays all approved events that are scheduled to happen in the future. These events remain visible until the end of their event period to ensure users are well-informed and have enough time to participate or prepare. Once the event date passes, the system automatically removes it from the upcoming list, maintaining a clean and relevant event feed for users, see figure 30.

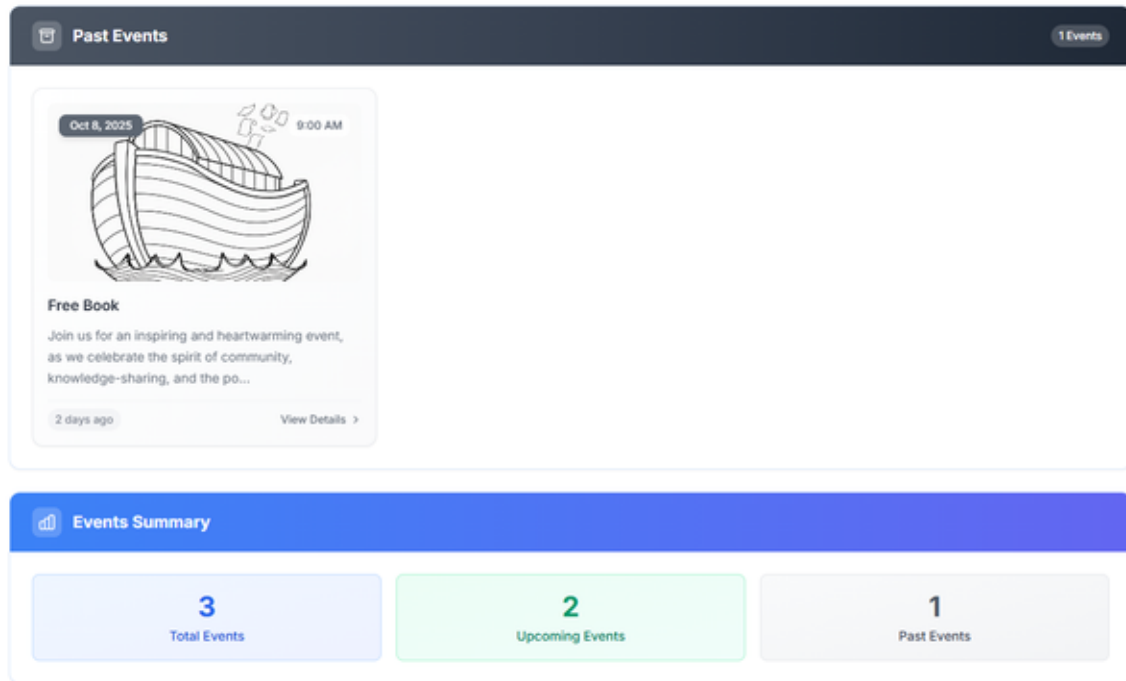


Figure 4.30. Library Past Events on User dashboard

Meanwhile, the **Past Events** section automatically collects events that have already taken place. For instance, if an event like *Free Books Giveaway* is scheduled for October 16, it will move to the Past Events section at midnight on October 17. This ensures that users can still view records of completed activities while keeping the upcoming section focused on future and ongoing events.

EVENT SUMMARY

The **Event Summary** section (Figure 4.31) presents a comprehensive overview of all library-related activities conducted within the system, displaying summarized statistics such as the total number of events, upcoming events, and

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past events. This section provides users and administrators with a clear visualization of the library's activity level and engagement in hosting educational and community-centered programs. By consolidating event data, the summary serves as an indicator of how active the library is in organizing events that promote literacy, knowledge sharing, and public participation.



Figure 4.31. Library Event Summary Dashboard

4. To evaluate the developed system using ISO 25010:2023.

To evaluate the developed ARK system, the ISO/IEC 25010:2023 standard is adopted as a benchmark for software quality. This standard provides a comprehensive framework to assess the overall performance and effectiveness of the system, ensuring it meets internationally recognized quality criteria.



System Evaluation

Statement	SD	D	MA	A	SA	MR	VI
1.1 Functional Completeness The ARK system provides all the key features that make the library experience smoother and more effective. It includes tools like an AI-powered smart search for quickly finding books, personalized book recommendations suited to each user, and an event proposal submission feature for sharing ideas or activities. These functions ensure that users can easily access and manage what they need within the system.	0	0	1	8	31	4.75	SA
1.2 Functional Appropriateness The ARK system provides useful and meaningful results that truly fit what user need inside the library. When you search for a topic, it gives relevant and accurate materials that match your studies. The features are designed to help you with your academic work, research, or participation in library events — making it not just functional but also practical for everyday user use.	0	1	2	11	26	4.55	SA
Average Weighted Mean						4.65	SA

Table 4.1. Functional Suitability

Legends: SD – Strongly Disagree, D – Disagree, MA – Moderately Agree, A – Agree, SA – Strongly Agree, MR – Mean Rating, VI – Verbal Interpretation.



The data suggest that the participants strongly agreed that the functionality of the ARK system was deemed very suitable. Functional Completeness have a mean of 4.75, signifying that the ARK system was supplying the user with available and necessary tools for full operation to enable a positive library learning experience. Users made comments about the benefit of an AI-enhanced smart search, book recommendations tailored to individual users, and the capability for event proposal submission to fund external speakers, all of which contributed to making resources accessible and navigating library events. Functional Appropriateness has a mean of 4.55, thus reflecting that the navigation and information provided an accurate and pertinent result to their needs; academic research, studying or attending an event. Further demonstrating overall, that the general average of 4.65 (Strongly Agree) indicates that the ARK system is achieving its user functional capacity by providing Library users the complete, practical, and meaningful features that it had indicated, therefore appropriately meets its function as a modern academic Library management system.



Statement	SD	D	MA	A	SA	MR	VI
2.1 Time Behavior	0	0	4	17	19	4.38	A
The ARK system loads pages, search results, and other information quickly so that users do not experience long waiting times. Whether searching for a book, checking upcoming events, or updating records, the system responds within a few seconds. This fast performance helps both user complete tasks efficiently, creating a smooth, productive, and frustration-free library experience even during busy hours.							
2.2 Capacity	0	0	5	13	2	4.42	A
The ARK system remains stable and responsive even when many users access it at the same time. It can handle multiple tasks — such as searching for books or updating records — without lagging or crashing. This ensures a smooth and reliable experience for all users, even during busy library hours.							
Average Weighted Mean						4.4	A

Table 4.2. Performance Efficiency

Legends: SD – Strongly Disagree, D – Disagree, MA – Moderately Agree, A – Agree, SA – Strongly Agree, MR – Mean Rating, VI – Verbal Interpretation.



The findings suggest that respondents mostly concur that the ARK system is proficient and functions efficiently under the average use. The Time Behavior category, which had a mean of 4.38, indicates reasonable time frame when performing searches, page loads, and record updates; however, slight improvements during excess heavy use may candidate for a more responsive experience. The Capacity indicator obtained a mean of 4.42; confirming that the system remained reliable and steady during concurrent use by various users at the same time as noted. Thus, the concurrent usage of the system without significant delays and the inability to crash the system shows its reliability and level of scalability. Overall, the ARK System demonstrates a level of performance efficiency with an adequate balance of speed, responsiveness, and system capacity. The only recommendations is slight optimization which may be warranted for peak library use hours.



Statement	SD	D	MA	A	SA	MR	VI
3.1 Availability	0	0	2	10	28	4.65	SA
The ARK system should always be accessible and ready to use whenever needed. Users should be able to perform tasks such as searching for books, viewing events, or managing records without experiencing sudden shutdowns or server errors. This reliability ensures that the system can be depended on anytime for all library-related activities.							
3.2 Fault Tolerance	1	1	5	8	25	4.38	A
The ARK system should keep working smoothly even if small technical issues occur, such as weak internet connection, minor bugs, or short system interruptions. It must be able to recover quickly and continue operating without disrupting your tasks. This ensures that users can still accomplish what they need without losing progress or experiencing major inconvenience.							
Average Weighted Mean							4.51 SA

Table 4.3. Reliability

Legends: SD – Strongly Disagree, D – Disagree, MA – Moderately Agree, A – Agree, SA – Strongly Agree,

MR – Mean Rating, VI – Verbal Interpretation.



In terms of reliability, respondents fully agree that the ARK system provides reliable service and uninterrupted access to the system. For the Availability dimension, the mean is 4.65, indicating that the system is reliable and accessible, providing access 24/7, with a little to no opportunity for downtime and minimal technical issues. Users can accomplish tasks such as searching, browsing, or record management without being interrupted. The Fault Tolerance dimension, which obtained a mean of 4.38, indicates that the system will recover after a brief technical interruption when these issues are minor, such as likely connection issues or short bug, and importantly, user progress remains intact. When taken as a whole, these results demonstrate that the ARK system is a reliable system that offers users consistent service levels, systems recovery, and user convenience during operation. The reliability dimension rating serves to emphasize the ARK system's usefulness as a motivating factor in maintaining efficient library practice.



Statement	SD	D	MA	A	SA	MR	VI
4.1 Confidentiality and Access Control The ARK system ensures that all user information and activities are kept private and protected from unauthorized access. Only authorized users can view or manage specific data, so personal accounts and activities remain secure. This helps maintain the privacy and security of all information within the system.	0	0	1	14	25	4.6	SA
4.2 Data Protection and Integrity The ARK system keeps all stored data secure and protected from loss, corruption, or unauthorized modification. This means that important records — such as book lists, event proposals, and user profiles — remain accurate and safe from accidental deletion or hacking. It ensures that the system's information stays trustworthy and reliable for all users.	0	0	4	23	13	4.22	A
Average Weighted Mean						4.41	A

Table 4.4. Security

Legends: SD – Strongly Disagree, D – Disagree, MA – Moderately Agree, A – Agree, SA – Strongly Agree, MR – Mean Rating, VI – Verbal Interpretation.



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Results within the security dimension are indicative of majority views among users that the ARK system demonstrates safeguards which are adequate to protect against the loss of privacy or compromise of data. The mean rating for the Confidentiality and Access Control category is 4.60, which suggests the users feel they can be confident the system will restrict unauthorized access to disclosed information, and access to user activity. In other words, they feel the system will restrict access where it needs to in order to keep the user private. The mean rating for the Data Protection and Integrity dimension is 4.22, meaning users feel confident due to the way the system responds to questions about data accuracy and reliability so their data will be not corrupted or misused. Together, these findings suggest that the ARK system maintains strong data protection measures that foster trust and promote secure handling of user information. Therefore, the ARK system creates a secure environment for users so as to perform tasks with confidence that their data is protected and being managed properly.



Statement	SD	D	MA	A	SA	MR	VI
5.1 Learnability	1	0	2	8	29	4.6	SA
The ARK system is simple to learn and understand, even for first-time users. You should be able to navigate its features, search for books, view recommendations, or manage content without needing special training. Clear labels, instructions, and intuitive icons help users quickly get familiar with the system.							
5.2 Ease of Navigation	0	0	2	13	25	4.58	SA
The ARK system has a well-organized and user-friendly interface that makes it easy to find what you need. Menus, buttons, and text are clearly visible and logically placed, so users can browse without confusion or frustration. This design helps user efficiently access features and complete tasks.							
Average Weighted Mean	4.59						SA

Table 4.5. Usability

Legends: SD – Strongly Disagree, D – Disagree, MA – Moderately Agree, A – Agree, SA – Strongly Agree,
MR – Mean Rating, VI – Verbal Interpretation.



The usability evaluation received among the highest scores, indicating that participants viewed the ARK system as intuitive, simple, and feasible. With a score average of 4.6, the Learnability indicator indicates that, novice users are able to become oriented to the system and how to use it with minimal help, and have had some general experience with technology use, with sometimes no prior exposure at all. The interface, labels, and icons are designed and organized for users to become familiar with the ARK system and to complete their tasks effectively. The Ease of Navigation score of 4.58 shows the same logic for menus, buttons, and visuals and how they facilitate the process of browsing and locating the features that users need. This user-centered design allows for an ease of interactions that minimizes confusion, while in contrast allows for increased productivity. Ultimately, the ARK system is very usable and drives a comprehensible interface that lends itself to being accessible, efficient, and a pleasing user experience.

Statement	Weighted Mean	Verbal Interpretation
Functional Suitability	4.65	Strongly Agree
Performance Efficiency	4.4	Agree
Reliability	4.51	Strongly Agree
Security	4.41	Agree
Usability	4.59	Strongly Agree
Overall Weighted Mean	4.51	Strongly Agree

Table 4.6. General Evaluation

The results of the ARK system evaluation demonstrate that it is performing well in all areas assessed and received a positive overall rating from users. The



highest ratings were for Functional Suitability (4.65) and Usability (4.59) illustrating that the ARK has a full set of relevant, sensible, and usable features to accommodate a library setting. Reliability (4.51) also received a very high rating indicating the system is reliable in its operation, and minimizes down time, plus is able to recover from minor technical issues. Security (4.41) was rated acceptable, demonstrating that the ARK system adequately protects users' data and privacy. Performance Efficiency (4.40) was a little lower than the other areas, but still indicates effective system performance, with some opportunities for improvement for response time during heavy usage. In summary, the ARK system provides a reliable, safe, and efficient environment to facilitate library operations, while encouraging more patron engagement with a complete package of features, reliability, and user-friendly design.



Chapter 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter details the last sections of the study: summary of findings, conclusions for the attainment of the objectives of the study, and recommendations for further related studies or investigation

Summary

This research summary is presented according to the statement of objectives made in Chapter 1.

The study is entitled: *“Academic Resource and Knowledge (ARK): An AI-Driven Platform for Enhanced Learning and Knowledge Management”*. This work involves the design and development of an AI-driven library system for the Kaban ng Hiyas Congressional Library. The system aims to improve access to and organization of library resources through smart search and personalized recommendations. Additionally, it integrates an event management module to display upcoming library activities and allow users to submit event proposals. Administrators are provided with analytics and insights to monitor library usage and optimize resource management.

Objective 1: The smart search system integrates machine learning (ML), natural language processing (NLP), and artificial intelligence (AI) to deliver an intelligent and context-aware search experience. ML is implemented using the pre-trained



SentenceTransformer model (all-MiniLM-L6-v2), which transforms both user queries and library resources into dense vector embeddings, allowing the system to identify relevant materials even when queries do not exactly match keywords. NLP techniques process and interpret human language, applying tokenization, context understanding, and semantic encoding to capture meaning and relationships between words and phrases, while also supporting summarization of retrieved content into concise, human-readable outputs. AI orchestrates intelligent decision-making by evaluating similarity scores from the ML embeddings, ranking the most relevant resources, and generating coherent responses. When the library dataset lacks sufficient information, the AI module applies fallback reasoning using **GPT-4** to provide accurate answers while still recommending relevant books. The combination of these models, SentenceTransformer for semantic embeddings and GPT-4 for contextual reasoning, enables the system to go beyond traditional keyword searches, bridging the gap between user intent and available resources and effectively satisfying Objective 1.

Objective 2: The ARK system incorporates a comprehensive data-driven recommendation engine designed to enhance user engagement and streamline book discovery. This system combines multiple recommendation strategies including semantic similarity, category-based, trending within category, and author-based recommendations to deliver personalized suggestions. Semantic similarity recommendations leverage vector embeddings to identify contextually related books beyond exact title or keyword matches, while category-based and



trending recommendations highlight books thematically aligned with the user's current interests and popular choices within each category. Author-based recommendations provide continuity by suggesting other works from the same author, encouraging deeper exploration. All these strategies are aggregated in a combined recommendations method, ensuring that suggestions are diverse, relevant, and user-centered. Supporting this recommendation system, an analytics dashboard provides administrators with insights on user engagement and category usage. Visualizations, such as line graphs of the trending categories, allow library staff to monitor trends and make informed decisions to optimize library collections. By integrating personalized machine learning based recommendations with usage analytics, the ARK system aligns with modern hybrid recommender frameworks, improving both user satisfaction and operational management of library resources

Objective 3: This aimed to integrate an event calendar in the ARK Library System to display upcoming library activities and allow users to submit event proposals, thereby enhancing user engagement and supporting library outreach. The system provides a structured and interactive interface where users can view events in a clear layout, submit proposals with detailed information and attachments, and receive confirmation of their submissions. Proposal submissions are restricted to registered users to ensure accountability, and the system validates inputs such as event details, date, time, and file formats. On the administration side, librarians can review, accept, or manage proposals, and accepted events can be posted for public visibility. Automated email notifications keep users informed of the status of



their submissions, promoting transparency and communication between users and administrators. The system also differentiates upcoming and past events, maintaining an organized display for users and providing an event summary dashboard to visualize overall library activity. This workflow ensures efficient management, encourages active participation, and effectively fulfills Objective 3.

Objective 4: The ARK (Academic Resource and Knowledge) system was evaluated using ISO/IEC 25010:2023 standards to assess its software quality across multiple dimensions. The assessment focused on functional suitability, performance efficiency, reliability, security, and usability. Functional suitability considered whether the system provides complete and appropriate features, such as AI-powered smart search, personalized book recommendations, and an event proposal submission feature. Performance efficiency examined the system's responsiveness and ability to handle multiple users simultaneously. Reliability evaluated the system's availability and fault tolerance under minor technical disruptions. Security assessed confidentiality, access control, and data integrity, ensuring user information and library records are protected. Usability focused on how easily users can learn and navigate the system, highlighting intuitive design and clear interface organization. Forty library visitors participated in the evaluation, providing insights into the system's effectiveness in supporting library activities and academic work.



Conclusion

Objective 1: The successful implementation of Objective 1, developing an intelligent, context-aware search system, directly benefits library users by making information retrieval faster, more accurate, and less dependent on exact keyword queries. By leveraging semantic embeddings through SentenceTransformer and contextual reasoning with GPT-4, the system interprets user intent and presents relevant results even when queries are vague or incomplete. This enhances the overall library experience by reducing search time, minimizing frustration, and ensuring that users can efficiently access materials that truly match their academic or research needs. Consequently, the system not only improves task efficiency but also supports learning, research, and informed decision-making, making the library a more practical and user-friendly resource for all visitors.

Objective 2: The ARK system successfully achieves its goal of providing a data-driven recommendation engine that delivers personalized book suggestions. By combining semantic similarity, category-based trends, popularity within categories, and author-based recommendations, the system ensures that users receive relevant and contextually meaningful suggestions, making it easier to discover new resources and engage more deeply with the library's collection. The integrated recommendations approach unifies these strategies, offering a diverse and user-centered experience that goes beyond simple popularity rankings. Additionally, the analytics dashboard provides administrators with actionable insights into category usage and user interactions. Visualizations, such as line graphs of the most



accessed categories, support informed decisions for optimizing library collections. Overall, this implementation enhances the user experience by making resource discovery more intuitive while empowering library management with data-driven insights for efficient and targeted collection development.

Objective 3: The integration of the event calendar in the ARK Library System successfully meets Objective 3 by providing an interactive and organized platform for managing library activities and event proposals. Users can easily view upcoming and past events, submit detailed proposals with attachments, and receive automated email notifications about their submission status, promoting transparency and active engagement. Limiting submissions to registered users and implementing input validation for event details, dates, and file formats ensures data accuracy and accountability. Administrators benefit from an efficient workflow for reviewing, approving, and managing proposals, with accepted events automatically displayed for public visibility. The event summary dashboard further allows monitoring of overall library activity and engagement trends. Overall, this module enhances user participation, streamlines communication, and strengthens the library's outreach and community-building efforts, making the system more interactive and responsive to user needs.

Objective 4: The Academic Resource and Knowledge (ARK) system successfully meets Objective 4 by demonstrating strong software quality in accordance with ISO/IEC 25010:2023 standards. Evaluation by forty respondents showed high



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ratings across key quality factors: functional suitability (4.65, SA), performance efficiency (4.40, A), reliability (4.51, SA), security (4.41, A), and usability (4.59, SA). These results indicate that the system effectively supports its core functions—smart search, personalized recommendations, and event proposal management—while remaining responsive during peak usage, stable under varying conditions, secure in data handling, and easy to navigate. Overall, the ARK system provides a reliable, efficient, and user-friendly platform that enhances library management, facilitates academic engagement, and ensures a secure and seamless experience for all users.

The ARK (Academic Resource and Knowledge) system successfully achieves its four objectives, delivering a comprehensive and user-centered platform for library management and academic engagement. The intelligent smart search (Objective 1) allows users to find relevant resources quickly, even with vague or incomplete queries, enhancing research efficiency and reducing frustration. The data-driven recommendation engine (Objective 2) provides personalized and contextually meaningful book suggestions, improving resource discovery and keeping users engaged while offering administrators actionable insights through the analytics dashboard. The event management module (Objective 3) streamlines the submission, approval, and monitoring of library events, fostering transparency, user participation, and community-building. Finally, the system's adherence to ISO/IEC 25010:2023 software quality standards (Objective 4) ensures that it is reliable, secure, efficient, and easy to use, providing



a seamless and trustworthy experience for all users. Collectively, these features create a practical, efficient, and interactive library platform that meets both user needs and administrative goals, enhancing learning, engagement, and overall library services.

Recommendation

Based on the findings and conclusions of the study, the following recommendations were presented to improve and extend the capabilities of the ARK (Academic Resource and Knowledge) Library System. During the development process, certain limitations were encountered, such as limited time for large-scale testing, the use of a controlled dataset, and the absence of a mobile version for wider accessibility. These factors restricted the full assessment of the system's performance under real deployment conditions.

Future researchers and developers are encouraged to enhance the system by integrating advanced features such as a **scan-to-search function**, which would allow users to scan book covers or barcodes to find similar resources, and a **speech-to-search option** that enables voice-based queries for faster and more accessible searching. The addition of a **user content-sharing module** is also recommended, allowing students and faculty to upload and share their own research outputs, articles, or video lessons to promote knowledge exchange within the academic community.

Moreover, the development of a **mobile application version** is advised to



make the system more convenient and accessible, enabling users to access library services anytime and anywhere. Integrating an **AI-powered chat assistant** could also enhance user support by providing instant guidance, answering inquiries, and improving interaction between users and the system.

These recommendations are intended to guide future developers and researchers in improving the ARK Library System, ensuring that it continues to evolve as an intelligent, accessible, and user-centered platform that supports both academic learning and community engagement.



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APPENDIX A- LETTER TO CONDUCT RESEARCH

March 15, 2023

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Mr. Merry Mayo Capang
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Maynila Ctr., Mandaluyong, Metro Manila

To whom it may concern,

Dear Sir,

I hope this letter finds you well and in good spirits. I'm writing on behalf of a group of Information Technology students at Jose Rizal University. Our group is made up of dedicated students who are eager to participate in a meaningful research project that meets both our academic requirements, specifically Project 1, Thesis I, and the needs of your esteemed organization.

Our research group includes the following members:

CATAPANG, Alberto	Project Manager / Technical Writer
ALONZO, John Gabriel	Network Designer / UI Designer
BALONDO, Nathan	Software Engineer / Programmer
YORD, Leonardo Artero	Systems Analyst / Database Designer

With the guidance and approval of our subject adviser, Dr. Atanacio, we are eager to begin a research project that will include system development tailored to your organization's specific requirements and challenges. Our goal is to contribute to the improvement of processes, efficiency, and overall effectiveness by implementing innovative technological solutions.

Our proposed research plan includes, but is not limited to, the following activities:

1. Data Gathering: We intend to collect relevant data and information about your organization's current systems, workflow, and areas for improvement.

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2. Surveys: We intend to conduct surveys among stakeholders in your organization to gather insights, feedback, and suggestions that will help shape our system development strategy.
3. Research Activities: We will conduct rigorous analysis, design, development, testing, and evaluation phases to ensure the viability and effectiveness of the proposed system solutions.

We are committed to maintaining the highest levels of professionalism, confidentiality, and integrity throughout the research process. Furthermore, we guarantee that our research activities will be carried out in strict accordance with ethical principles and all applicable organizational policies and regulations. We would appreciate your cooperation and support in facilitating access to relevant resources, data, and personnel who can provide valuable insights and assistance during our research.

We believe that this collaborative effort will not only improve our learning experience, but will also provide tangible benefits to your organization. We are excited to contribute our skills, knowledge, and enthusiasm to this project and look forward to working closely with you.

Thank you for considering our proposal. If you have any questions or need more information, please do not hesitate to contact us at the following numbers or email addresses.

Yours sincerely,


Alfredo Capang
Project Manager for HexTech-JRU

Noted by


Dr. Marko Atanacio
Subject and Technical Adviser



APPENDIX B- EVALUATION QUESTIONS

1. Functional Suitability

Description (optional)

...

*

1.1 Functional Completeness

The ARK system provides all the key features that make the library experience smoother and more effective. It includes tools like an AI-powered smart search for quickly finding books, personalized book recommendations suited to each user, and an event proposal submission feature for sharing ideas or activities. These functions ensure that users can easily access and manage what they need within the system.

(Ang ARK system ay naglalaman ng lahat ng mahahalagang tampok na nagpapadali at nagpapahusay sa paggamit ng library. Kabilang dito ang AI-powered smart search para sa mabilis na paghahanap ng libro, personalized book recommendations na akma sa bawat user, at event proposal submission feature para sa pagpapasa ng mga ideya o aktibidad. Tinitiyak ng mga tampok na ito na users ay madaling makakagamit at makakapamahala sa system.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐

1.2 Functional Appropriateness

*

The ARK system provides useful and meaningful results that truly fit what user need inside the library. When you search for a topic, it gives relevant and accurate materials that match your studies. The features are designed to help you with your academic work, research, or participation in library events — making it not just functional but also practical for everyday user use.

(Nagbibigay ang ARK system ng makabuluhan at kapaki-pakinabang na mga resulta na talagang naaayon sa pangangailangan ng mga user sa loob ng library. Kapag naghahanap ka ng isang paksa, nagbibigay ito ng mga aklat at materyales na may kaugnayan at eksakto sa iyong pag-aaral. Ang mga feature nito ay idinisenyo upang makatulong sa iyong mga academic requirements, research, at maging sa mga aktibidad ng library — kaya ito ay hindi lamang gumagana, kundi kapaki-pakinabang din sa araw-araw na paggamit ng mga user.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐



2. Performance Efficiency

Description (optional)

★

2.1 Time Behavior

The ARK system loads pages, search results, and other information quickly so that users do not experience long waiting times. Whether searching for a book, checking upcoming events, or updating records, the system responds within a few seconds. This fast performance helps both user complete tasks efficiently, creating a smooth, productive, and frustration-free library experience even during busy hours.

(Mabilis mag-load ng mga pahina, search results, at iba pang impormasyon ang ARK system upang maiwasan ang mahabang paghihintay. Kapag naghahanap ng libro, tiningnan ang mga paparating na events, o nag-update ng records, agad mong makikita ang resulta. Ang mabilis na takbo ng system ay tumutulong sa user na matapos ang kanilang mga gawain nang mabilis at maayos, na nagreresulta sa mas produktibo at walang abalang karanasan sa library.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐

2.2 Capacity

The ARK system remains stable and responsive even when many users access it at the same time. It can handle multiple tasks – such as searching for books or updating records – without lagging or crashing. This ensures a smooth and reliable experience for all users, even during busy library hours.

(Ang ARK system ay nananatiling matatag at mabilis kahit sabay-sabay itong ginagamit ng maraming user. Kaya nitong magpatakbo ng maraming gawain tulad ng paghahanap ng libro o pag-update ng records nang walang pagkaantala o error. Tinitiyak nito ang maayos at maaasahang karanasan para sa lahat ng gumagamit kahit sa oras ng kasagsagan sa library.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐

**3. Reliability**

Description (optional)

*

3.1 Availability

The ARK system should always be accessible and ready to use whenever needed. Users should be able to perform tasks such as searching for books, viewing events, or managing records without experiencing sudden shutdowns or server errors. This reliability ensures that the system can be depended on anytime for all library-related activities.

(Ang ARK system ay dapat laging accessible at handang gamitin anumang oras na kailangan ito. Dapat ay maayos na maisagawa ng user ang mga gawain tulad ng paghahanap ng libro, pagtingin sa mga events, o pamamahala ng records nang walang biglaang pagsasara o error. Tinitiyak nito ang maaasahang paggamit ng system para sa lahat ng aktibidad sa library.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐

3.2 Fault Tolerance

The ARK system should keep working smoothly even if small technical issues occur, such as weak internet connection, minor bugs, or short system interruptions. It must be able to recover quickly and continue operating without disrupting your tasks. This ensures that users can still accomplish what they need without losing progress or experiencing major inconvenience.

(Ang ARK system ay dapat patuloy na gumana nang maayos kahit magkaroon ng maliliit na teknikal na problema tulad ng mahina ang internet connection, maliit na bug, o panandaliang error sa system. Dapat nitong agad na maibalik ang normal na takbo nang hindi naaapektuhan ang ginagawa ng user. Sa ganitong paraan, magpapatuloy pa rin ang mga gawain nang walang abala o pagkawala ng progreso.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐



4. Security

Description (optional)

...

4.1 Confidentiality and Access Control

The ARK system ensures that all user information and activities are kept private and protected from unauthorized access. Only authorized users can view or manage specific data, so personal accounts and activities remain secure. This helps maintain the privacy and security of all information within the system.

(Tinitiyak ng ARK system na ang lahat ng impormasyon at gawain ng mga user ay nananatiling pribado at protektado laban sa hindi awtorisadong pag-access. Tanging mga awtorisadong user lamang ang makakakita o makakapamahala ng tiyak na data, kaya nananatiling ligtas ang mga personal na account at aktibidad. Sa ganitong paraan, napapanatili ang privacy at seguridad ng lahat ng impormasyon sa system.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐

4.2 Data Protection and Integrity

The ARK system keeps all stored data secure and protected from loss, corruption, or unauthorized modification. This means that important records — such as book lists, event proposals, and user profiles — remain accurate and safe from accidental deletion or hacking. It ensures that the system's information stays trustworthy and reliable for all users.

(Pinapanatili ng ARK system na ligtas at protektado ang lahat ng nakaimbak na data laban sa pagkawala, pagkasira, o hindi awtorisadong pagbabago. Ibig sabihin, ang mga mahalagang record gaya ng listahan ng libro, event proposals, at user profiles ay nananatiling tama at buo, at hindi madaling mabura o ma-hack. Tinitiyak nito na ang lahat ng impormasyon sa system ay mapagkakatiwalaan at maaasahan ng mga user.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐

**5. Usability**

Description (optional)

...

*

5.1 Learnability

The ARK system is simple to learn and understand, even for first-time users. You should be able to navigate its features, search for books, view recommendations, or manage content without needing special training. Clear labels, instructions, and intuitive icons help users quickly get familiar with the system.

(Madaling matutunan at maintindihan ang ARK system kahit bago ka pa lang gumagamit nito. Dapat ay kaya mong mag-navigate sa mga features, maghanap ng libro, tingnan ang recommendations, o mag-manage ng content nang hindi kailangan ng espesyal na training. Ang malinaw na labels, instructions, at intuitive na icons ay tumutulong sa mabilis na pagka-familiar ng user sa system.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐

5.2 Ease of Navigation

The ARK system has a well-organized and user-friendly interface that makes it easy to find what you need. Menus, buttons, and text are clearly visible and logically placed, so users can browse without confusion or frustration. This design helps user efficiently access features and complete tasks.

(May maayos at user-friendly na interface ang ARK system na nagpapadali sa paghahanap ng kailangan mo. Ang menus, buttons, at text ay malinaw at maayos ang pagkakalagay, kaya hindi malilito o mahihirapan ang user sa pag-browse. Ang disenyo na ito ay tumutulong sa parehong staff na mabilis na ma-access ang features at matapos ang kanilang mga gawain.)

Strongly Disagree 1 2 3 4 5 Strongly Agree

☐ ☐ ☐ ☐ ☐



APPENDIX C- EVALUATION RESULT

Timestamp	Educational Level	Are you from Mandakuyang City?	3.1 Functional Completeness The ARK system provides all the key features (Ang ARK system ay naglalaman ng lahat ng ito)	3.2 Functional Appropriateness The ARK system provides useful and means (Nagkakaape ang ARK system ng makabuluhan)	3.3 Time Behavior The ARK system loads pages, search results, (Mabilis mag-load ng mga pahina, search results)	3.4 Capacity The ARK system remains stable and responsive (Ang ARK system ay nananatili nang matatag at mabilis)	3.5 Availability The ARK system should always be accessible (Ang ARK system ay dapat laging accessible sa)	3.6 Fault Tolerance The ARK system should be able to handle errors (Ang ARK system ay dapat magaling sa pagpapaligay ng mga error)
10/10/2025 17:12:22	College	Yes	5	5	4	4	4	5
10/10/2025 18:26:24	College	San Juan City	5	4	5	5	5	5
10/10/2025 20:04:52	College	Quezon City	5	4	4	5	5	5
10/10/2025 20:05:21	College	Yes	5	5	5	4	5	5
10/10/2025 20:13:33	Graduate	Yes	5	5	5	5	5	5
10/10/2025 20:16:00	College	Manila	5	5	5	5	5	5
10/10/2025 20:26:13	College	Taguig City	5	4	4	5	5	5
10/10/2025 20:28:27	College	no	4	3	5	5	4	5
10/10/2025 20:32:14	College	Calamba	4	4	4	5	5	5
10/10/2025 20:33:41	College	Taguig City	5	5	3	5	5	5
10/10/2025 20:47:53	Senior High School	Makati City	5	4	5	4	5	5
10/10/2025 20:59:22	Professional	makati city	5	5	5	4	5	5
10/10/2025 21:15:30	College	Yes	4	5	4	3	4	5
10/10/2025 21:17:31	College	Yes	5	5	5	5	5	5
10/10/2025 21:19:06	Senior High School	Yes	3	4	1	1	1	5
10/10/2025 21:19:31	Senior High School	Yes	5	5	5	3	5	5
10/10/2025 21:20:32	College	Yes	5	4	3	5	5	5
10/10/2025 21:20:51	Senior High School	Yes	5	5	4	5	5	5

3.1 Availability The ARK system should always be accessible (Ang ARK system ay dapat laging accessible sa)	3.2 Fault Tolerance The ARK system should keep working smooth (Ang ARK system ay dapat patuloy na gumana)	4.1 Confidentiality and Access Control The ARK system ensures that all user roles (Tatitiyak ng ARK system na ang lahat ng mga)	4.2 Data Protection and Integrity The ARK system keeps all stored data secure (Pinapanatili ng ARK system na ligtas at protektado)	5.1 Usability The ARK system is simple to learn and use (Mababasa matutuhan at maintutuhan ang ARK)	5.2 Ease of Navigation The ARK system has a well-organized and up-to-date (May maayos at user-friendly na interface ang ARK)	Column 15	Spreadsheets
4	5	5	5	4	5	4	
5	5	5	5	4	5	5	
5	5	4	5	5	4	5	
4	5	5	5	5	5	5	
5	5	5	5	4	5	4	
5	5	5	5	5	5	5	
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4	5	3	4	4	5	3	
3	4	4	5	4	5	4	
5	5	5	5	5	5	5	
4	4	5	4	5	5	4	
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5	5	5	5	5	5	5	
5	5	5	5	4	5	5	
5	5	5	5	5	5	5	
4	5	5	5	5	5	5	
5	5	5	4	4	5	4	
3	3	2	4	5	4	4	



1. Functional Suitability

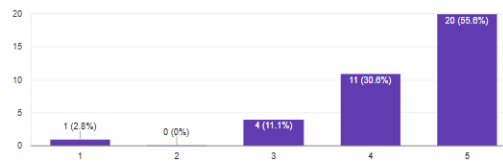
1.1 Functional Completeness

[Copy chart](#)

The ARK system provides all the key features that make the library experience smoother and more effective. It includes tools like an AI-powered smart search for quickly finding books, personalized book recommendations suited to each user, and an event proposal submission feature for sharing ideas or activities. These functions ensure that users can easily access and manage what they need within the system.

(Ang ARK system ay naglalaman ng lahat ng mahalagang tampok na nagpapadali at nagpapahusay sa paggamit ng library. Kabilang dito ang AI-powered smart search para sa mabilis na paghahanap ng libro, personalized book recommendations na akma sa bawat user, at event proposal submission feature para sa pagpapasa ng mga ideya o aktibidad. Tinitiyak ng mga tampok na ito na users ay madaling makakagamit at makakapamahala sa system.)

36 responses



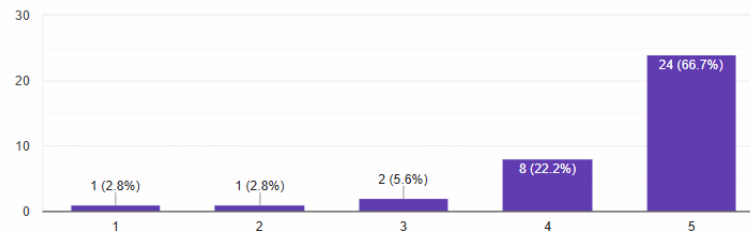
1.2 Functional Appropriateness

[Copy chart](#)

The ARK system provides useful and meaningful results that truly fit what user need inside the library. When you search for a topic, it gives relevant and accurate materials that match your studies. The features are designed to help you with your academic work, research, or participation in library events – making it not just functional but also practical for everyday user use.

(Nagbibigay ang ARK system ng makabuluhan at kapaki-pakinabang na mga resulta na talagang naaayon sa pangangailangan ng mga user sa loob ng library. Kapag naghahanap ka ng isang paksa, nagbibigay ito ng mga aklat at materyales na may kaugnayan at eksakto sa iyong pag-aaral. Ang mga feature nito ay idinisenyo upang makatulong sa iyong mga academic requirements, research, at maging sa mga aktibidad ng library – kaya ito ay hindi lamang gumagana, kundi kapaki-pakinabang din sa araw-araw na paggamit ng mga user.)

36 responses





2. Performance Efficiency

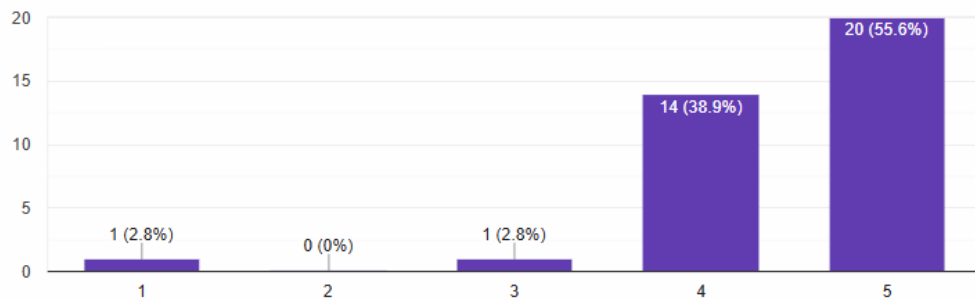
[Copy chart](#)

2.1 Time Behavior

The ARK system loads pages, search results, and other information quickly so that users do not experience long waiting times. Whether searching for a book, checking upcoming events, or updating records, the system responds within a few seconds. This fast performance helps both user complete tasks efficiently, creating a smooth, productive, and frustration-free library experience even during busy hours.

(Mabilis mag-load ng mga pahina, search results, at iba pang impormasyon ang ARK system upang maiwasan ang mahabang paghihintay. Kapag naghahanap ng libro, tiningnan ang mga paparating na events, o nag-update ng records, agad mong makikita ang resulta. Ang mabilis na takbo ng system ay tumutulong sa user na matapos ang kanilang mga gawain nang mabilis at maayos, na nagreresulta sa mas produktibo at walang abalang karanasan sa library.)

36 responses



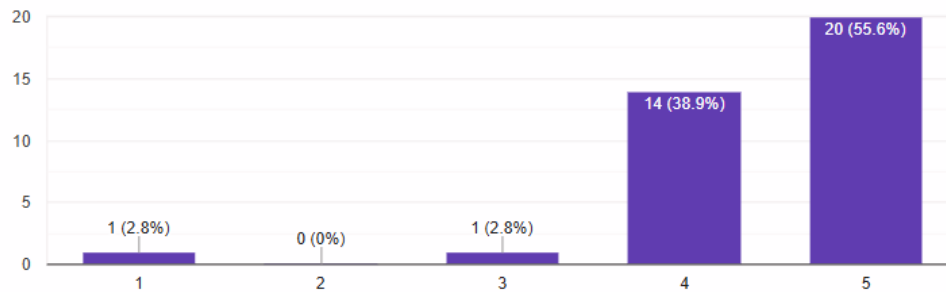
[Copy chart](#)

2.2 Capacity

The ARK system remains stable and responsive even when many users access it at the same time. It can handle multiple tasks — such as searching for books or updating records — without lagging or crashing. This ensures a smooth and reliable experience for all users, even during busy library hours.

(Ang ARK system ay nananatiling matatag at mabilis kahit sabay-sabay itong ginagamit ng maraming user. Kaya nitong magpatakbo ng maraming gawain tulad ng paghahanap ng libro o pag-update ng records nang walang pagkaantala o error. Tinitiyak nito ang maayos at maaasahang karanasan para sa lahat ng gumagamit kahit sa oras ng kasagsagan sa library.)

36 responses





3. Reliability

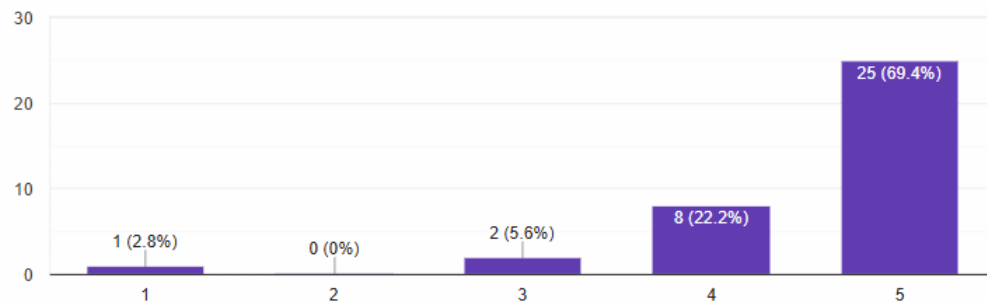
3.1 Availability

[Copy chart](#)

The ARK system should always be accessible and ready to use whenever needed. Users should be able to perform tasks such as searching for books, viewing events, or managing records without experiencing sudden shutdowns or server errors. This reliability ensures that the system can be depended on anytime for all library-related activities.

(Ang ARK system ay dapat laging accessible at handang gamitin anumang oras na kailangan ito. Dapat ay maayos na maisagawa ng user ang mga gawain tulad ng paghahanap ng libro, pagtingin sa mga events, o pamamahala ng records nang walang biglaang pagsasara o error. Tinitiyak nito ang maaasahang paggamit ng system para sa lahat ng aktibidad sa library.)

36 responses



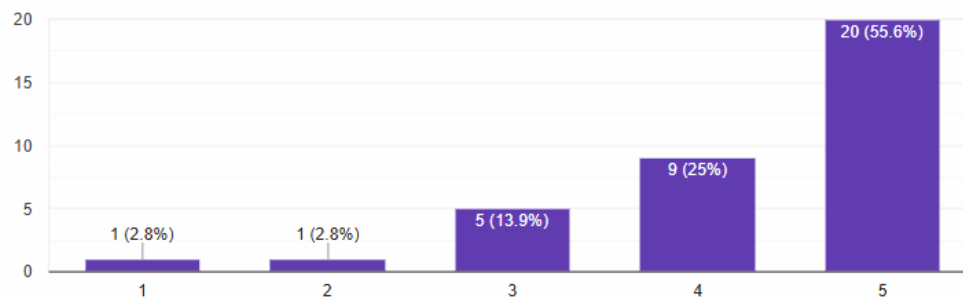
[Copy chart](#)

3.2 Fault Tolerance

The ARK system should keep working smoothly even if small technical issues occur, such as weak internet connection, minor bugs, or short system interruptions. It must be able to recover quickly and continue operating without disrupting your tasks. This ensures that users can still accomplish what they need without losing progress or experiencing major inconvenience.

(Ang ARK system ay dapat patuloy na gumana nang maayos kahit magkaroon ng maliliit na teknikal na problema tulad ng mahina ang internet connection, maliit na bug, o panandaliang error sa system. Dapat nitong agad na maibalik ang normal na takbo nang hindi naaapektuhan ang ginagawa ng user. Sa ganitong paraan, magpapatuloy pa rin ang mga gawain nang walang abala o pagkawala ng progreso.)

36 responses





4. Security

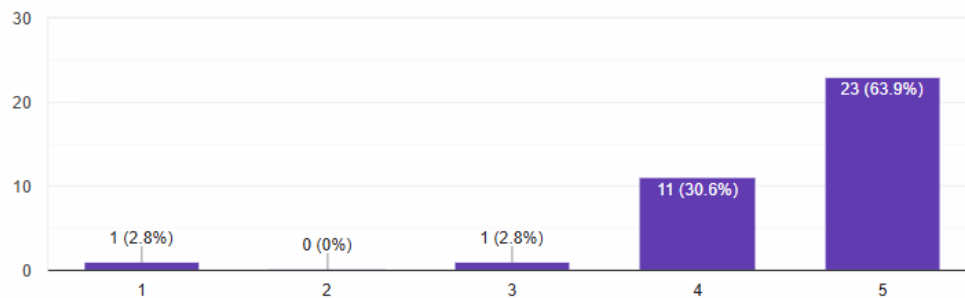
4.1 Confidentiality and Access Control

[Copy chart](#)

The ARK system ensures that all user information and activities are kept private and protected from unauthorized access. Only authorized users can view or manage specific data, so personal accounts and activities remain secure. This helps maintain the privacy and security of all information within the system.

(Tinitiyak ng ARK system na ang lahat ng impormasyon at gawain ng mga user ay nananatiling pribado at protektado laban sa hindi awtorisadong pag-access. Tanging mga awtorisadong user lamang ang makakakita o makakapamahala ng tiyak na data, kaya nananatiling ligtas ang mga personal na account at aktibidad. Sa ganitong paraan, napapanatili ang privacy at seguridad ng lahat ng impormasyon sa system.)

36 responses



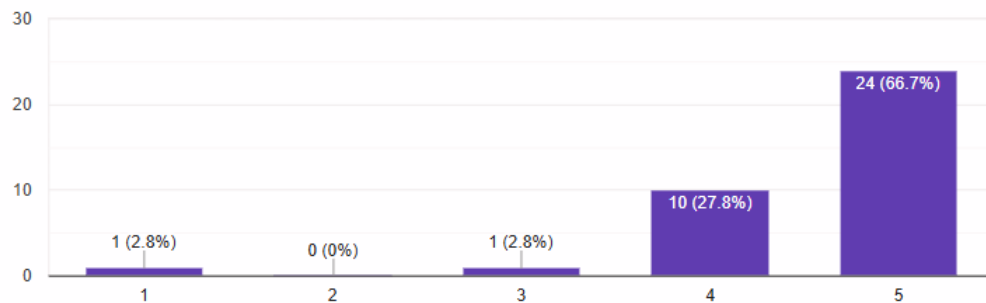
[Copy chart](#)

4.2 Data Protection and Integrity

The ARK system keeps all stored data secure and protected from loss, corruption, or unauthorized modification. This means that important records — such as book lists, event proposals, and user profiles — remain accurate and safe from accidental deletion or hacking. It ensures that the system's information stays trustworthy and reliable for all users.

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36 responses





5. Usability

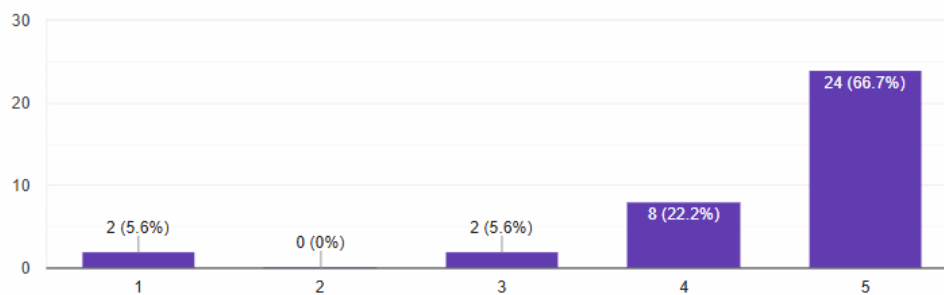
5.1 Learnability

[Copy chart](#)

The ARK system is simple to learn and understand, even for first-time users. You should be able to navigate its features, search for books, view recommendations, or manage content without needing special training. Clear labels, instructions, and intuitive icons help users quickly get familiar with the system.

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36 responses



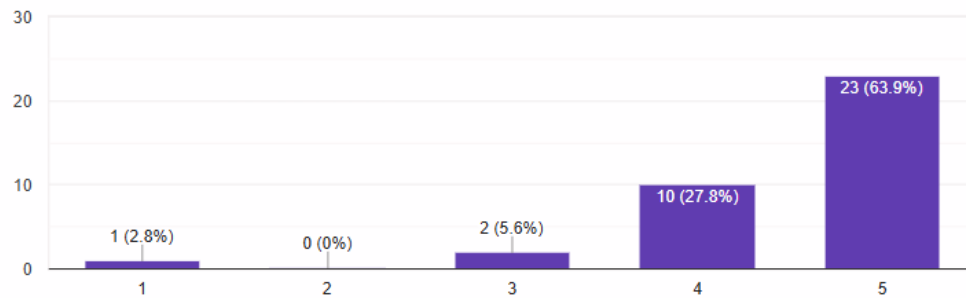
[Copy chart](#)

5.2 Ease of Navigation

The ARK system has a well-organized and user-friendly interface that makes it easy to find what you need. Menus, buttons, and text are clearly visible and logically placed, so users can browse without confusion or frustration. This design helps user efficiently access features and complete tasks.

(May maayos at user-friendly na interface ang ARK system na nagpapadali sa paghahanap ng kailangan mo. Ang menus, buttons, at text ay malinaw at maayos ang pagkakalagay, kaya hindi malilito o mahihirapan ang user sa pag-browse. Ang disenyo na ito ay tumutulong sa parehong staff na mabilis na ma-access ang features at matapos ang kanilang mga gawain.)

36 responses





APPENDIX D- SYSTEM SURVEY WITH LIBRARY STAFF





APPENDIX E- Adviser Acceptance Form

F-COL-011
06-15-2013

JOSÉ RIZAL UNIVERSITY

ADVISER ACCEPTANCE FORM

TO: DR. MARLON ATANACIO
Name of Faculty

Hereby accepts the proponent's project study:

Hextech
Group Name

1. Alonzo, John Gabriel
2. Balondo, Naithan
3. Catapang, Alberto
4. Yoro, Leonardo Antero
5. _____

as project study advisees for their study.

Academic Resource and Knowledge (ARK): An AI-Driven Platform for
Enhanced Learning and Knowledge Management
Title of the Project Study

AI-Powered Search and Recommendation
Sub-System 1

User Engagement and Knowledge Contribution
Sub-System 2

Content Management and Administration
Sub-System 3

For the School Year: 2024-2025

CONFORME:


DR. MARLON ATANACIO
Signature of Adviser Over Printed Name



Bio-Data of Authors

ALBERTO CATAPANG

As an IT student with a solid technical foundation, I am eager to expand my knowledge, particularly in data analytics. Passionate about learning data analysis, visualization, and machine learning, I aim to contribute to data-driven projects and enhance decision-making processes.



EDUCATION

COLLEGE

Jose Rizal University
Mandaluyong City
2022-Present
Bachelor of Science in Information
Technology

SENIOR HIGH SCHOOL

Jose Rizal University
Mandaluyong City
2020-2022-Present
Humanities and Social Sciences

SECONDARY

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Colleges Inc.
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2016-2020

PRIMARY

Nasugbu West Central School
Nasugbu, Batangas
2010-2016

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- ✉ alberto.catapang@my.jru.edu
- 📍 Sgt. Bumatay street, Mandaluyong City

SKILLS

- Python (Fundamentals knowledge in programming and data analysis)
- PHP (Fundamentals knowledge in web development and server-side scripting)
- Database Management (Fundamentals knowledge in database design and SQL queries)
- HTML (Fundamentals knowledge in front-end web development and page structure)
- CSS (Fundamentals knowledge in styling web pages and responsive design)



NAITHAN BALONDO

Driven by a passion for technology, I am an IT student eager to expand my knowledge in areas such as data analytics, software development, and system optimization. I aim to apply my skills to solve complex problems and contribute to innovative projects



EDUCATION

SENIOR HIGH SCHOOL/COLLEGE

Jose Rizal University 80 Shaw Boulevard, Mandaluyong, 1552 Metro Manila 2019 - Present Bachelor of Science in Information Technology

SECONDARY

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PRIMARY

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- Database Management (Fundamentals knowledge in database design and SQL queries)
- HTML (Fundamentals knowledge in front-end web development and page structure)
- CSS (Fundamentals knowledge in styling web pages and responsive design)



John Gabriel Alonzo

Being an Information Technology student with well-structured technical knowledge, having a passion for web developing and cybersecurity, I am willing to improve, learn new skills, and gather more information that I can utilize for future works. I am setting a goal to contribute to developing web-based platforms and securing digital information in our innovating digital world.



EDUCATION

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Mandaluyong City, Metro Manila
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2020 School Year Culinary Arts

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PRIMARY

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Pangasinan 2009 - 2015 School Year

AWARDS AND RECOGNITION

CompTIA ITF+ Certified Candidate

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- johngabriel.alonzo@my.jru.edu
- Alaminos City, Pangasinan

SKILLS

- HTML (Fundamentals knowledge in front-end web development and pages structure)
- CSS (Fundamentals knowledge in styling web pages and responsive design)
- Java (Fundamentals knowledge in programming)
- Python (Fundamentals knowledge in programming and data analysis)
- PHP (Fundamentals knowledge in web development and sever-side scripting)
- Database Management (Fundamentals knowledge in database design and SQL queries)



LEONARDO ANTERO YORO

As an IT student with a strong technical foundation, I am excited to deepen my knowledge in areas like data analytics, database management, and machine learning. My passion lies in mastering data analysis, visualization, and machine learning techniques, with the goal of contributing to data-driven projects and improving decision-making processes.



EDUCATION

COLLEGE

Jose Rizal University 80 Shaw Boulevard, Mandaluyong, 1552 Metro Manila 2019 - Present Bachelor of Science in Information Technology

SENIOR HIGH SCHOOL

Mataas na Paaralang Neptali A. Gonzales Nueve de Pebrero, Brgy. Mauway, Mandaluyong City Graduated S.Y 2021-2022

SECONDARY

Mandaluyong High School Aglipay St., Brgy. Poblacion, Mandaluyong City Finished S.Y. 2019-2020

PRIMARY

Doña Pilar C. Gonzaga Elementary School M. Gonzaga, Mandaluyong St., Mandaluyong City Graduated S.Y. 2015-2016

AWARDS AND RECOGNITION

- CompTIA ITF+ Certified Candidate
- Pearson database certification passer
- Academic year 2023 - 2024 Dean's Lister
- Academic year 2024 - 2025 Dean's Lister
- DOST-SEI scholar

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SKILLS

- Python (Fundamentals knowledge in programming and data analysis)
- PHP (Fundamentals knowledge in web development and server-side scripting)
- Database Management (Fundamentals knowledge in database design and SQL queries)
- HTML (Fundamentals knowledge in front-end web development and page structure)
- CSS (Fundamentals knowledge in styling web pages and responsive design)
- Java (Fundamentals knowledge in object-oriented programming and application development)
- VB.NET (Fundamentals knowledge in Windows application development and programming)



JOSÉ RIZAL UNIVERSITY COLLEGE DIVISION

Grammarian Certificate

Jose Rizal University
College of Computer Studies and Engineering
Department of Information Technology

LANGUAGE EDITING CERTIFICATE

This document certifies that the manuscript listed below was edited for proper English Language, grammar, punctuation, spelling, and over-all style by **Dr. Maria Katrina Rodriguez** and this thesis is ready for final bookbinding.

Manuscript Title: **“Academic Resource and Knowledge (ARK): An AI-Driven Platform for Enhanced Learning and Knowledge Management”**

Authors:

JOHN GABRIEL ALONZO
NAITHAN BALONDO
ALBERTO CATAPANG
LEONARDO ANTERO YORO

Date Issued: December 1, 2025


MARIA KATRINA P. RODRIGUEZ, Ed.D.

Language Editor



Certificate of Turnover and Acceptance

Jose Rizal University
College of Computer Studies and Engineering
Department of Information Technology

CERTIFICATE OF TURNOVER & ACCEPTANCE

This is to certify that the Academic Resource and Knowledge (ARK): An AI-Driven Platform for Enhanced Learning and Knowledge Management has been officially turned over to the client, Kaban ng hiyas congressional library, by Alberto Catapang, Leonardo Antero Yoro, Naithan Balondo, and John Gabriel Alonzo, of the group Hextech, on December 9, 2025 at Maysilo Cir, Mandaluyong City, Metro Manila

As part of the turnover process:

1. The system has been fully delivered to the client in its complete and operational form.
2. A detailed demonstration of the system's features, functionalities, and usage was conducted for the client, ensuring clarity and understanding of its operations.
3. The client has acknowledged the receipt of the system and expressed satisfaction with its presentation and functionalities.

For the Client:

Name: MERCEDITH M. CAPANG

Signature: _____

Position: HEAD-Of-C

For the Developer/Team:

Name: Alberto Catapang

Signature: _____

Position: Project Manager





Acknowledgment

The researchers would like to express their heartfelt gratitude to the following individuals and institutions who have contributed their support, guidance, and encouragement throughout the development and completion of this capstone project.

First and foremost, we thank God Almighty for granting us wisdom, strength, and perseverance. His divine guidance has been our constant source of inspiration and clarity in every stage of this research.

Our deepest appreciation is extended to Dr. Marlon Atanacio, our Technical Adviser, for his invaluable time, patience, and expert guidance. His insights and constructive feedback greatly shaped and strengthened the direction and quality of this study.

We are sincerely grateful to our client, Mrs. Mercy Mago Capanang, Officer-in-Charge of the Kaban ng Hiyas Congressional Library in Mandaluyong City, for granting us access to essential resources and for supporting our data-gathering activities. Her cooperation and assistance made this project possible.

Our sincerest thanks also go to the faculty members who supported our academic journey, especially Dr. Ronel Bilog and Dr. Sharmaine Maglapuz, whose helpful suggestions, consistent encouragement, and professional guidance contributed significantly to the improvement of this research.

The researchers extend their individual acknowledgments as follows:

Yoro expresses profound gratitude to the Department of Science and Technology Science Education Institute (DOST SEI) for providing financial assistance and thesis allowance. Deepest thanks are also given to his parents, Mrs. Mary Jane Yoro and Mr. Denise Yoro, his younger sister Dominique Yoro, and his significant other, Ms. January Baldomero, for their unwavering support and encouragement throughout this journey.

Catapang wishes to sincerely thank his parent, especially his mother Mrs. Sonia Catapang, for their love, guidance, and motivation. He also extends heartfelt appreciation to his significant other, Ms. Stacey Alemaña, for her constant support, and to his organization, the Rizalian Ambassadors (RAMS), for their friendship and inspiration.

Balondo conveys heartfelt gratitude to his parents, especially his mother Ms. Gemma Balondo, for their sacrifices, love, and support, as well as to his



JOSÉ RIZAL UNIVERSITY COLLEGE DIVISION

significant other, Ms. Via Bianca Mercado, for her patience, motivation, and unwavering encouragement.

Alonzo extends his deep appreciation to his family, especially to his mother, Mrs. Victoria Alonzo; to his father, Mr. Leonardo Alonzo; and his siblings, Cielo Alonzo & Carlene Alonzo, for their steadfast support and guidance. He also thanks his peers from the Rizalian Ambassadors (RAMS) for the camaraderie and inspiration that helped him persevere through this research.

To our friends and classmates who offered moral support, shared insights, and motivated us along the way, thank you for reminding us that we were never alone in this journey.

Finally, our profound gratitude goes to everyone who, in one way or another, contributed to the success of this study.

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Dedication

This research is dedicated to the people who inspired, guided, and strengthened us throughout this journey.

To our families, thank you for your endless love, patience, and support. Your belief in our abilities motivated us to keep striving, even in moments of doubt and difficulty. Your sacrifices and encouragement became our foundation and driving force.

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Above all, this work is dedicated to God Almighty, who granted us wisdom, resilience, and clarity. His guidance has been our constant light throughout this endeavor.

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